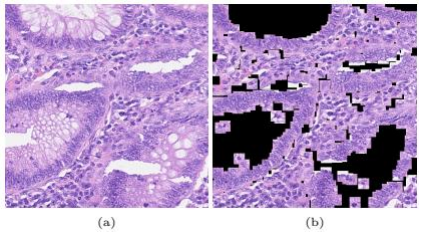
ELEC 5690 – Advanced Topics in Artificial Intelligence for Medical Image Analysis



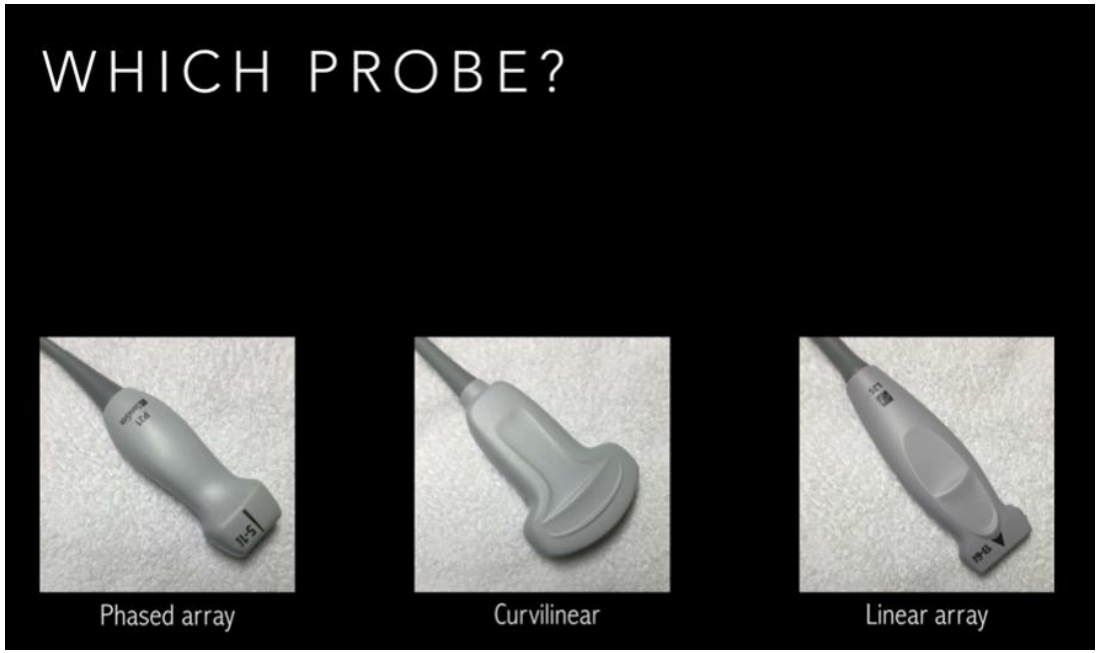


Lecture 06: Medical Images (Ultrasound and Pathology)

Oct 13, 2026

Outline

Ultrasound and pathology





Ultrasound physics and task-specific models
Multi-view learning and EchoPrime

Whole-slide learning and pathology experiments
TITAN and medical model post-training

1. Ultrasound

- How is ultrasound used?
- How are images generated?
- What are the main limitations?





The ultrasound probe contains a rectangular grid of 'piezoelectric' crystals, which 'deform' when a charge is applied.

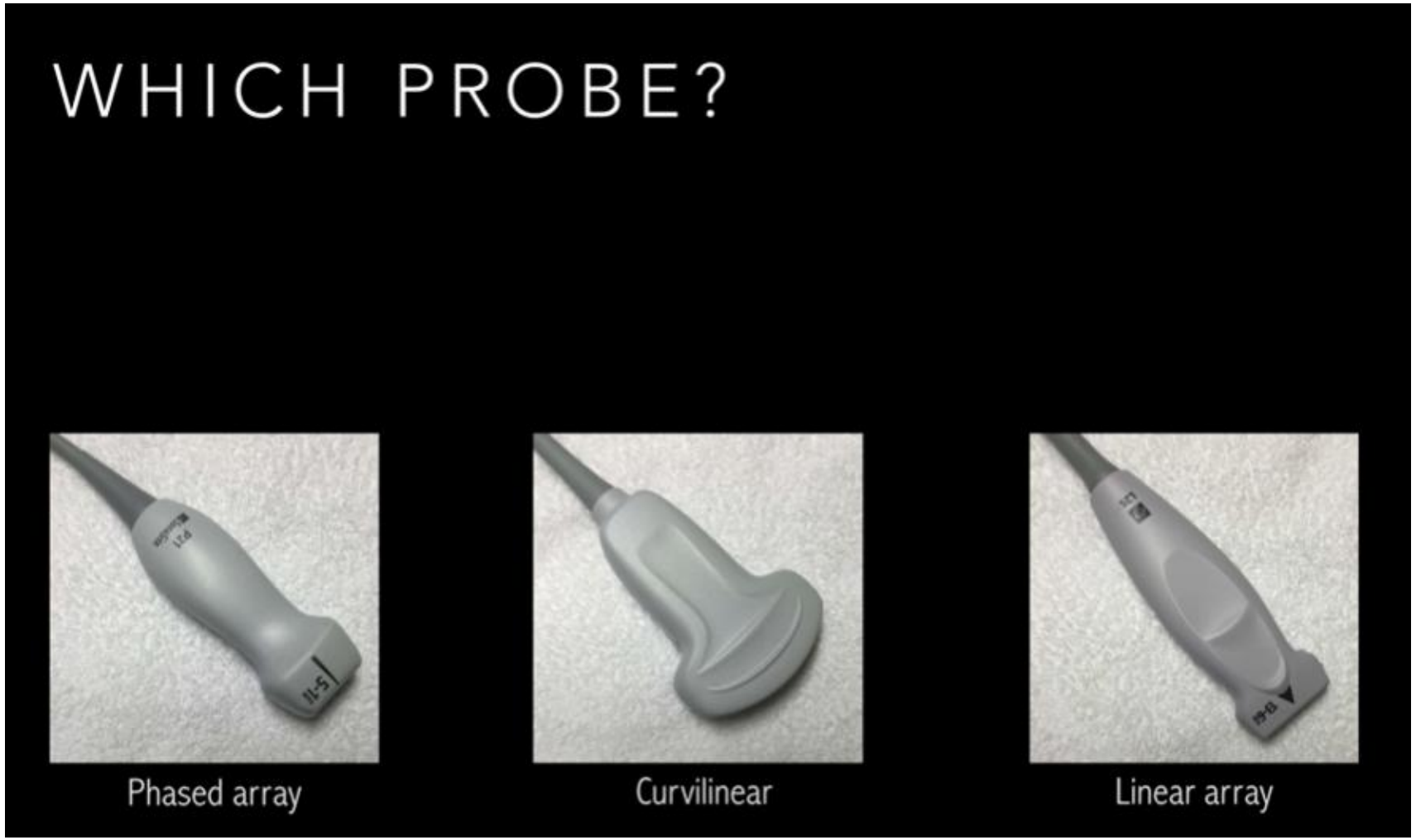
Applying an alternating current causes these crystals to alternately expand and contract.

This movement - rapid expansion and contraction - is transmitted to the patient's tissues when in contact with the ultrasound probe.



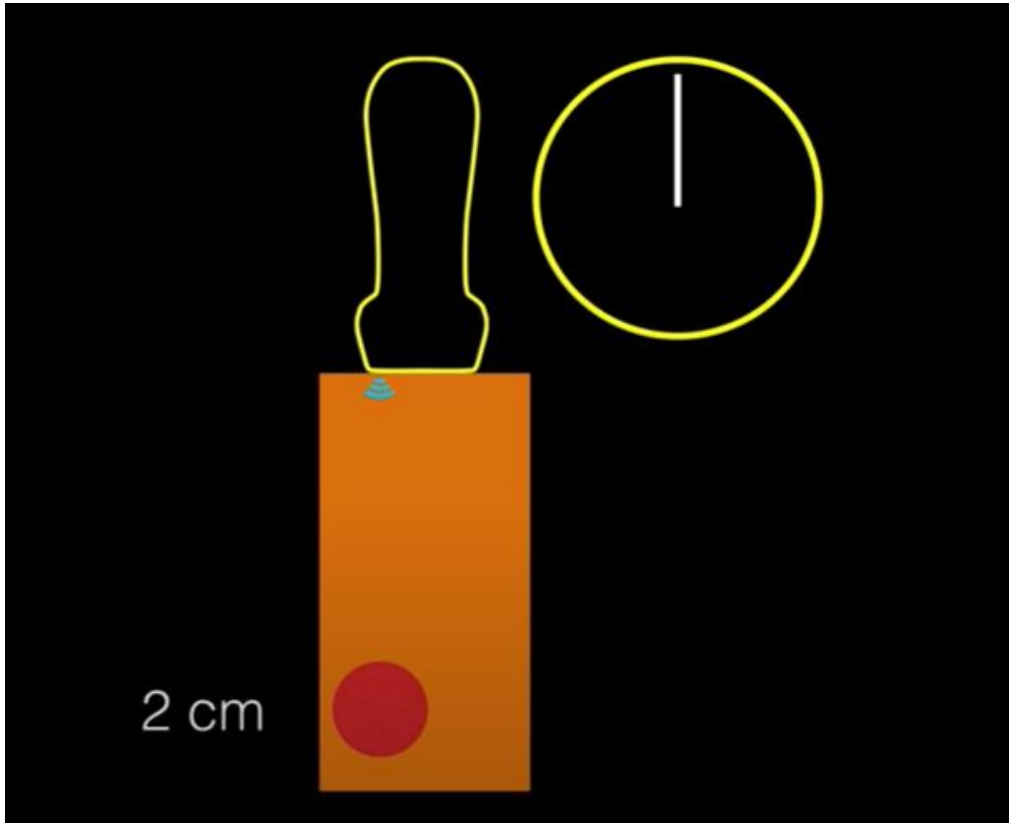
1. Ultrasound

Depends on what kind of scan you are planning to do



1. Ultrasound

Frequency, resolution and imaging depth



$$\lambda = c / f \qquad d = ct / 2$$

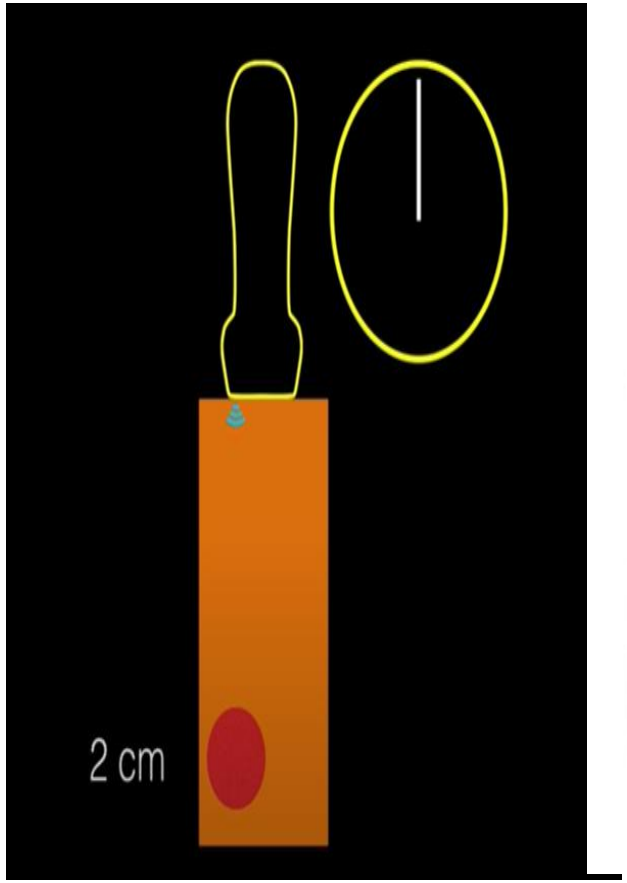
- Higher frequency gives a shorter wavelength
- Greater attenuation limits penetration depth
- The round-trip travel time determines depth

Which probe would you choose for a deep target?

1. Ultrasound

Basic principle

- A transducer converts electrical signals to acoustic signals, generates pulses of ultrasound, which are sent through a patient's body.
- Organ boundaries and complex tissues produce echoes (by reflection or scattering) that return back to and are detected by the transducer, which converts the acoustic signal to an electrical signal.
- The ultrasound imaging system then processes the echoes and presents a grayscale image of human anatomy on a display. Each point in the image corresponds to the anatomic location of an echo-generating structure and its brightness corresponds to the echo strength.
- Obstetrics, oncology, cardiology and gastroenterology





Figures from 2015 - Medical Imaging Signals and Systems from Jerry and Jonathan

1. Ultrasound

Reflection and refraction at a plane interface
A wave travels from fat (1,450 m/s) to liver (1,570 m/s).
For 45° incidence, find the reflected and transmitted angles.

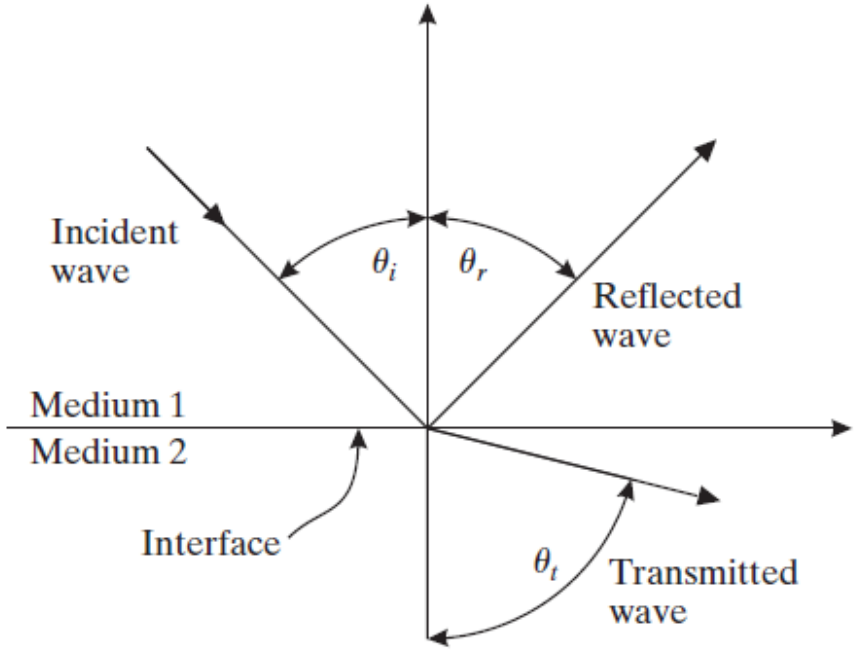
Reflection: 45°
sin θt = (1,570 / 1,450) sin 45°
Transmission: θt ≈ 49.96°

$$\theta_i = \theta_r$$

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{c_1}{c_2},$$

Snell's law

Speeds of sound in medium 1 and 2.

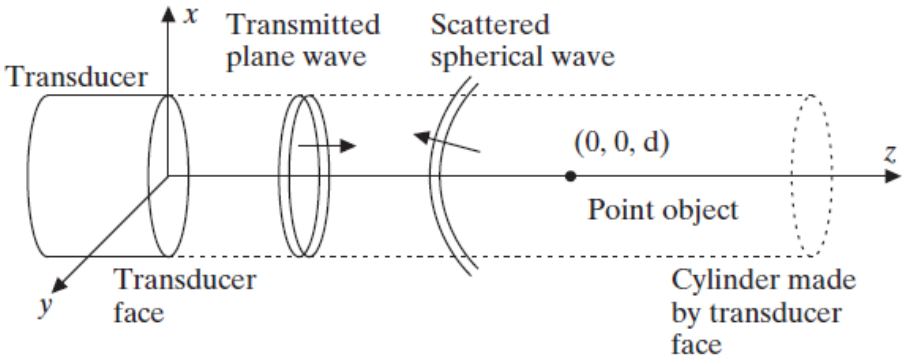


An incident plane wave reflects some energy and transmits the rest at a plane interface.

1. Ultrasound

Attenuation and Scattering

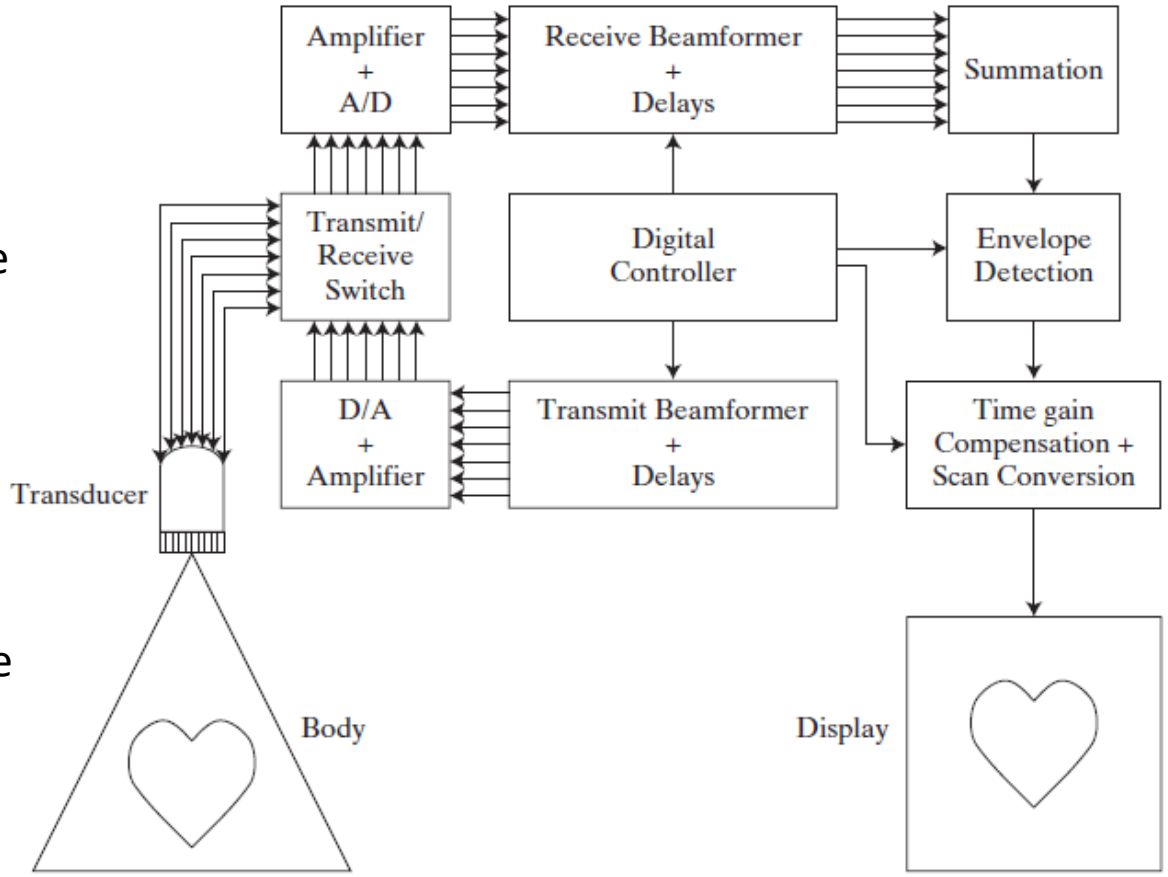
- Attenuation refers to loss of wave amplitude due to all mechanisms including absorption, scattering, and mode conversion.
- Absorption is the process by which the wave energy is converted to thermal energy, which is then dissipated in the medium.
- Scattering is the process by which secondary spherical waves are generated as the wave propagates.
- Mode conversion is the process by which longitudinal waves are converted to transverse shear waves (and back again).



Basic pulse-echo experiment

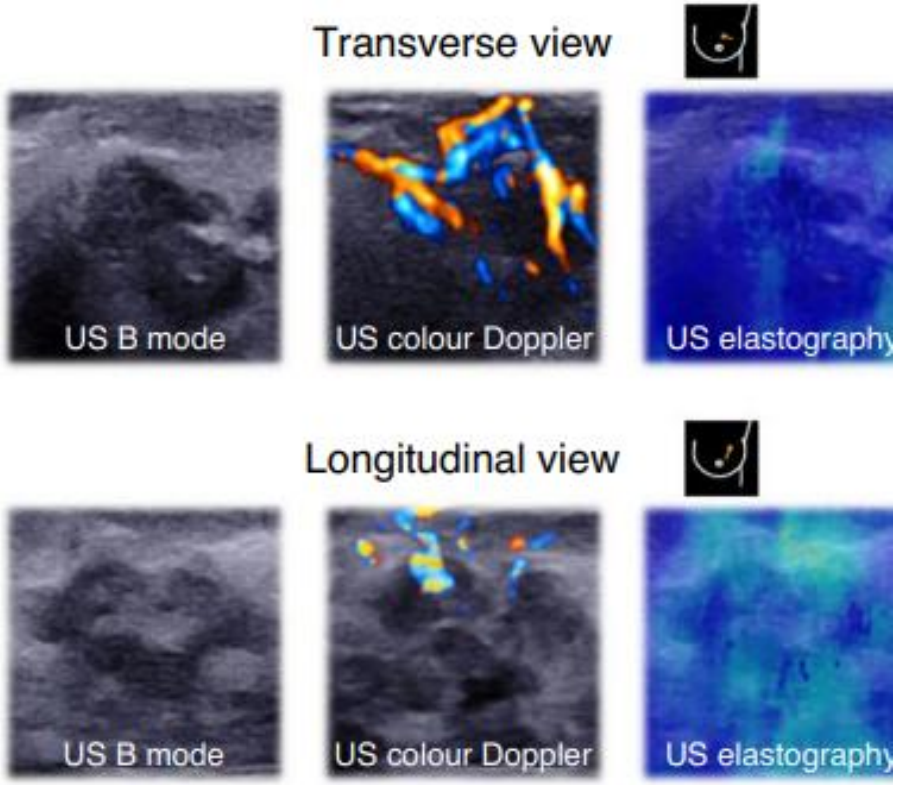
1. Ultrasound

- The vast majority of medical ultrasound imaging systems use the same transducer for both generation and reception of ultrasound.
- The transducer is coupled to the body using an ‘‘acoustic gel,’’ and a brief pulse-like acoustic wave is generated.
- This wave propagates into the body, where it encounters reflecting surfaces and small scatterers. These objects reflect or scatter the sound, a part of which returns to the transducer.
- The transducer then converts the acoustic wave sensed at its face to an electrical signal that can be amplified, stored, and displayed.
- The simplest transducer assembly, also called an ultrasound probe.



2. Ultrasound: case studies

Breast ultrasound: modalities and views

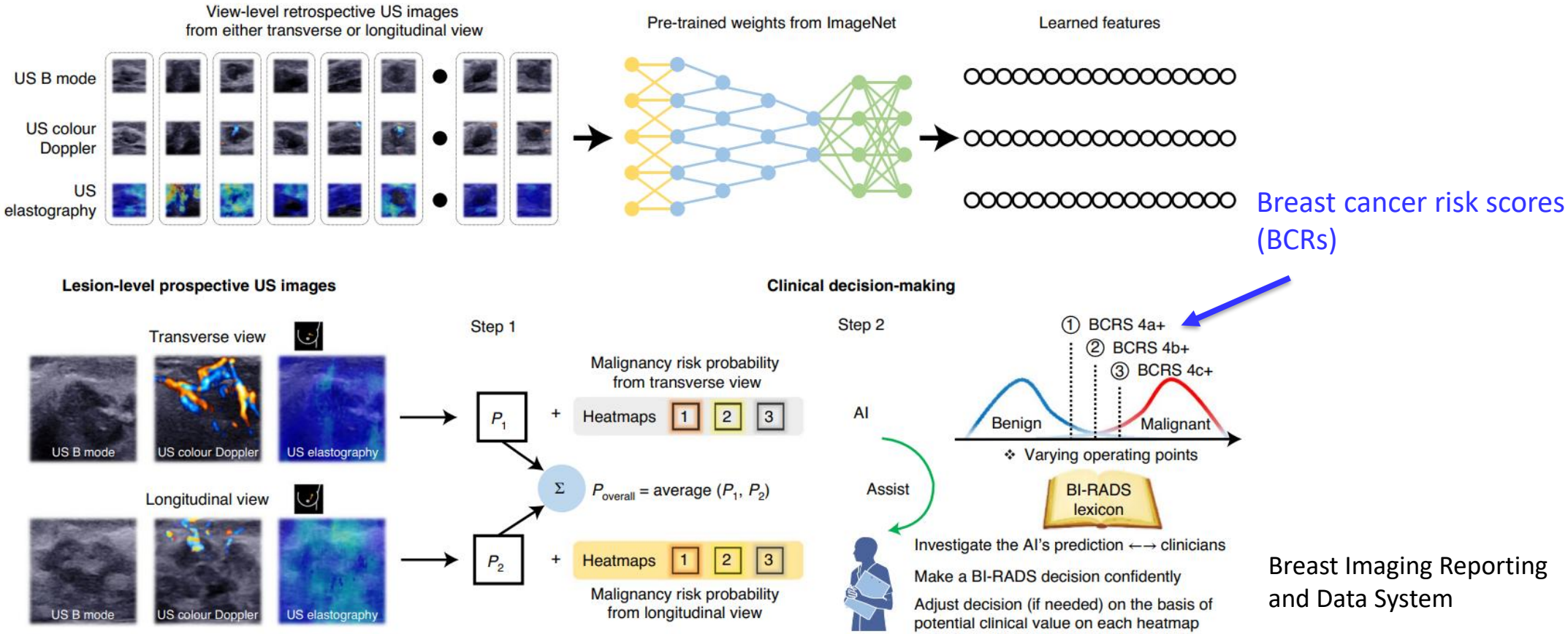


- B-mode shows tissue appearance
- Colour Doppler adds flow information
- Elastography adds stiffness information
- Two views sample different lesion planes

Qian et al., Nature Biomedical Engineering 2021. Input detail from the original method figure.

2. Ultrasound: case studies

Prospective assessment of breast cancer risk from multimodal multiview ultrasound images via clinically applicable deep learning

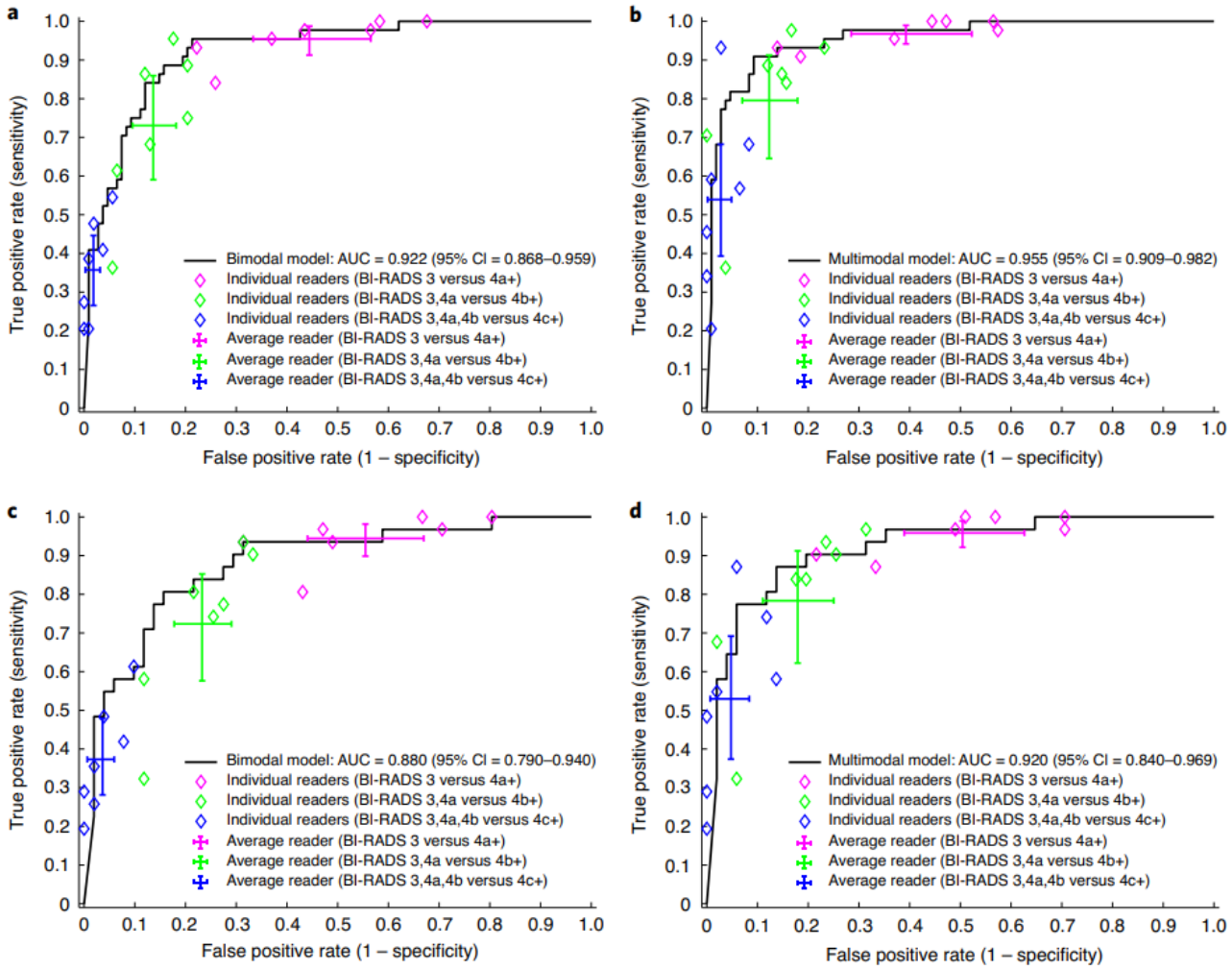


Breast Imaging Reporting and Data System

Qian et al. Prospective assessment of breast cancer risk from multimodal multiview ultrasound images via clinically applicable deep learning, Nature biomedical engineering, 2021

2. Ultrasound: case studies

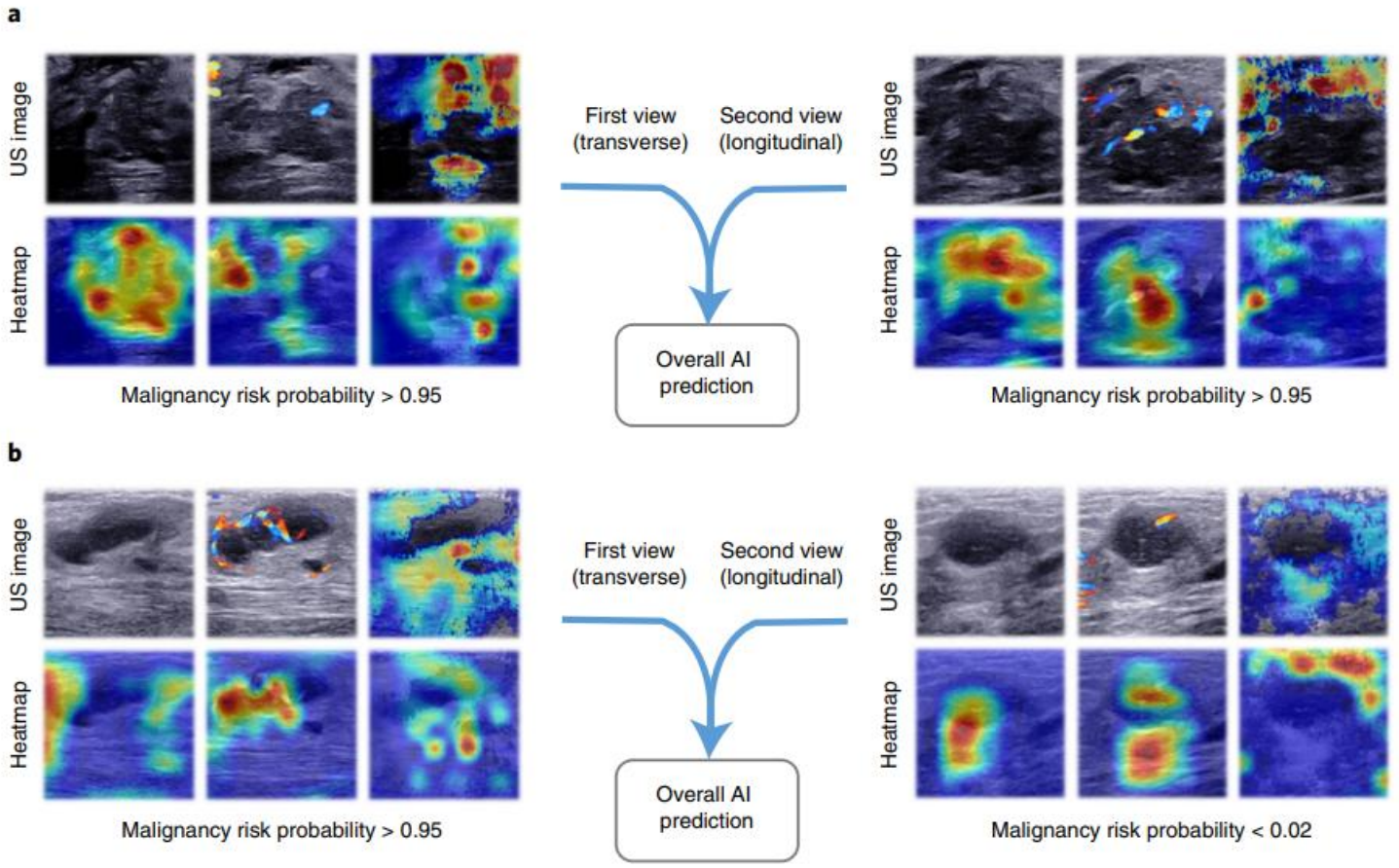
Prospective assessment of breast cancer risk from multimodal multiview ultrasound images via clinically applicable deep learning



The performance of our AI system was compared with each of the seven readers and the averaged performance of the seven readers at three decision modes

2. Ultrasound: case studies

Prospective assessment of breast cancer risk from multimodal multiview ultrasound images via clinically applicable deep learning

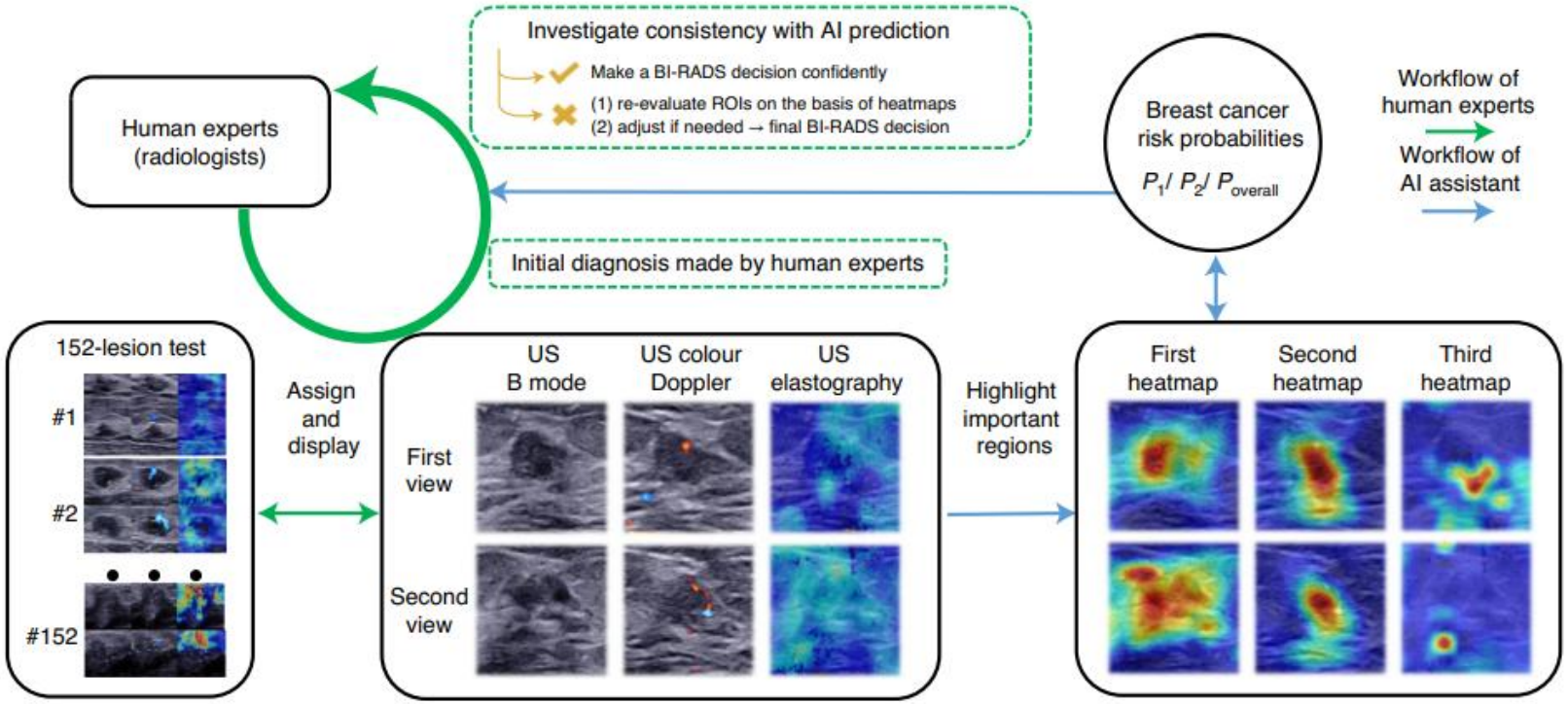


Color-coded heatmaps overlaid with the corresponding US images were generated from the final convolution layer using the Grad-CAM approach

Despite that there is a disagreement between the false positive from the transverse view and true negative from the longitudinal view, combing orthogonal views may mitigate overall prediction bias.

2. Ultrasound: case studies

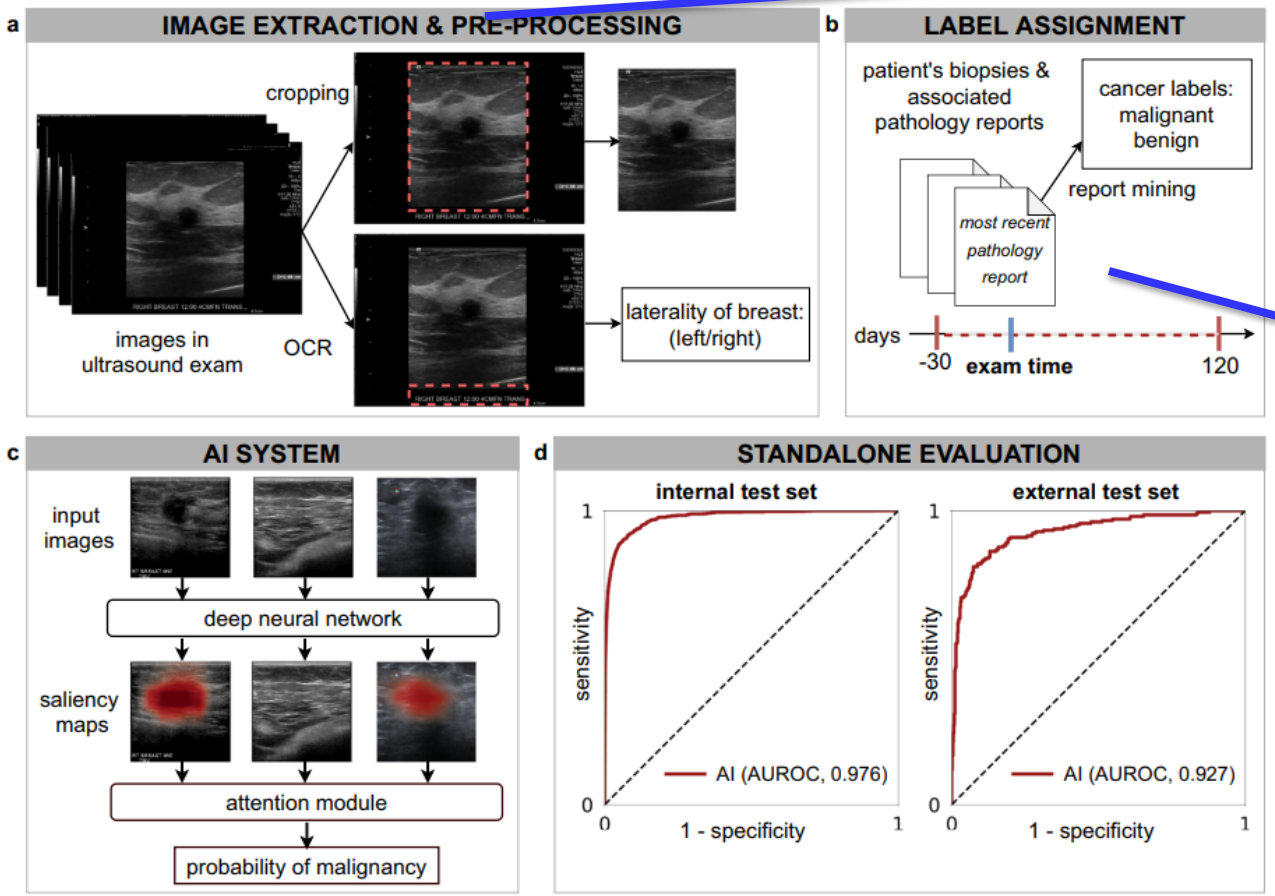
Prospective assessment of breast cancer risk from multimodal multiview ultrasound images via clinically applicable deep learning



The workflow of the AI-assisted reader study is shown. For each randomly assigned and displayed test lesion, the reader examined information from three aspects, including original multimodal multiview US images, the corresponding highlighted heatmaps and the AI prediction on malignancy risk probability, then finally determined the BI-RADS categorization.

2. Ultrasound: case studies

Artificial intelligence system reduces false-positive findings in the interpretation of breast ultrasound exams



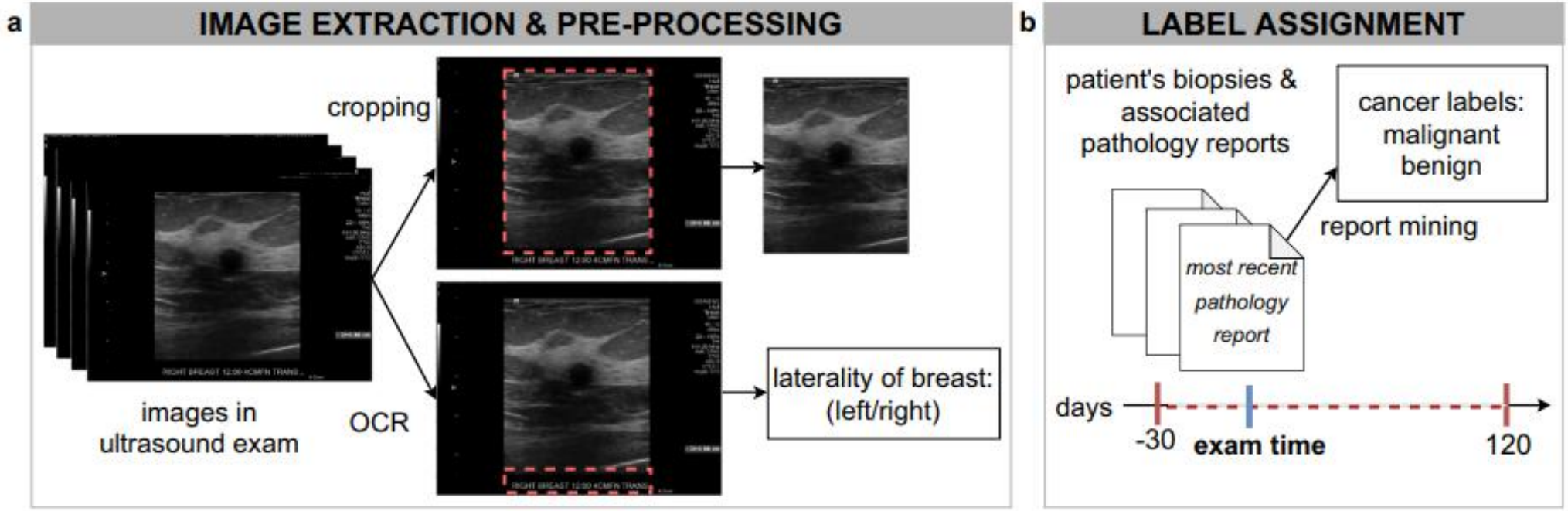
US images were pre-processed to extract the breast laterality (i.e., left or right breast) and to include only the part of the image which shows the breast (cropping out the image periphery which typically contains textual metadeta about the patient and US acquisition technique)

we assigned a cancer label using the recorded pathology reports for the respective patient within −30 and 120 days from the time of the US examination.

Shen et al. Artificial intelligence system reduces false-positive findings in the interpretation of breast ultrasound exams, Nature communications, 2021

2. Ultrasound: case studies

Breast ultrasound: label construction

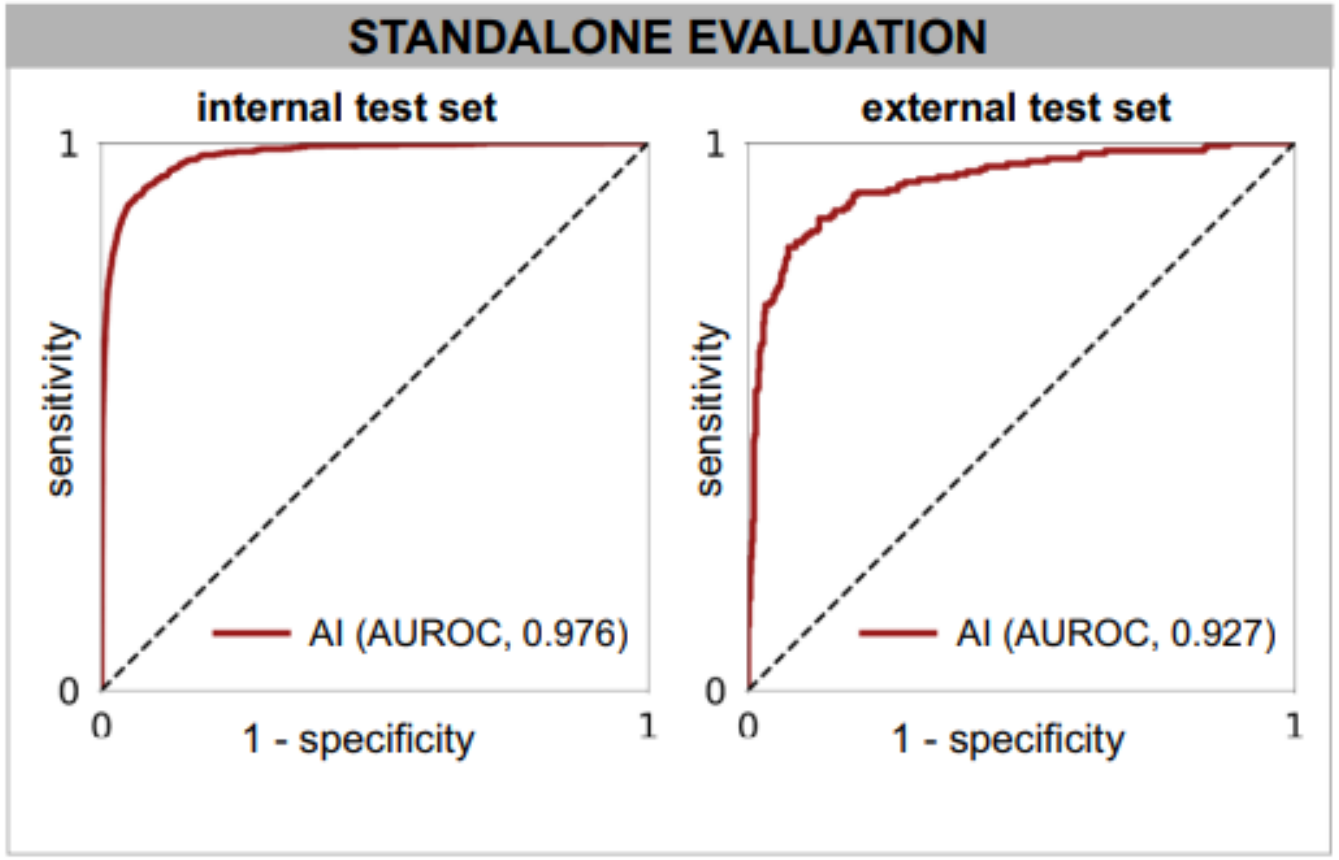


Patient-level splitting and the pathology time window determine what the model can learn.

Shen et al., Nature Communications 2021. Fig. 1a,b, enlarged.

2. Ultrasound: case studies

Breast ultrasound: internal and external tests



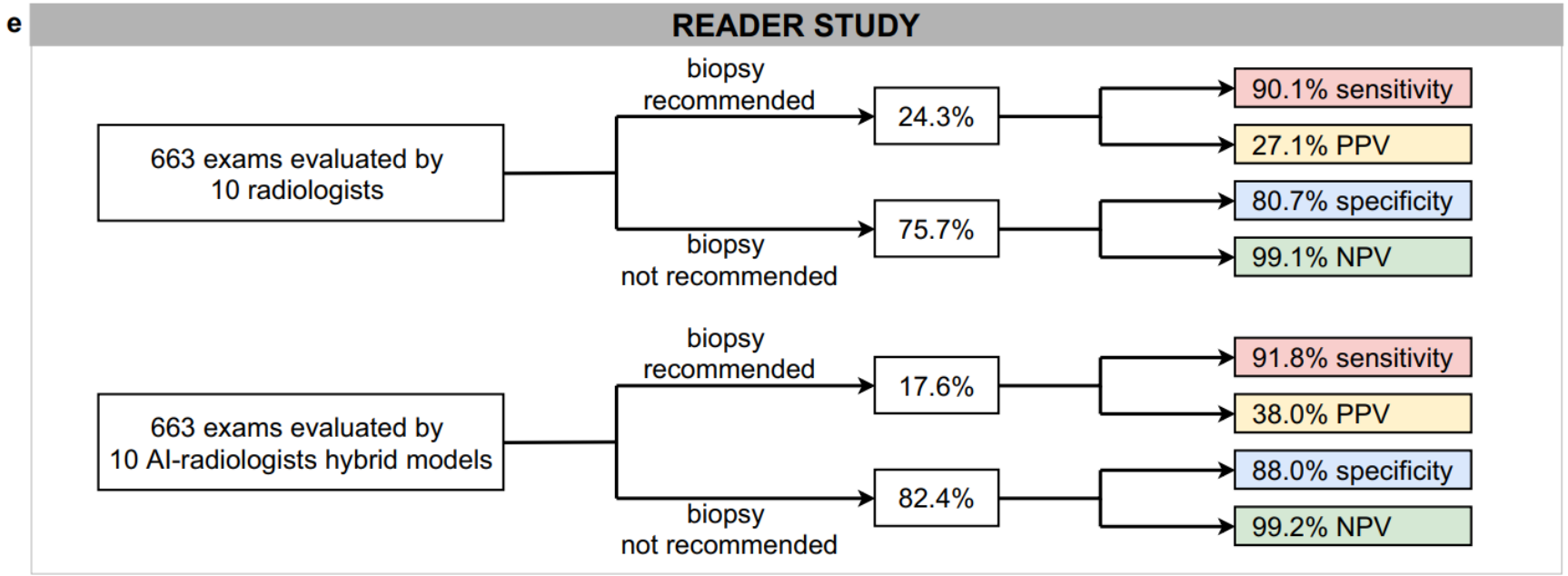
Test cohort	AUROC (95% CI)
NYU internal	0.976 (0.972–0.980)
BUSI external	0.927 (0.907–0.959)

External performance decreased despite a strong internal result.

Shen et al., Nature Communications 2021. Fig. 1d and Results.

2. Ultrasound: case studies

Breast ultrasound: retrospective reader comparison



Sensitivity = TP / (TP + FN)
Fewer false negatives

Specificity = TN / (TN + FP)
Fewer false positives

663 exams (1,024 breasts), 10 readers.
Hybrid predictions combine AI scores with reader ratings.

2. Ultrasound: case studies

Artificial intelligence system reduces false-positive findings in the interpretation of breast ultrasound exams

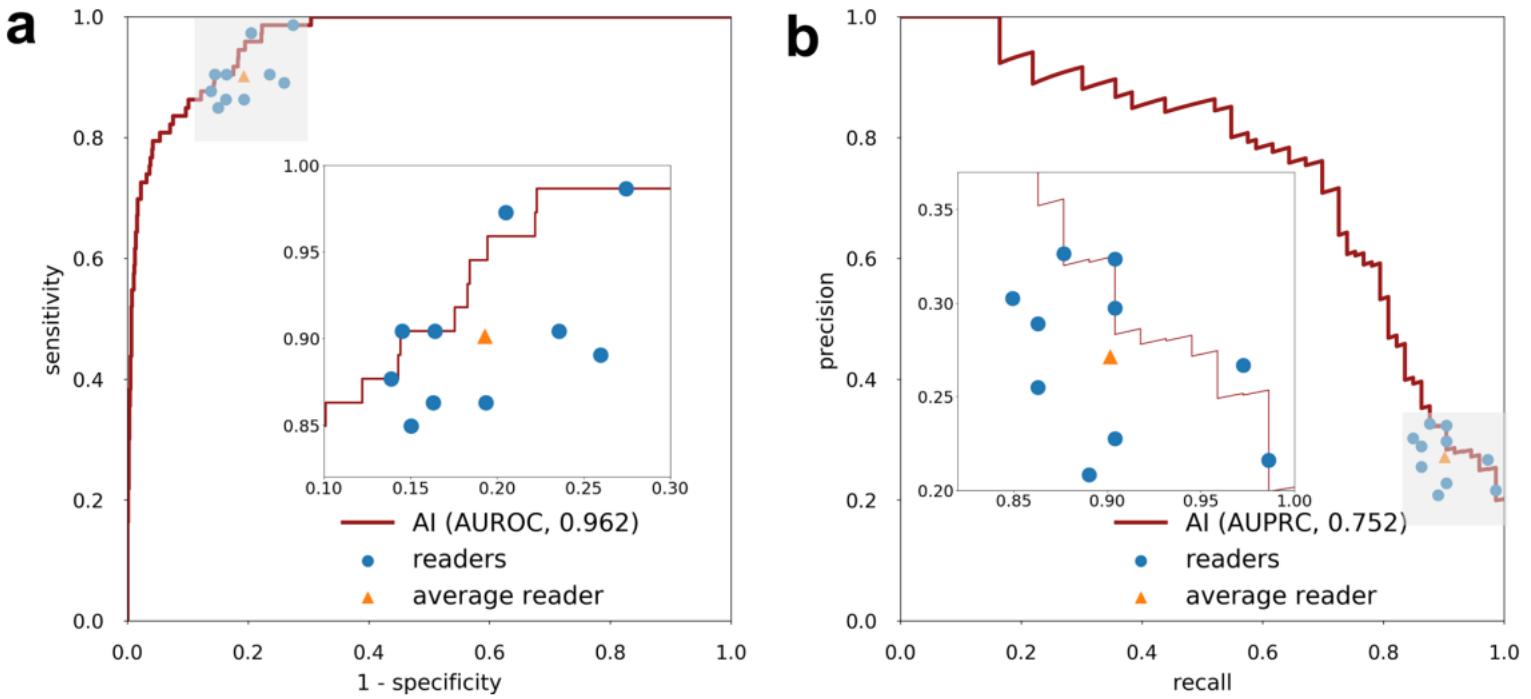
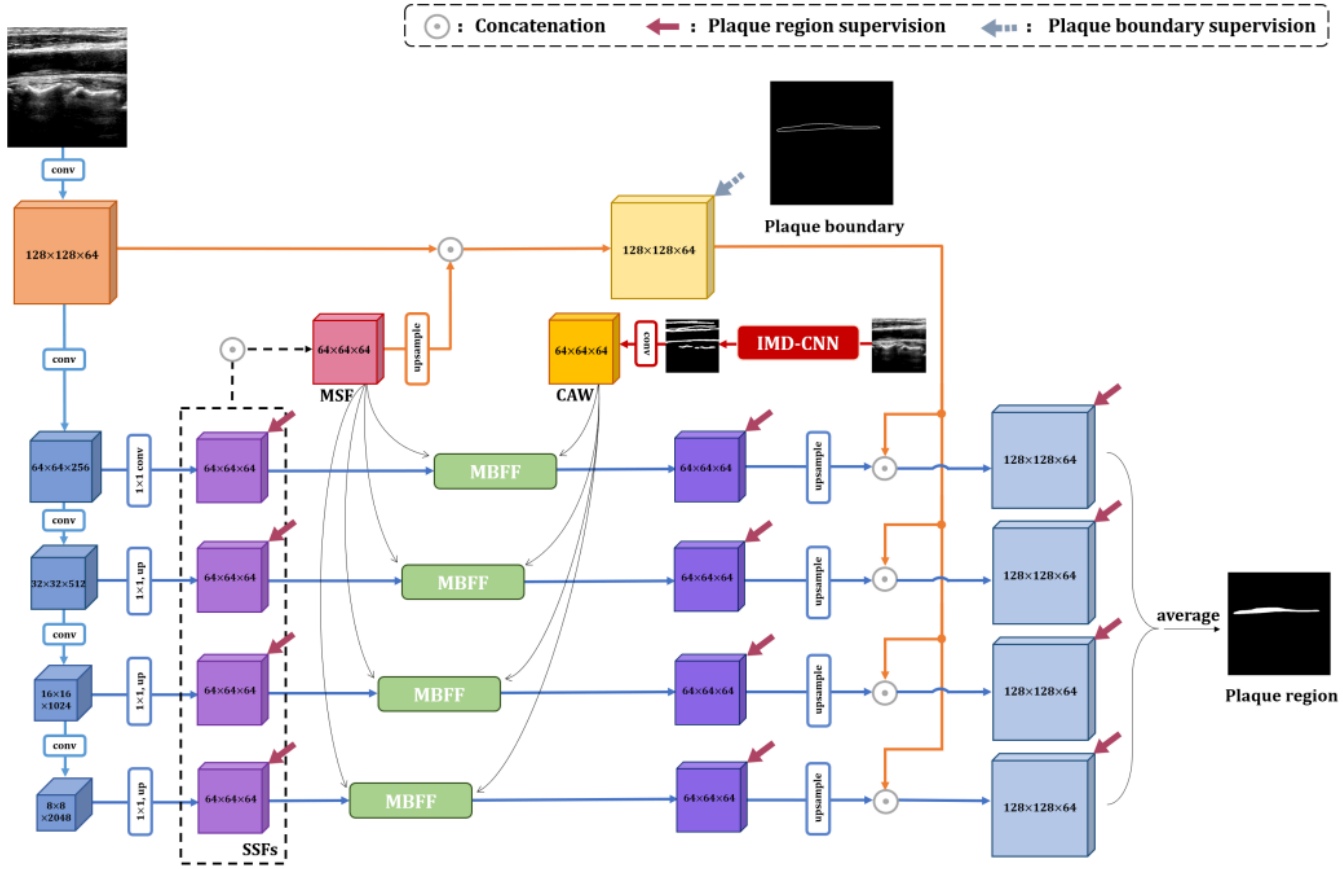


Fig. 2 Reader study results. The performance of the AI system on the reader study population ($n = 1024$ breasts) using ROC curve (**a**) and precision-recall curve (**b**). The AI achieved 0.962 (95% CI: 0.943, 0.979) AUROC and 0.752 (95% CI: 0.675, 0.849) AUPRC. Each data point represents a single reader and the triangles correspond to the average reader performance. The inset shows a magnification of the gray shaded region.

Shen et al. Artificial intelligence system reduces false-positive findings in the interpretation of breast ultrasound exams, Nature communications, 2021

2. Ultrasound: case studies

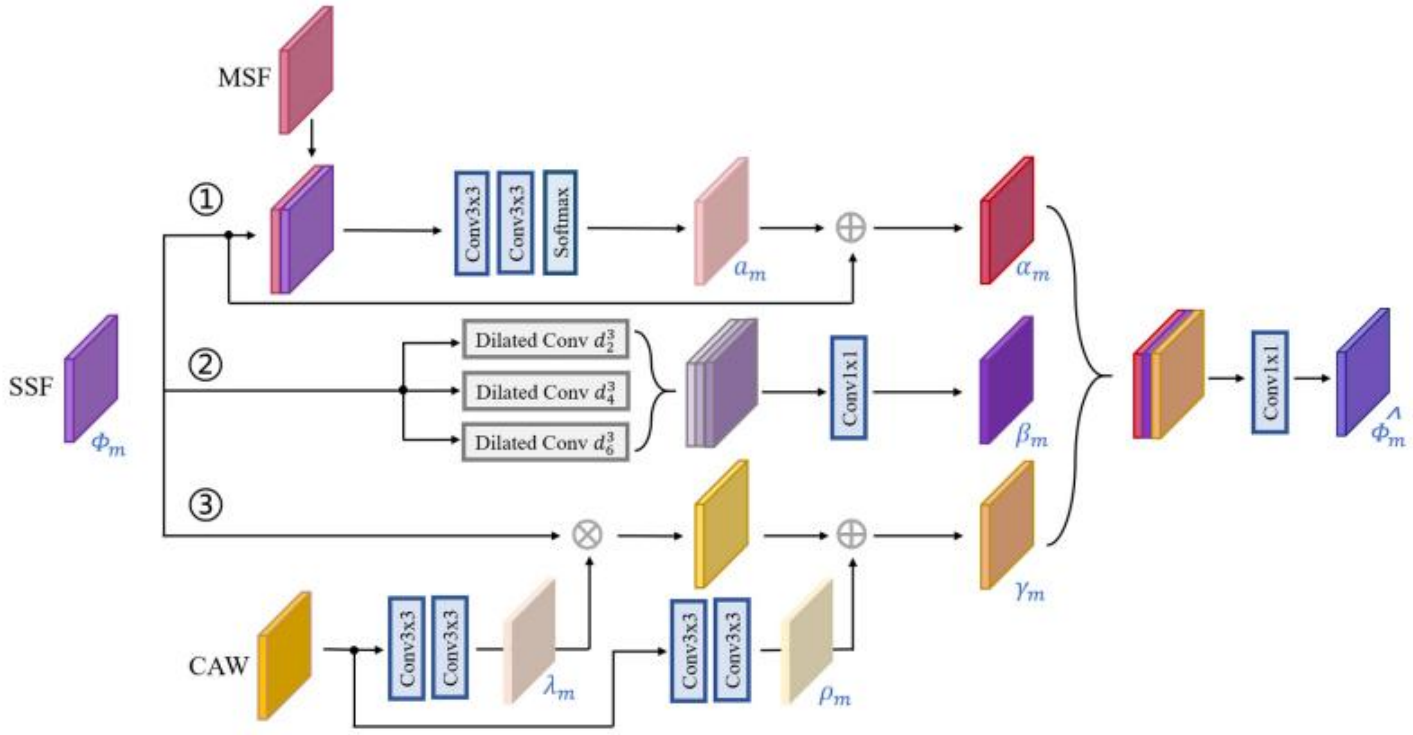
MBFF-Net: overall architecture



Mi et al. MBFF-Net: Multi-Branch Feature Fusion Network for Carotid Plaque Segmentation in Ultrasound, MICCAI, 2021

2. Ultrasound: case studies

MBFF-Net: the feature fusion module



- The module combines several feature pathways
- Multi-scale context and attention enter different branches
- Concatenation and a convolution fuse the outputs

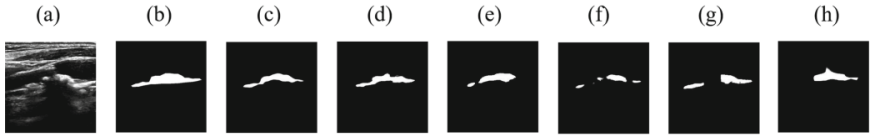
Mi et al., MICCAI 2021. Module diagram enlarged from the supplied lecture.

2. Ultrasound: case studies

MBFF-Net: quantitative comparisons and ablation

Method	Dice	IoU	Precision	Recall
PSPNet	.585	.472	.760	.586
ResUNet	.621	.521	.755	.635
U-Net	.630	.552	.817	.658
SegNet	.687	.593	.856	.676
DenseUNet	.778	.690	.872	.763
MBFF-Net	.780	.702	.849	.797

Components	Dice	IoU
Baseline	.666	.577
+ DC	.683	.597
+ CAW	.688	.611
+ MSF	.710	.625
All three	.736	.647
All three + BPS	.780	.702



430 images: 330 training, 100 test.
DenseUNet → MBFF: Dice +0.002; Precision −0.023.
What evidence would establish a reliable gain?

Mi et al., MICCAI 2021. Original Tables 1–2 and Fig. 4 detail. DOI: 10.1007/978-3-030-87240-3_30.

2. Ultrasound: case studies

MBFF-Net: Multi-Branch Feature Fusion Network for Carotid Plaque Segmentation in Ultrasound

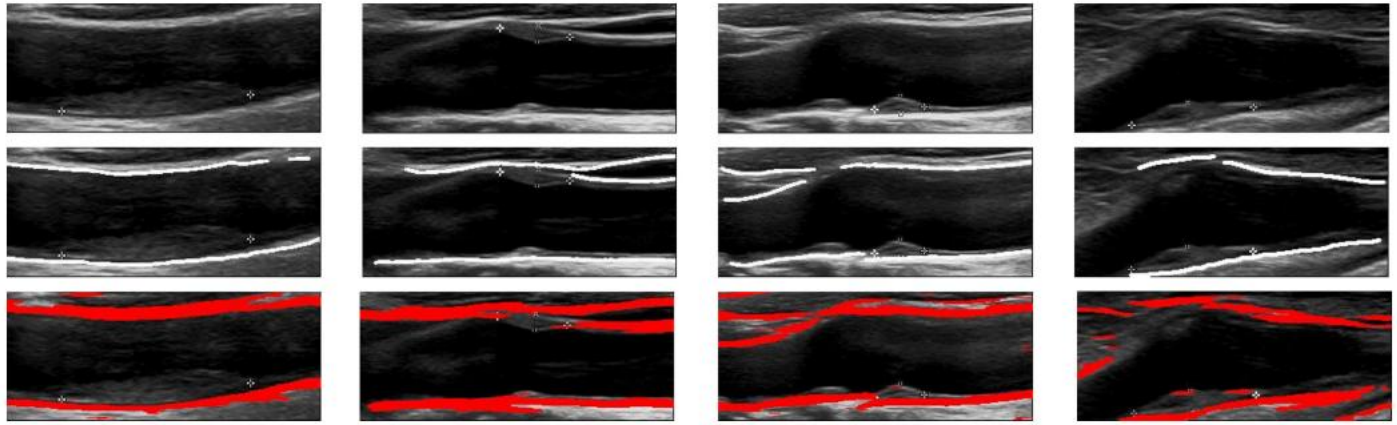


Fig. 2. Examples of the IM detection results. The first row shows the original images, the second row shows the manually labeled ground truth masks, and the third row shows the outputs of the IMD-CNN (the IM regions are shown in red color). (Color figure online)

Mi et al. MBFF-Net: Multi-Branch Feature Fusion Network for Carotid Plaque Segmentation in Ultrasound, MICCAI, 2021

2. Ultrasound: case studies

MBFF-Net: Multi-Branch Feature Fusion Network for Carotid Plaque Segmentation in Ultrasound

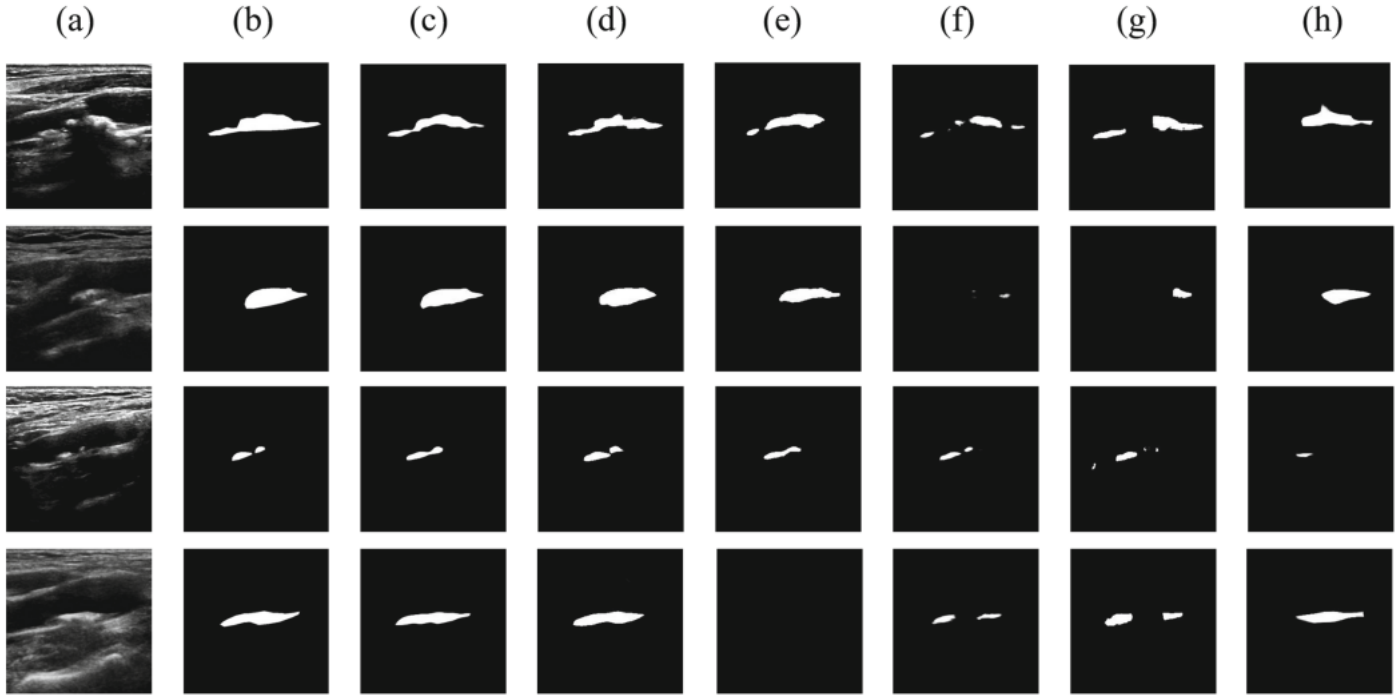


Fig. 4. Examples of plaque segmentation results produced by different methods. (a): CLS ultrasound images; (b): ground truth masks; (c)–(h): corresponding segmentations from our method, DenseUNet [18], SegNet [17], U-Net [16], ResUNet [15] and PSPNet [14], respectively.

Mi et al. MBFF-Net: Multi-Branch Feature Fusion Network for Carotid Plaque Segmentation in Ultrasound, MICCAI, 2021

2. Ultrasound: case studies

Rethinking Ultrasound Augmentation: A Physics-Inspired Approach

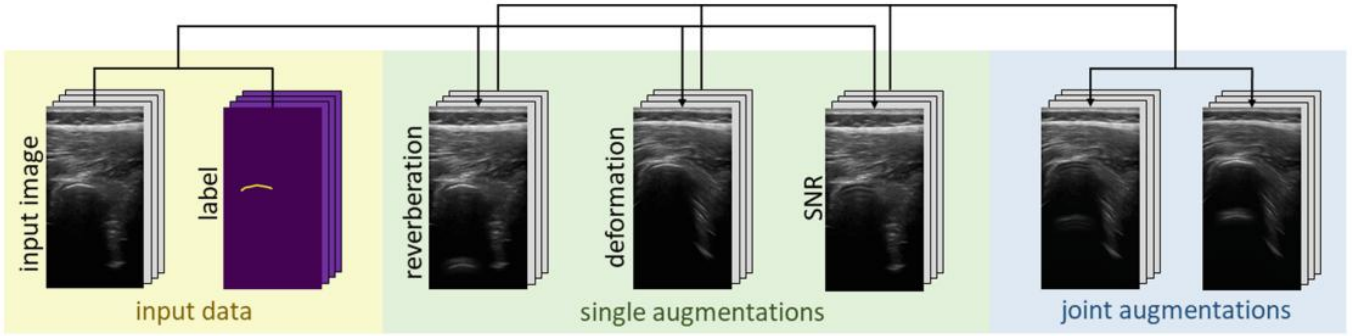


Fig. 1. Our proposed data augmentation pipeline: from US images and corresponding bone masks (yellow), different augmentations are generated (green) and can be merged in the same image (blue). (Color figure online)

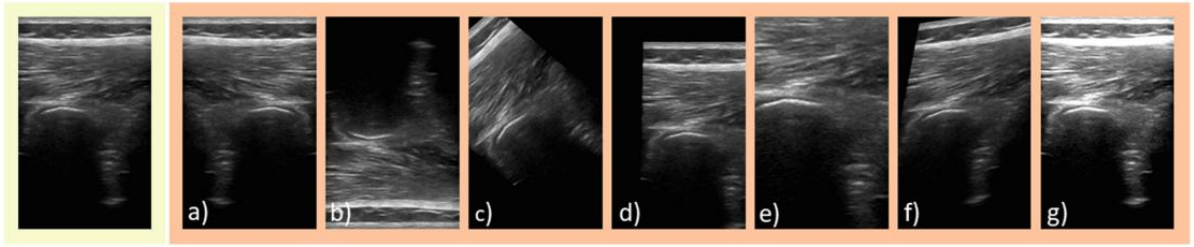


Fig. 2. Classical augmentations applied to an US image: a) horizontal flip, b) vertical flip, c) rotation, d) translation, e) scaling, f) shearing, g) brightness adaption.

Tirindelli et al. Rethinking Ultrasound Augmentation: A Physics-Inspired Approach, MICCAI, 2021

2. Ultrasound: case studies

Rethinking Ultrasound Augmentation: A Physics-Inspired Approach

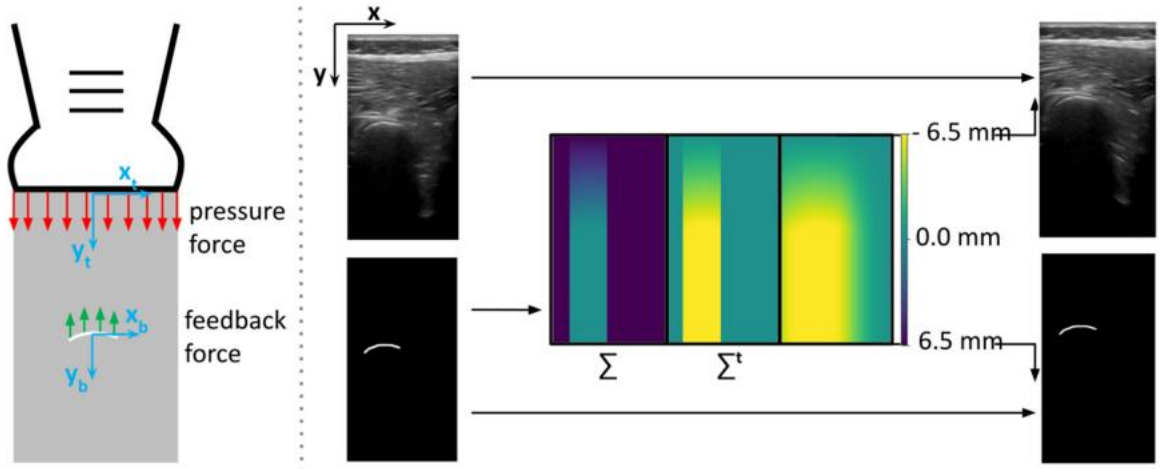


Fig. 3. Deformation Pipeline: Left: The forces applied by the US probe (red arrows) and bones (green arrows), and the coordinate systems centered in the bone (x_b, y_b) and the probe centroids (x_p, y_p), respectively. Right: The original B-mode image and label. The label is used to compute the DF (Σ) induced by the probe pressure. To avoid unrealistic gap above the skin layer, the DF is expressed in the probe coordinate system (Σ^t) and blurred. Σ^t is finally applied to both image and label. (Color figure online)

The proposed augmentations are inspired by the physical model of wave propagation in medical US. These augmentations use the identification of regions of high-echogenicity or attenuation in order to model and augment the wave propagation and resulting B-mode image. We propose augmenting training data based on deformation due to probe pressure, reverberation artifacts and signalto-noise-ratio, which have been tailored to linear US imaging.

Fig.3 simulates deformations from variations in the contact pressure of the US probe on interrogated tissues.

Tirindelli et al. Rethinking Ultrasound Augmentation: A Physics-Inspired Approach, MICCAI, 2021

2. Ultrasound: case studies

Rethinking Ultrasound Augmentation: A Physics-Inspired Approach

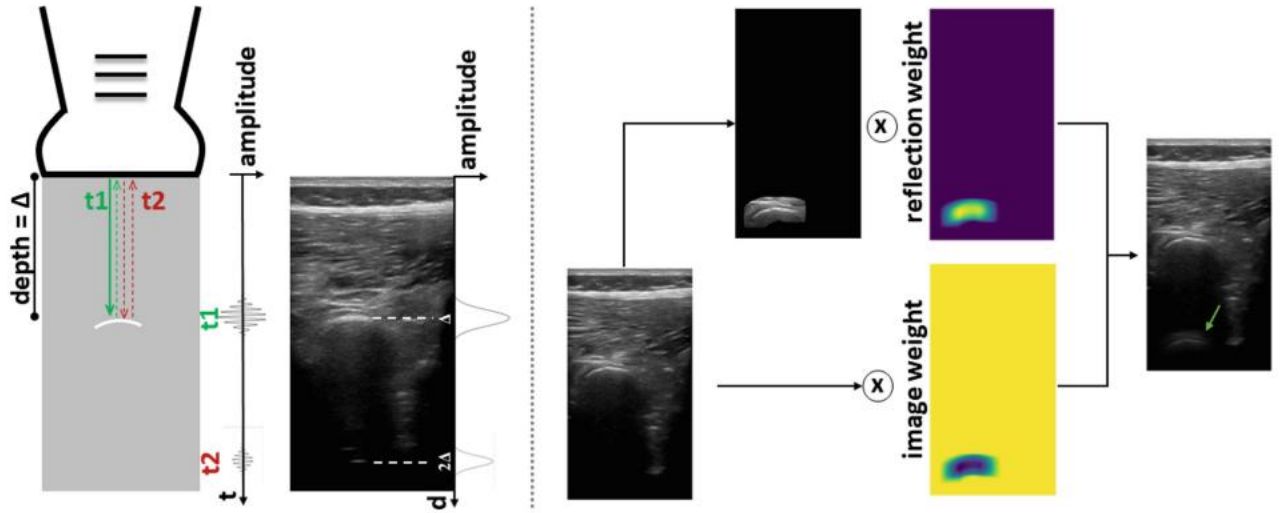


Fig. 4. Reverberation Pipeline. Left: The transmitted signal and the echos generated by a bone at depth Δ. In the time-amplitude plot the signals generated by the first echo (green arrow) and reverberation echo (red arrow) are reported. The first echo generates a bright area in the B-mode image at Δ, while the reverberation echo generates an artifact at 2Δ. Right: The generation of reverberation artifacts on B-mode data. The patch containing the bone is extracted and replicated at multiples of Δ. The replicated patch is weighted and summed up to the original image. (Color figure online)

Tirindelli et al. Rethinking Ultrasound Augmentation: A Physics-Inspired Approach, MICCAI, 2021

2. Ultrasound: case studies

Rethinking Ultrasound Augmentation: A Physics-Inspired Approach

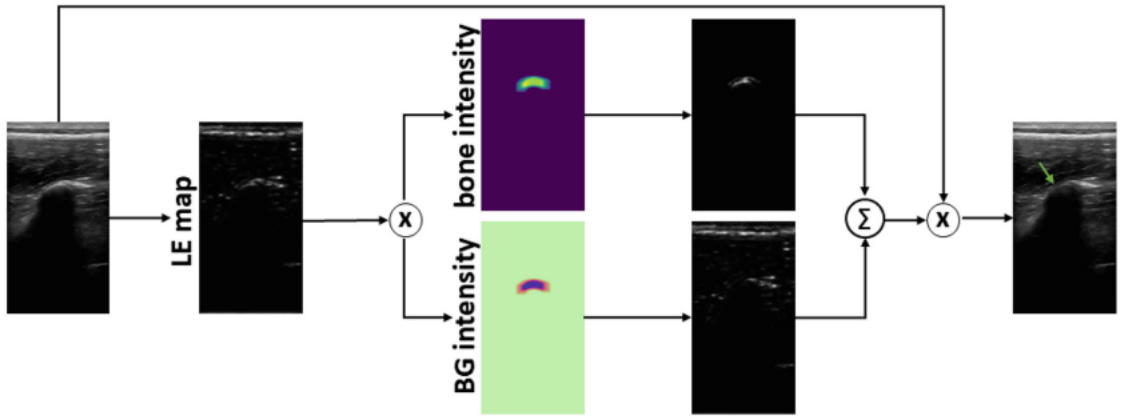


Fig. 5. SNR augmentation pipeline. The local energy (LE) map is computed based on the input image. This image is then multiplied with the label or its inverse and the intensity values to obtain tuned local energy maps for bone and background (BG) regions. The overall tuned local energy is computed by adding the bone and background parts and the augmented image is obtained by multiplication of the input image with the tuned local energy.

we propose to tune the SNR in BMode data by first extracting and scaling the signal in the image to enhance or reduce the SNR, e.g. the visibility of bones compared to the background.

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2. Ultrasound: case studies

Rethinking Ultrasound Augmentation: A Physics-Inspired Approach

Table 2. Baseline comparisons for bone segmentation with UNet and bone classification with DenseNet. We report the average metrics over 5-fold cross validation along with their respective standard deviation (±).

	Segmentation		Classification	
	DSC	HDF	ACC	F1
None	0.589 ± 0.07	20.72 ± 3.84	0.876 ± 0.06	0.770 ± 0.15
Classical	0.625 ± 0.03	17.76 ± 3.17	**0.883 ± 0.04**	0.780 ± 0.09
Reverb	0.604 ± 0.03	19.71 ± 2.20	**0.883 ± 0.03**	**0.802 ± 0.04**
Deformation	**0.626 ± 0.01**	19.06 ± 3.63	0.865 ± 0.04	0.759 ± 0.11
SNR	**0.626 ± 0.02**	**17.24 ± 1.83**	0.877 ± 0.06	0.764 ± 0.16
All	0.600 ± 0.02	17.32 ± 2.97	0.834 ± 0.02	0.742 ± 0.04

Tirindelli et al. Rethinking Ultrasound Augmentation: A Physics-Inspired Approach, MICCAI, 2021

Importance in Echocardiogram

Cardiac Structure Segmentation from Echocardiogram Videos

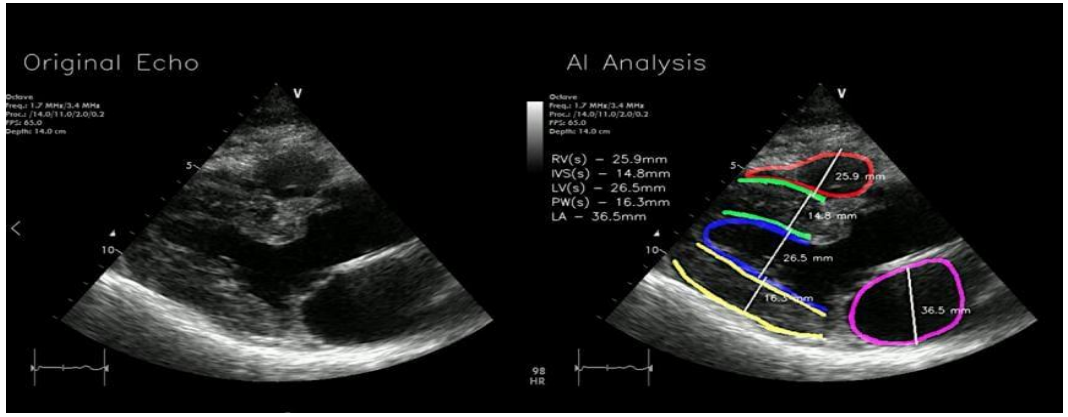


Figure 1. Examples of echocardiogram images [1].

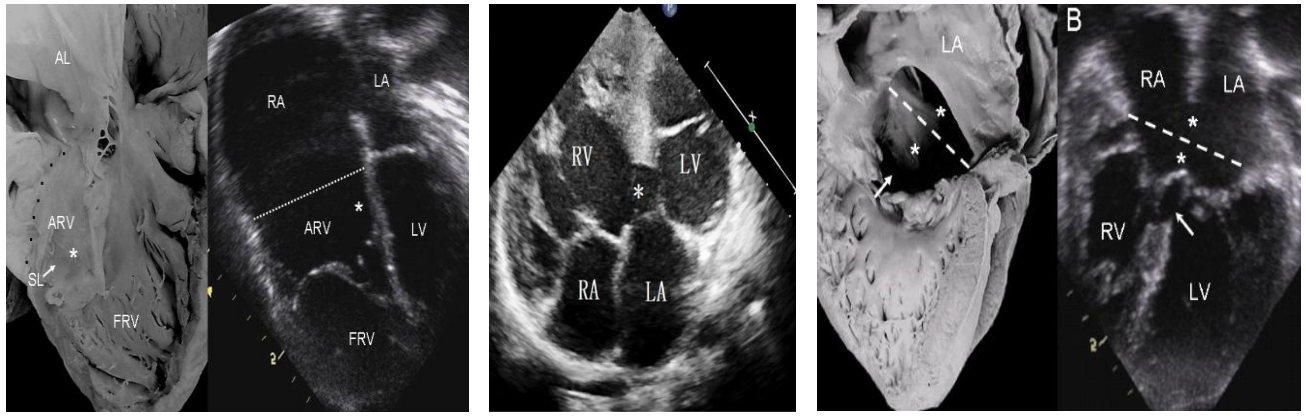


Figure 2. Example of Ebstein's Anomaly [2], Double-outlet Right Ventricle [3], and complete atrioventricular septal defect [4] from left to right.

Echocardiography is a widely used medical imaging technique that plays a crucial role in assessing cardiac functional parameters and structures.

- Cardiac structure segmentation refers to segment left ventricle (LV), right ventricle (RV), left atrium (LA) and right atrium (RA).
- It can help clinicians identify many cardiovascular diseases, such as congenital heart defects (CHDs).

[1] https://drjollydiagnostics.com/facilites/cardiology/ecocardiography/
[2] https://en.wikipedia.org/wiki/Ebstein%27s_anomaly
[3] https://journals.sagepub.com/doi/10.1177/2150135117692973
[4] https://cardiovascularultrasound.biomedcentral.com/articles/10.1186/1476-7120-6-33

Importance in Echocardiogram

Existing Challenges

- In the 2023 study, acquisition differences motivated cross-device evaluation.
- CardiacUDA and GraphEcho addressed video structure segmentation in that study setting.

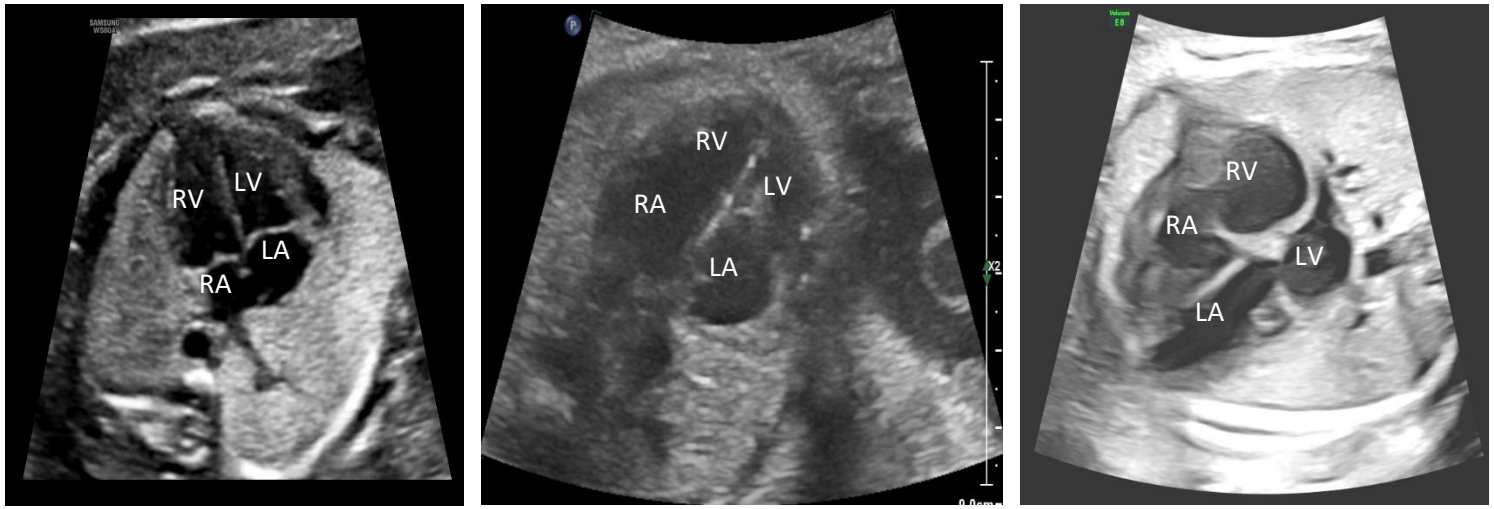
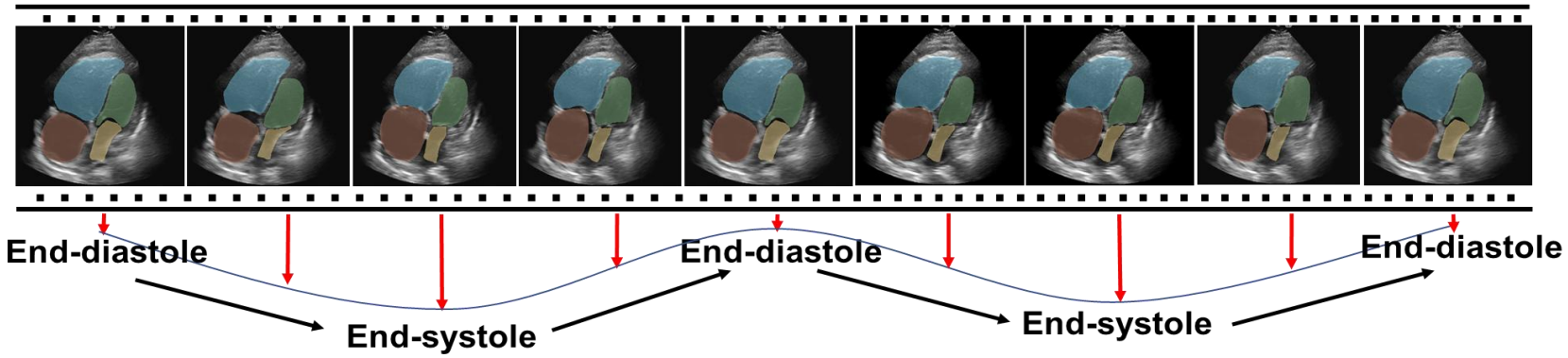


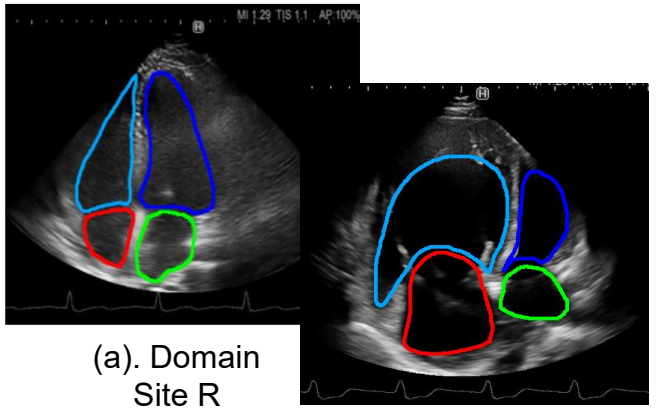
Figure 2: Samples collected from three difference devices (SAMSUNG, Philips & Voluson).

Echocardiogram Datasets

CardiacUDA – We propose a new benchmark dataset



An example video of our CardiacUDA.



(a). Domain Site R

(a). Domain Site B

(a) and (b) indicate echocardiogram videos from two different centers in CardiacUDA.

Dataset comparisons depend on the prediction task, available structure annotations, and cross-center splits.

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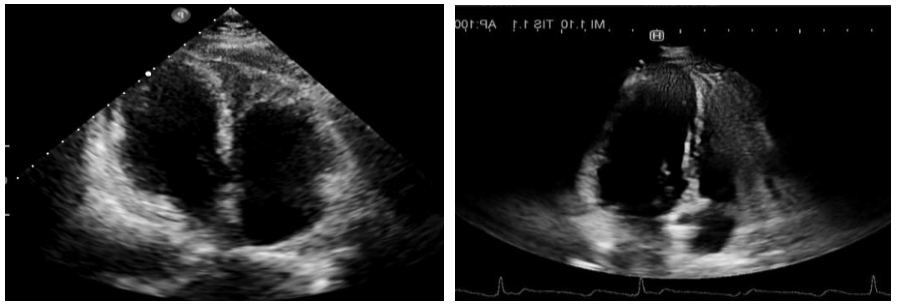
Problems in UDA

Existing UDA methods designed for natural images

- Usually considers domain shifting problem globally, having difficult in **modelling the irregular shape locally.**
- Have not considered both spatial and temporal characteristic according to the **morphology characteristic in Echocardiogram**;



(a). Domain difference in nature images.

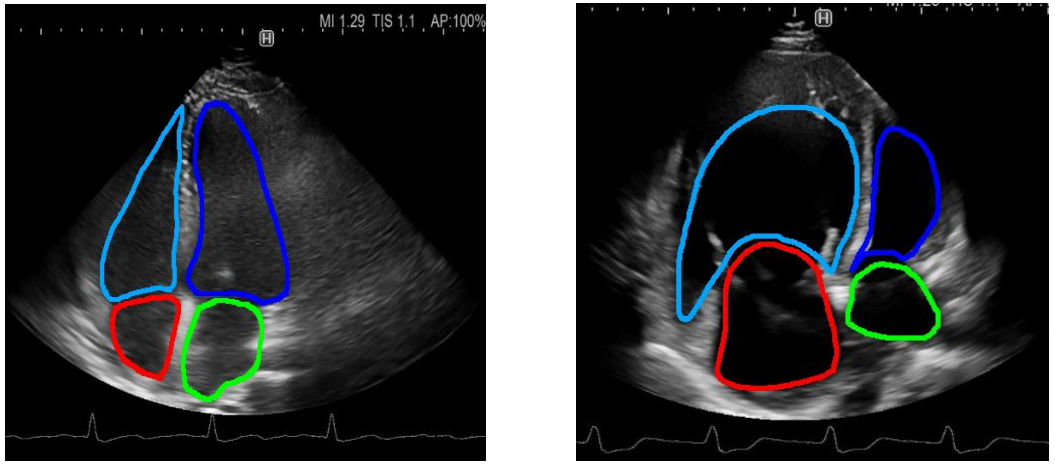


(b). Domain difference in Echocardiogram video.

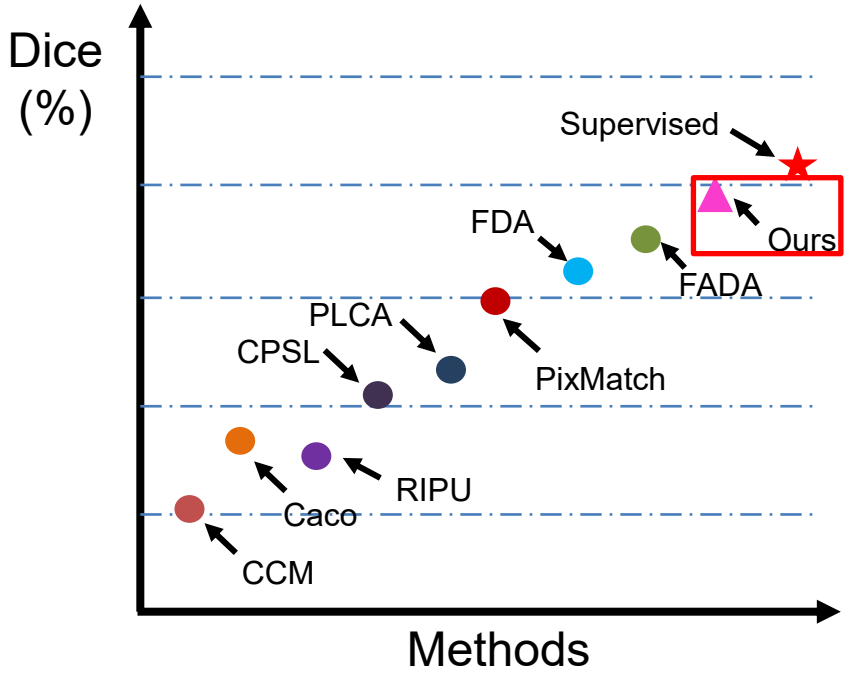
Motivation of Graph-Echo Design

Motivations of our Graph-Echo

1. Echocardiography exist both **spatial and temporal morphology**;
2. Graphical representations can **describe irregular objects;**
3. Graphs can iteratively **aggregate the knowledge** based on the broadcasting;



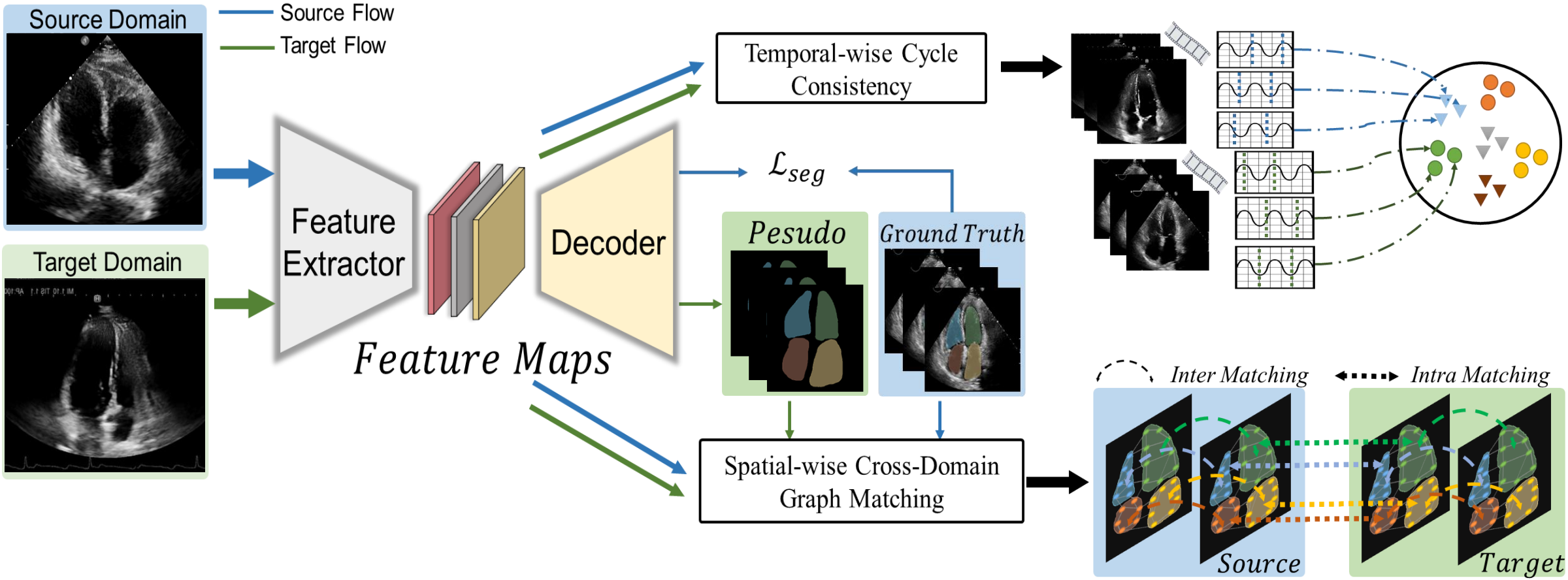
Example of echocardiogram videos with segmentation annotations.



The performance between existed methods and ours.

Graph-Driven UDA Segmentation

Overview of our proposed framework Graph-Echo



The Details of Modules

Temporal-wise Cycle Consistency modules

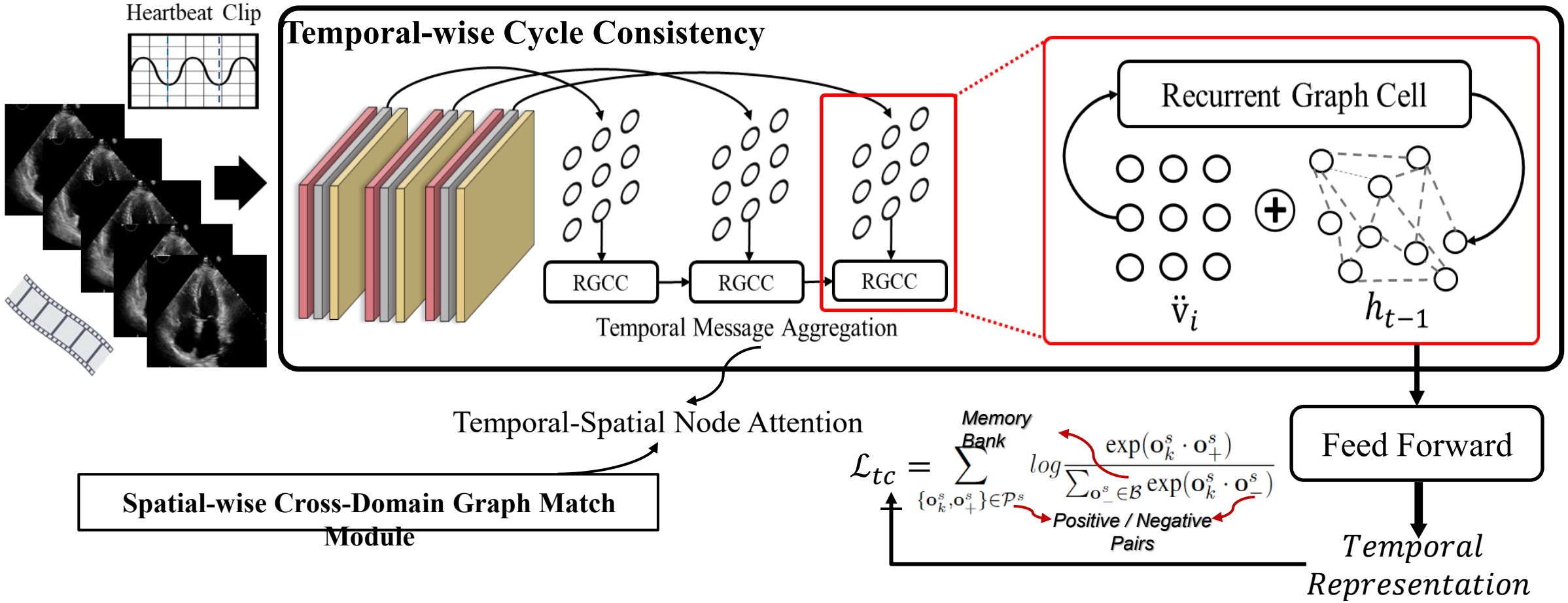


Figure 9. Temporally, we aggregate temporal messages by recurrent graph cells. Since each clip exist more than two heartbeat cycle, thus each clip that is extracted from video should be the same in temporal representation.

The Details of Modules

Spatial graph matching

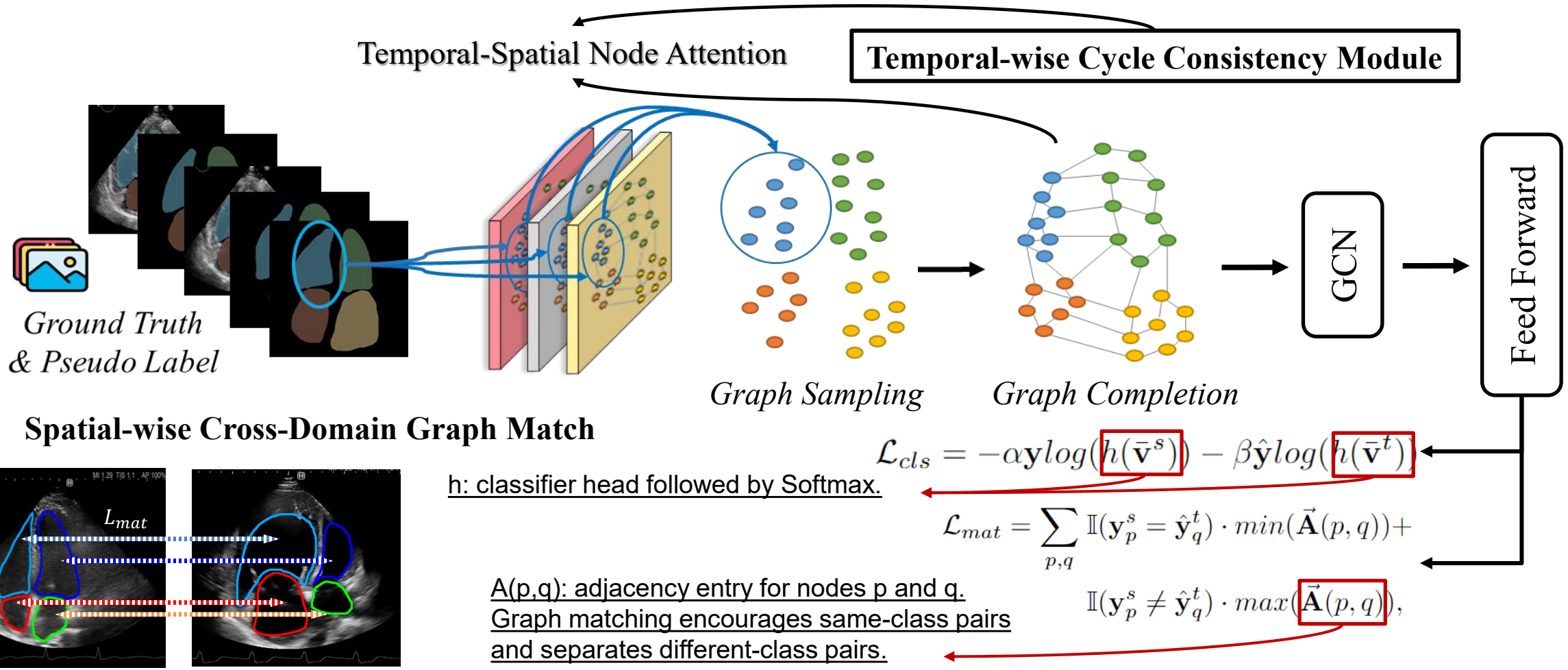


Figure 10. Spatially, we construct complete semantic graphs suitable for intra-domain and inter-domain matching of cardiac structures;

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The Main Result

Method	site G (*source*) ⟶ site R (*target*)					site R (*source*) ⟶ site G (*target*)				
	LVLA	PALA	LVSA	A4C	Avg.	LVLA	PALA	LVSA	A4C	Avg.
Semi-Supervised Segmentation Methods										
CPS CVPR' 21	63.2±0.9	60.5±1.1	57.0±0.8	64.2±1.0	61.2±0.7	64.9±1.4	63.3±1.1	59.8±0.6	61.0±1.9	62.2±1.4
PC^{2}Seg ICCV' 21	64.1±0.8	58.2±0.6	70.2±0.9	63.9±1.2	64.1±1.1	63.7±1.0	64.4±1.0	65.2±1.3	67.0±0.9	65.1±0.8
U^{2}PL CVPR' 22	66.2±1.2	62.9±1.5	69.1±0.9	64.4±1.4	65.6±1.6	62.1±1.2	61.5±1.1	61.9±1.0	65.0±0.5	62.7±0.9
Unsupervised Domain Adaptation Methods										
Without DA	53.5±0.4	42.8±0.7	47.6±0.6	50.1±1.0	48.5±0.7	55.2±0.5	47.4±0.6	49.1±0.3	52.8±0.9	51.1±0.5
PLCA NIPS' 20	58.2±1.7	21.0±4.9	40.2±3.6	60.3±2.3	44.9±3.1	60.1±2.9	38.8±3.0	42.9±2.7	59.4±2.6	50.3±2.9
PixMatch CVPR' 21	60.8±1.8	52.7±1.6	56.0±1.8	66.5±1.5	59.0±1.7	62.9±2.0	49.0±3.2	63.2±2.1	69.9±1.8	61.3±1.8
FDA CVPR' 20	67.3±2.0	65.5±1.5	54.8±2.3	64.3±1.9	63.0±1.5	65.8±1.7	63.2±1.8	61.9±2.1	64.5±1.5	63.9±1.8
FDA-MBT CVPR' 20	64.4±0.9	65.1±0.8	61.7±1.1	70.1±1.3	65.3±1.2	66.3±0.9	64.9±1.4	67.2±0.6	71.3±0.9	67.4±0.8
FADA ECCV' 20	70.1±1.2	68.3±1.4	76.1±0.7	72.4±0.6	71.7±0.8	69.9±0.9	67.7±1.0	74.5±1.4	70.0±0.5	70.5±1.1
Ours	**73.9±1.2**	**75.5±1.3**	**76.8±0.4**	**76.3±0.7**	**75.6±0.9**	**73.3±1.0**	**74.9±1.2**	**80.2±0.3**	**78.2±0.5**	**76.7±0.5**
Upper Bound	79.1±0.4	82.4±0.6	82.1±1.0	81.4±1.2	81.3±0.5	80.5±1.4	79.2±0.8	83.3±1.6	83.9±0.2	81.7±0.8

Table 3. Results across domain Site G and Site R, shows that results of different UDA method applied in our CardiacUDA dataset, where ours reach the highest Dice score with 4% and 6% improvement across two centers in domain adaptation.

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The Main Result

Result Across Different Datasets

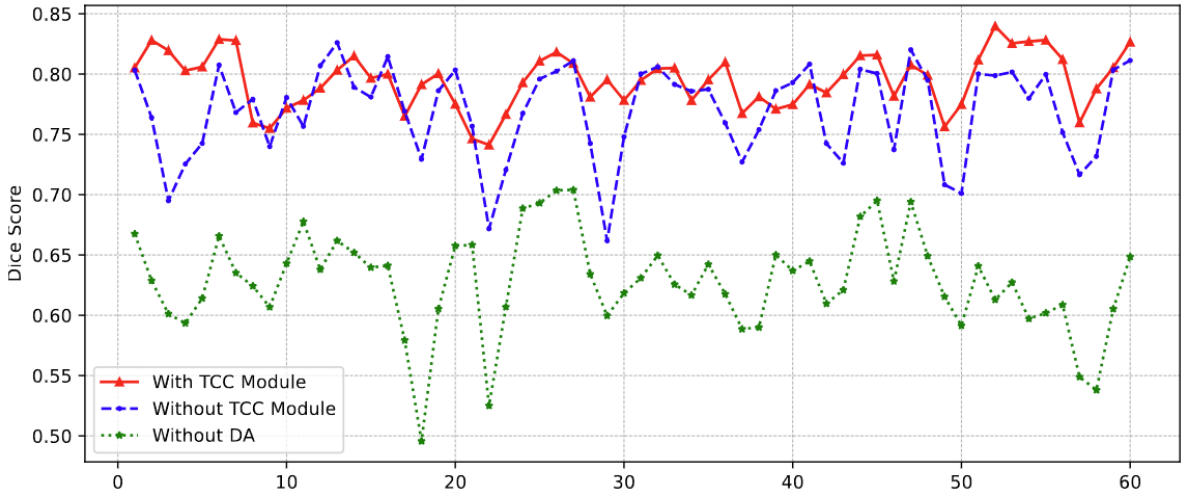
Method	CAMUS→Echo		Echo→CAMUS		Ours→Echo		Ours→CAMUS		Echo→Ours	CAMUS→Ours
	EDV	ESV	EDV	ESV	EDV	ESV	EDV	ESV	Avg.	Avg.
Without DA.	69.2±1.4	66.2±2.2	64.3±0.9	59.6±1.7	34.1±1.4	33.8±2.2	31.0±0.9	32.4±1.7	22.9±1.6	19.2±1.4
CVPR' 22 U^2PL	63.2±0.9	67.8±0.9	57.2±1.1	60.1±1.2	49.5±0.9	51.3±0.7	43.1±0.9	46.7±1.2	36.5±1.0	34.3±0.8
CVPR' 22 Caco	55.9±1.5	56.0±1.3	47.3±1.3	49.0±1.2	38.6±3.4	40.8±2.6	46.1±1.9	45.5±2.2	29.6±1.8	26.8±2.3
CVPR' 22 RIPU	64.3±1.2	67.7±1.0	70.2±0.6	68.2±0.9	46.9±0.7	47.4±0.7	56.0±1.2	51.5±1.4	36.2±1.7	31.7±2.0
NIPS' 20 PLCA	71.1±0.4	69.3±0.5	72.9±0.8	68.3±0.7	51.9±0.9	49.7±0.9	52.4±1.4	52.1±1.2	35.3±1.1	36.1±0.8
CVPR' 20 FDA	78.8±1.1	75.4±1.0	76.2±0.4	74.1±0.5	55.6±0.4	54.0±0.6	56.8±0.5	56.1±0.3	38.4±1.5	37.9±1.2
ECCV' 20 FADA	77.5±0.8	76.5±0.5	78.6±0.8	76.6±0.6	54.1±0.6	52.0±1.0	57.4±1.0	55.2±0.8	41.7±1.1	39.0±0.9
Ours	**83.4±0.7**	**81.8±0.9**	**87.6±0.4**	**82.4±1.0**	**61.2±0.5**	**61.8±0.7**	**66.3±0.3**	**64.9±0.4**	**46.2±1.3**	**44.0±1.6**
Upper Bound	93.4±0.6	90.5±1.3	89.3±1.1	87.9±0.8	93.4±0.6	90.5±1.3	89.3±1.1	87.9±0.8	81.3±0.9	81.3±0.6

Table 4. We also evaluate our method across dataset EchoNet, CAMUS and CardiacUDA, and report all results. Our method can reach highest score compare with other UDA methods.

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Ablation Studies

Figure 7. Segmentation result for each frame in a video example.



The Effectiveness of Different Modules & Loss

Table 5 : Effectiveness of SCGM and TCC.

	SCGM	TCC	Dice Scores (%)	
			G→R	R→G
Base	✗	✗	$48.5_{\pm 0.7}$	$51.1_{\pm 0.5}$
Base + SCGM	✓	✗	$74.3_{\pm 1.0}$	$71.3_{\pm 0.7}$
Base + TCC	✗	✓	$55.3_{\pm 0.8}$	$53.0_{\pm 1.2}$
Ours	✓	✓	$\mathbf{75.6}_{\pm \mathbf{0.9}}$	$\mathbf{76.7}_{\pm \mathbf{0.5}}$

Table 5: classification and matching losses.

$\mathcal{L}_{cls}$	$\mathcal{L}_{mat}$	Averaged Dice Score (%)	
		G→R	R→G
✗	✗	$48.5_{\pm 0.7}$	$51.1_{\pm 0.5}$
✓	✗	$51.9_{\pm 1.2}$	$53.7_{\pm 1.3}$
✗	✓	$53.4_{\pm 1.6}$	$54.0_{\pm 1.5}$
✓	✓	$74.3_{\pm 1.0}$	$71.3_{\pm 0.7}$

Table 6: temporal and adversarial losses.

$\mathcal{L}_{tc}$	$\mathcal{L}_{adv}$	Averaged Dice Score (%)	
		G→R	R→G
✗	✗	$74.3_{\pm 0.7}$	$71.3_{\pm 0.5}$
✓	✗	$74.4_{\pm 1.1}$	$75.6_{\pm 0.9}$
✗	✓	$74.1_{\pm 0.9}$	$73.5_{\pm 1.0}$
✓	✓	$\mathbf{75.6}_{\pm \mathbf{0.9}}$	$\mathbf{76.7}_{\pm \mathbf{0.5}}$

The Main Result

Visualization Results

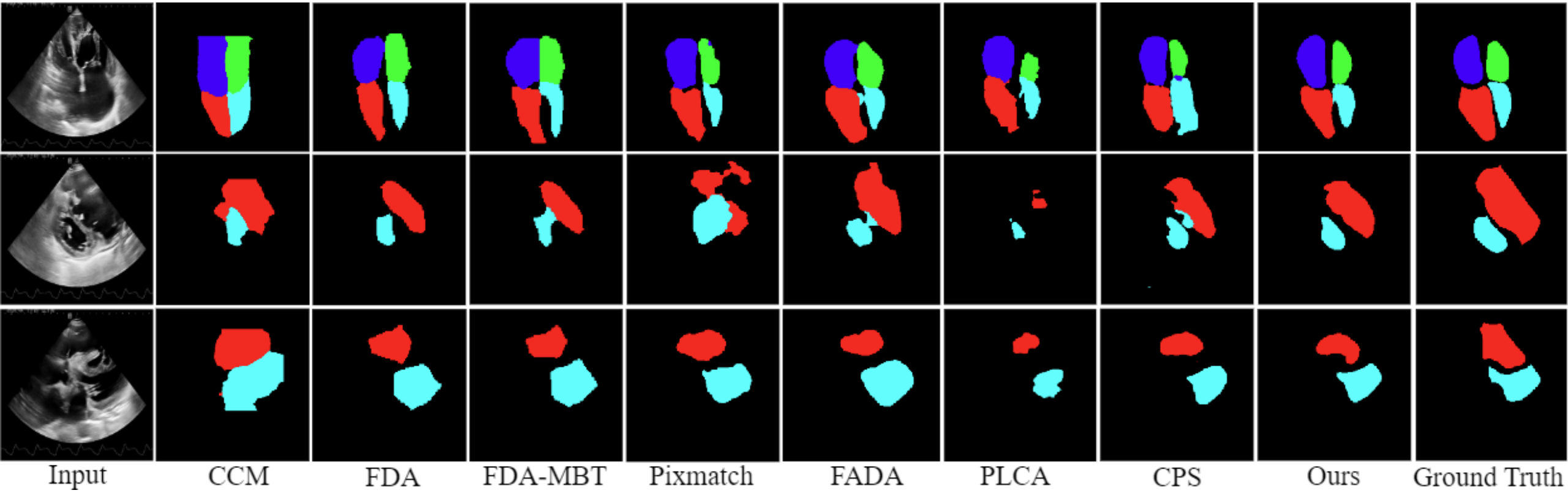
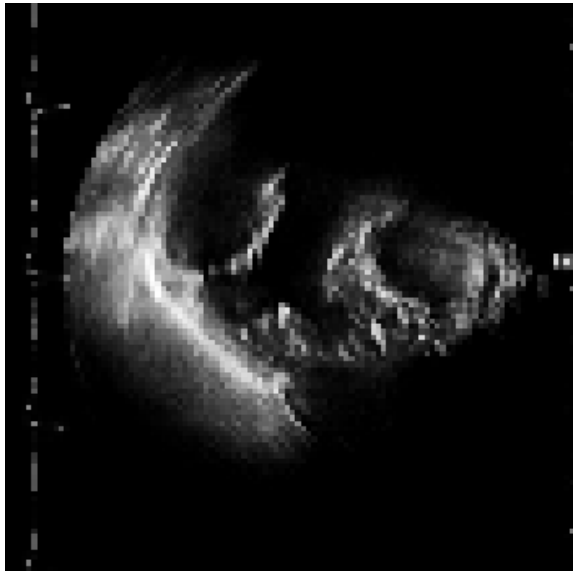


Figure 11. Three-view segmentation comparison. Red, green, blue and cyan represent RA, LV, RV and LA, respectively.

2. Ultrasound: case studies

An echocardiogram uses sound waves to create pictures of the heart. This common test can show blood flow through the heart and heart valves.

Accurate segmentation of the cardiac structure from echocardiogram videos is integral to several analysis tasks and has a significant impact on clinical practice.

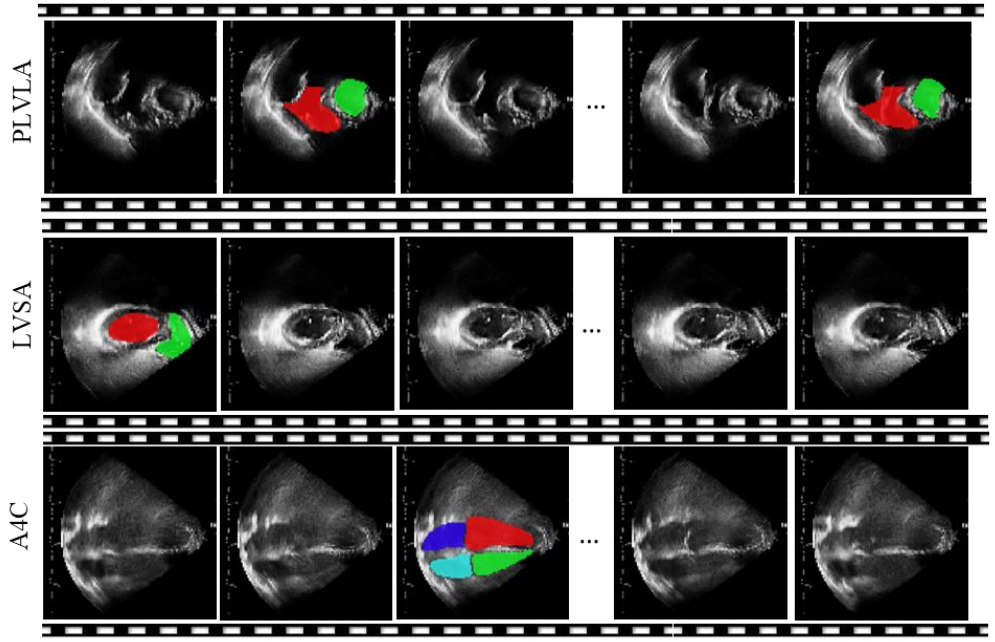


echocardiogram

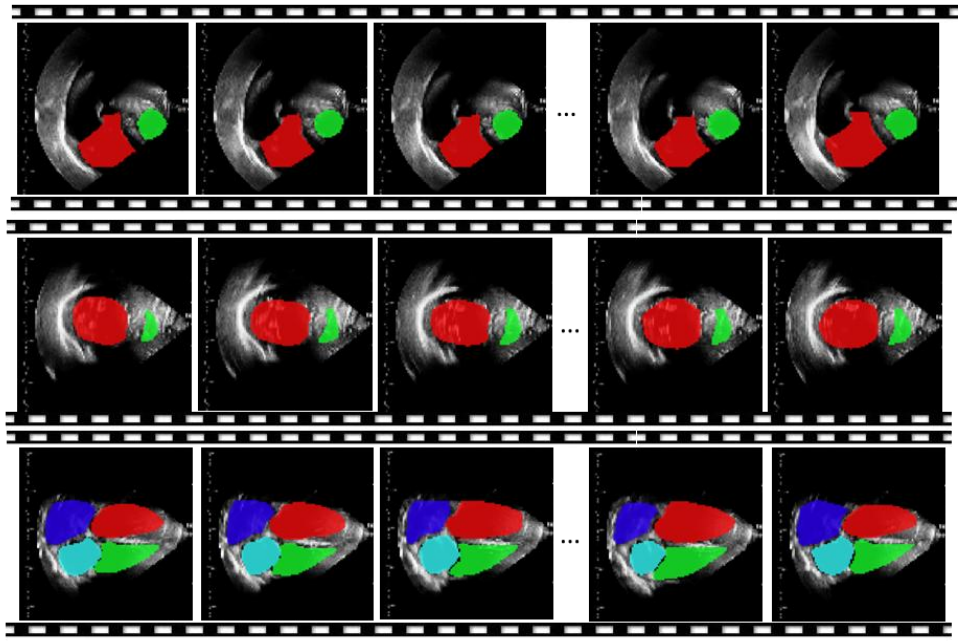
2. Ultrasound: case studies

Dataset

- We collect a large multi-view echocardiogram video dataset from one medical institution
  - **254 sparsely annotated videos** and **10 fully annotated videos** across three cardiac views (PLVLA, LVSA and A4C view)
  - Each sparsely annotated video includes **5 annotated frames**.
  - The average length of each video is **larger than 100 frames** that are able to cover more than one cardiac cycle.



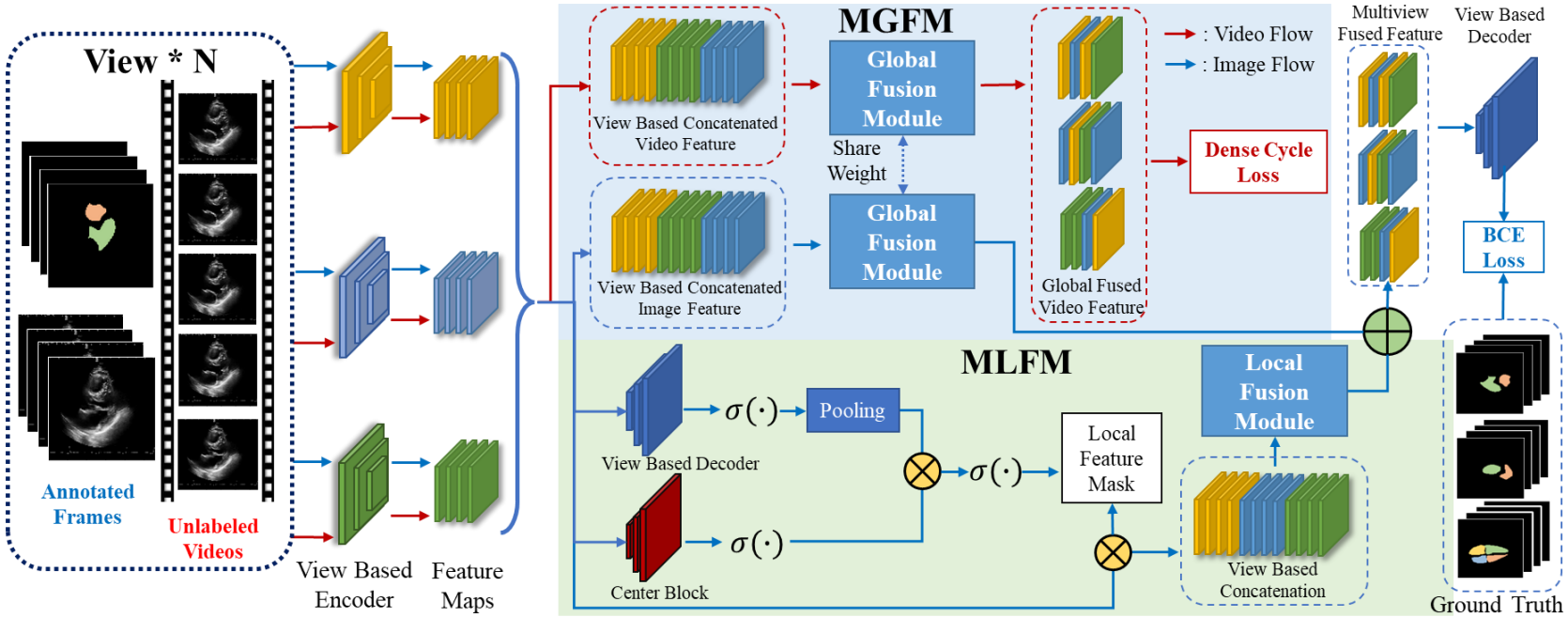
Multi-view Train Dataset

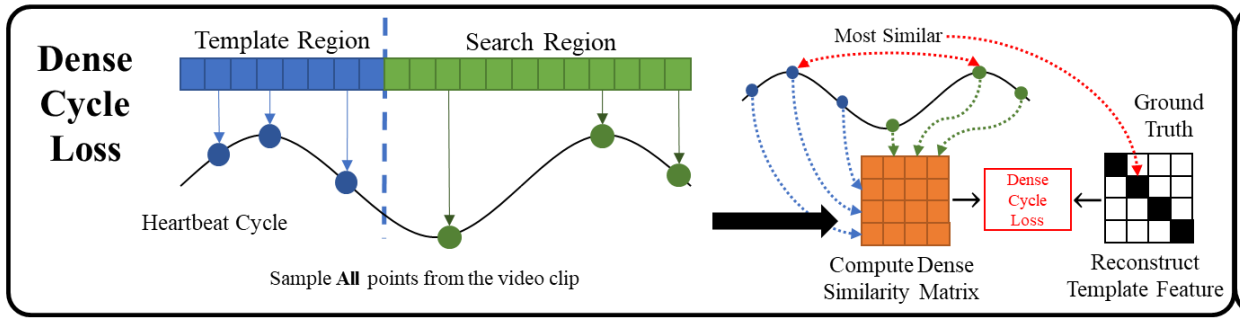


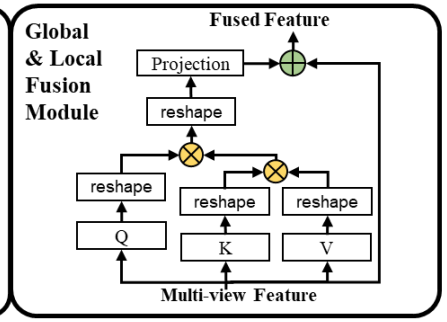
Multi-view Validation & Test Dataset

2. Ultrasound: case studies

Our Solution: GL-Fusion







Key Idea: Aggregate global-local information from different views and utilize video information based on temporal cyclicality.

Contribution:

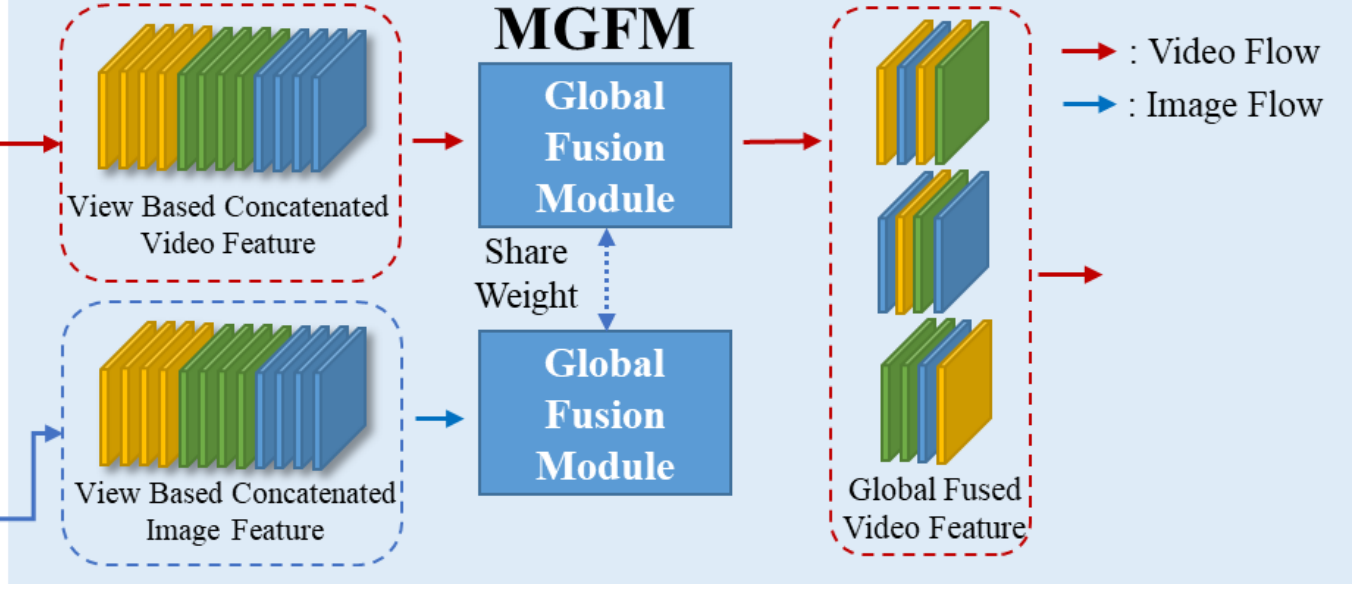
- Multi-view global-based fusion module (MGFM)
- Multi-view local-based fusion module (MLFM)
- Dense Cycle Loss

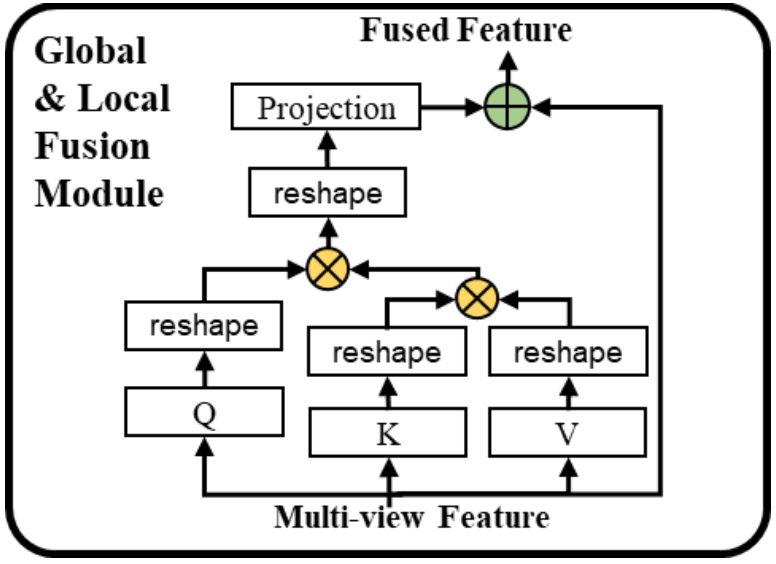
2. Ultrasound: case studies

MGFM

Features from each view enter the global fusion module.

1. Concatenate the view-specific features along the view dimension.
2. Use view-wise attention to exchange information across views.
3. Send the fused features to dense cycle learning and the segmentation decoder.



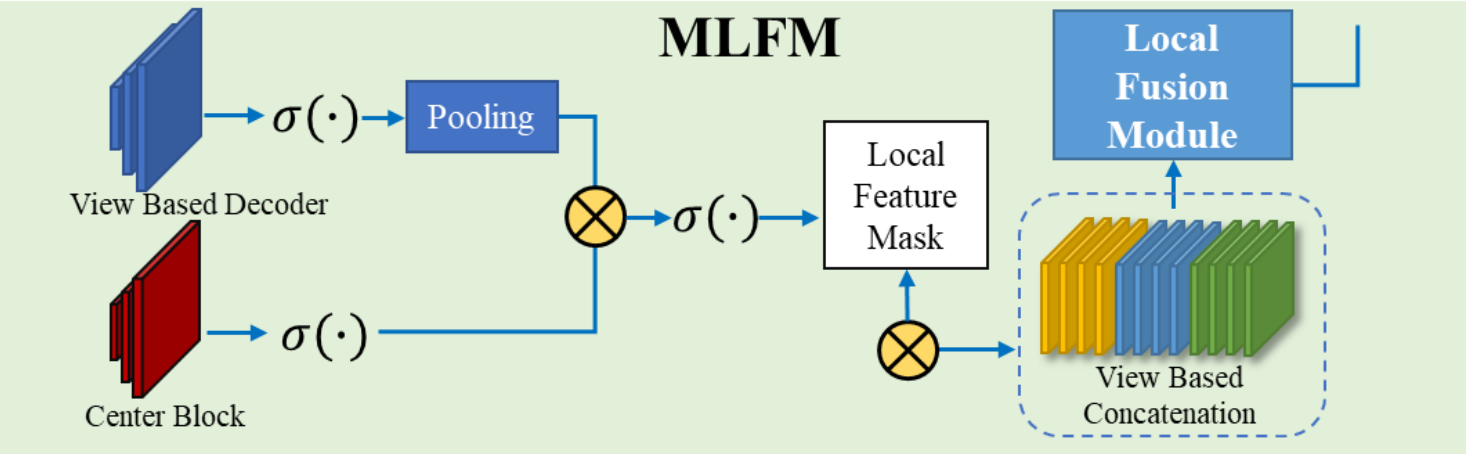


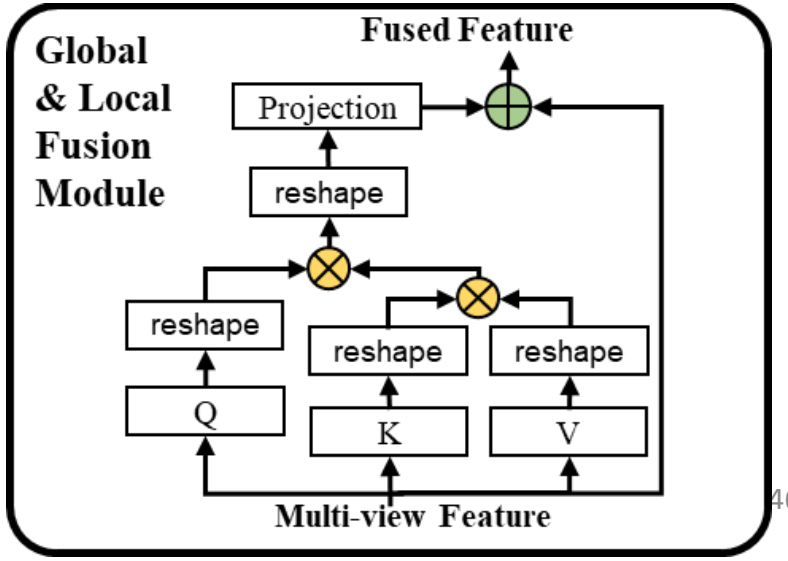
2. Ultrasound: case studies

MLFM

Features from each view enter the local fusion module.

1. The decoder and centre block produce a local mask M_i.
2. Form local features by element-wise gating: $F_i(\text{local}) = M_i \odot F_i$.
3. Use view-wise attention, as in MGFM, to fuse the local features.

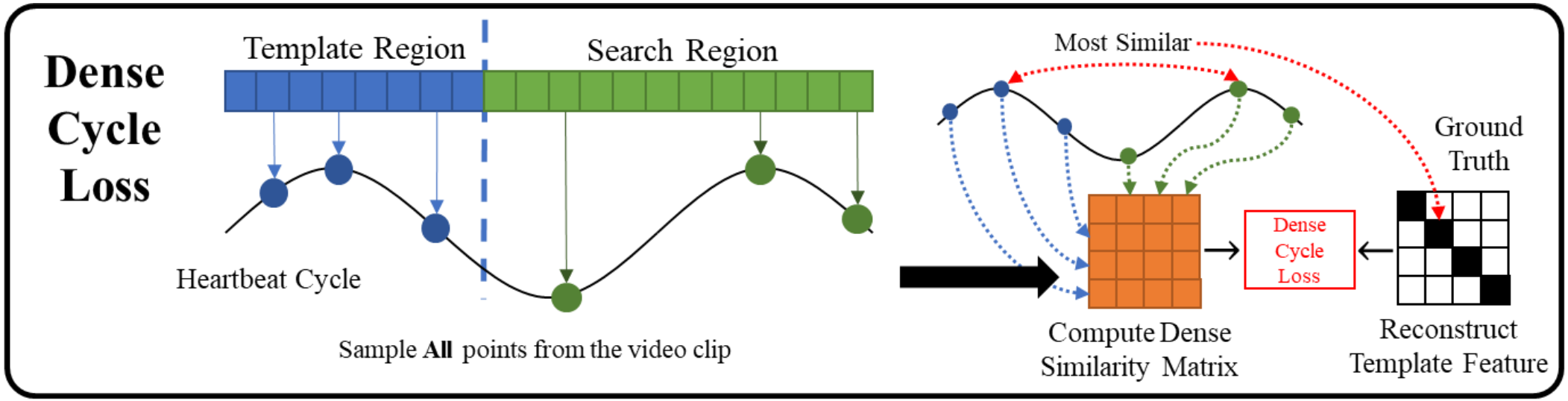




Dense Cycle Loss

Give extracted feature $\{F^i\}_{i=1}^{V}$ from different view:

1. Split the global-fused feature of each video into template region and search region
2. Densely sample all feature intervals and compute the similarity between template region and search region
3. Reconstruct the feature interval in template region based on the similarity in step2
4. Compute the similarity of reconstructed feature and the feature of search region, and consider the index as ground-True label



2. Ultrasound: case studies

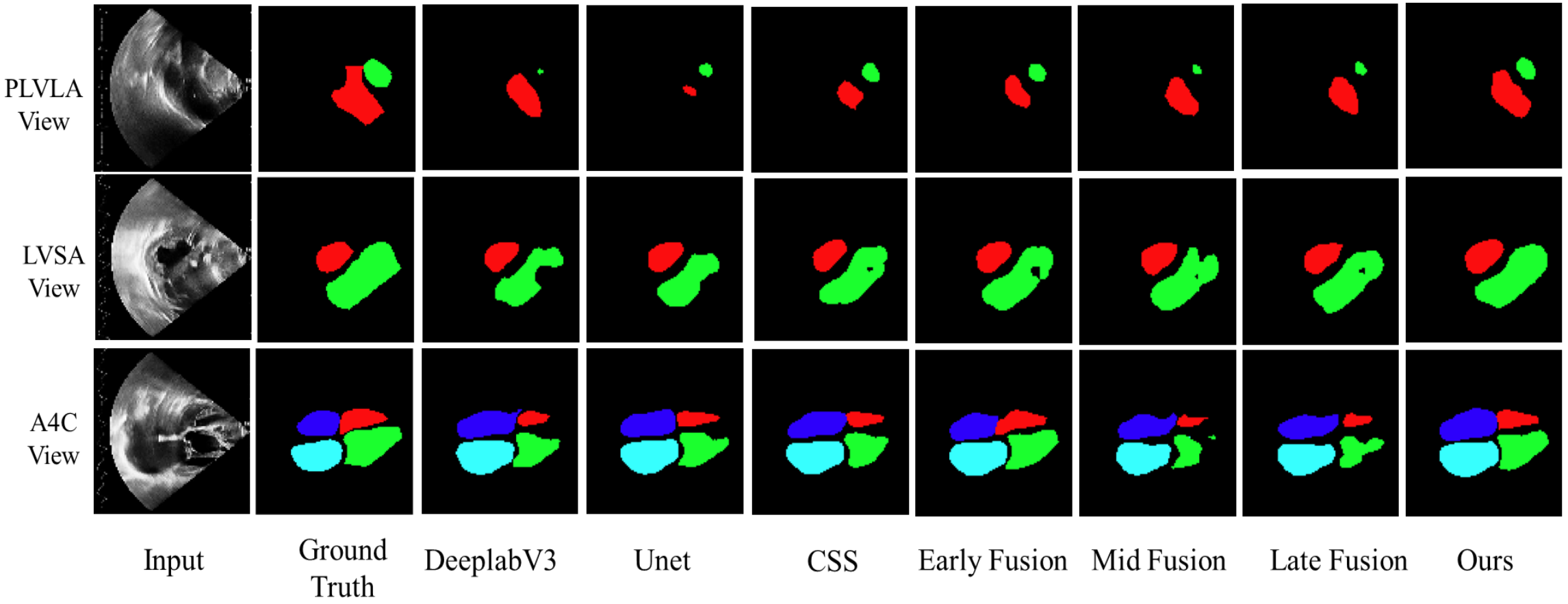
Experiments

Table 1. The comparison with other methods. all results are reported in Dice Score.

	Method	PLVLA	LVSA	A4C	Average Dice (%)
Single-view	DeeplabV3 [5]	70.93	75.14	77.33	74.46
	U-Net [19]	73.35	77.57	76.60	75.84
	CSS [7]	79.09	79.70	77.71	78.83
Fusion-based	Early-fusion	79.78	77.07	77.58	78.14
	Mid-fusion	77.89	76.75	72.44	75.69
	Late-fusion	71.62	75.31	74.68	73.87
	TransFusion [15]	78.79	80.23	59.31	72.78
	Ours	**83.84**	**81.76**	**81.28**	**82.29**

2. Ultrasound: case studies

Experiments



2. Ultrasound: case studies

Experiments

Result: Ablation Study

Table 2. Effectiveness of MGFM and MLFM. This table shows the performance of the Global and Local fusion modules

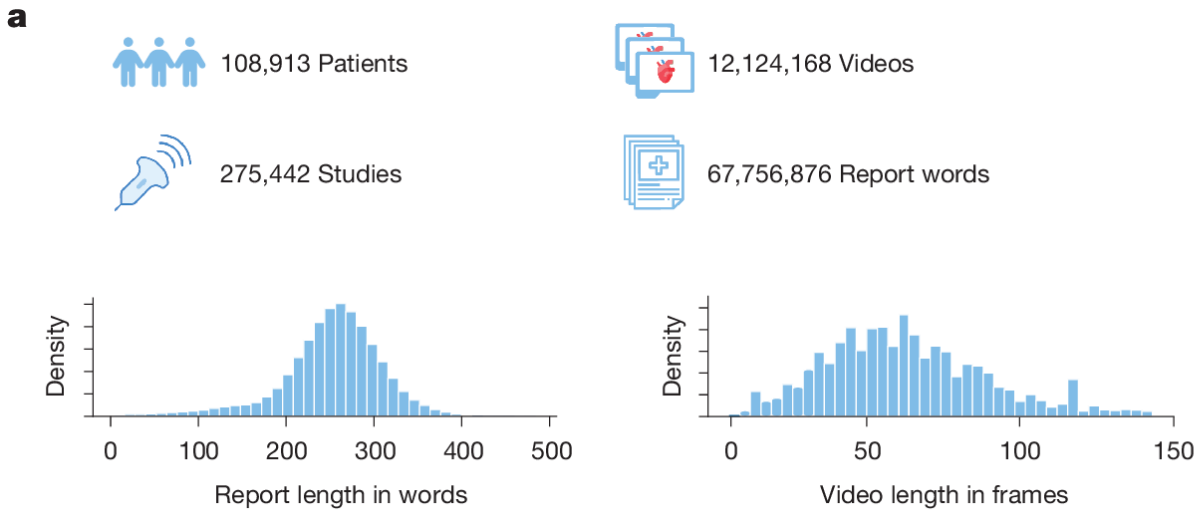
	MGFM	MLFM	Avg. Dice(%)
Base	✗	✗	74.46
Base+MGFM	✓	✗	80.20
Base+MLFM	✗	✓	78.41
Ours	✓	✓	**82.29**

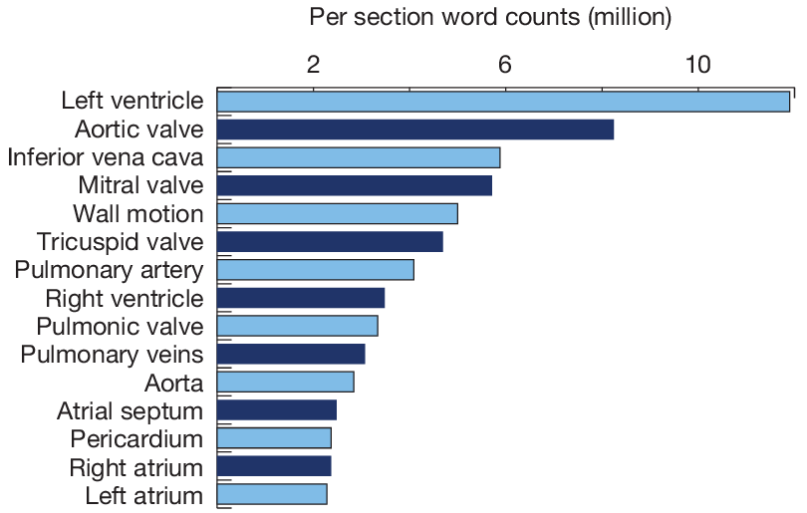
Table 3. Effectiveness of Cyc. and Dense Cyc.. This table shows the effectiveness of vanilla cycle loss [7] (Noted by Cyc.) and our proposed dense cycle (Noted by Dense Cyc.).

	Cyc.	Dense Cyc.	Avg. Dice(%)
Fusion-only	✗	✗	80.36
Fusion+Cyc.	✓	✗	79.33
GL-Fusion	✓	✓	**82.29**

2. Ultrasound: case studies

EchoPrime (1/8): training data



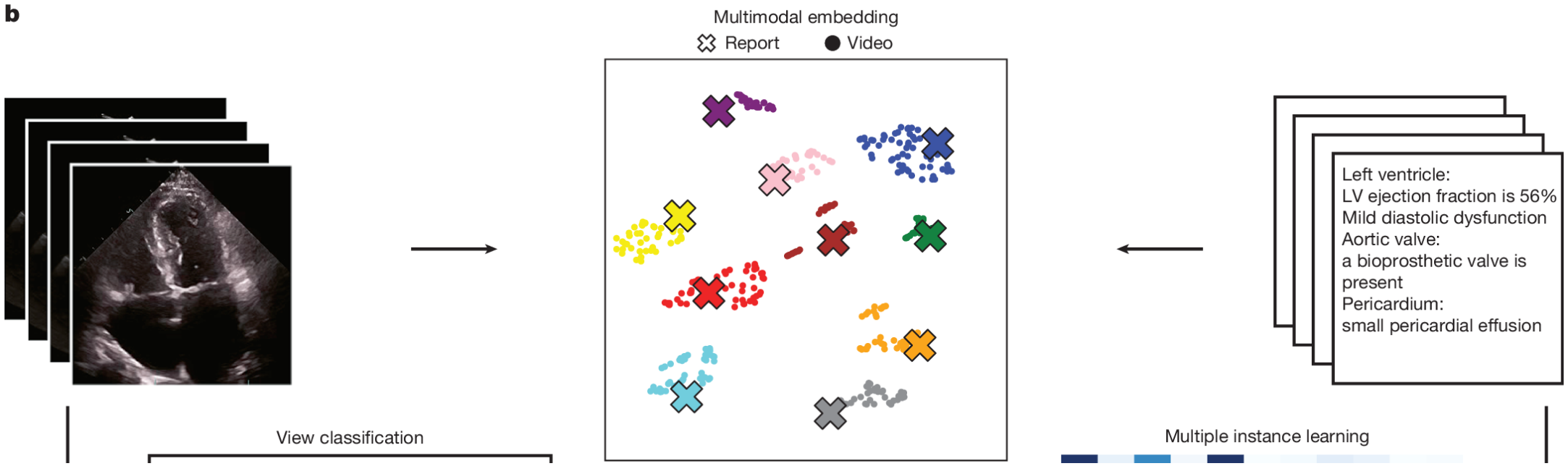


Split at CSMC	Patients	Studies	Videos
Training	107,663	272,256	11,984,170
Validation	250	565	25,167
Internal test	1,000	2,621	114,831

Vukadinovic et al., Nature 2026 (online 2025). Fig. 1a and Methods. Figures reproduced for teaching, CC BY-NC-ND 4.0.

2. Ultrasound: case studies

EchoPrime (2/8): video–report alignment



$$s(i,j) = \cos(v_i, t_j) / \tau$$

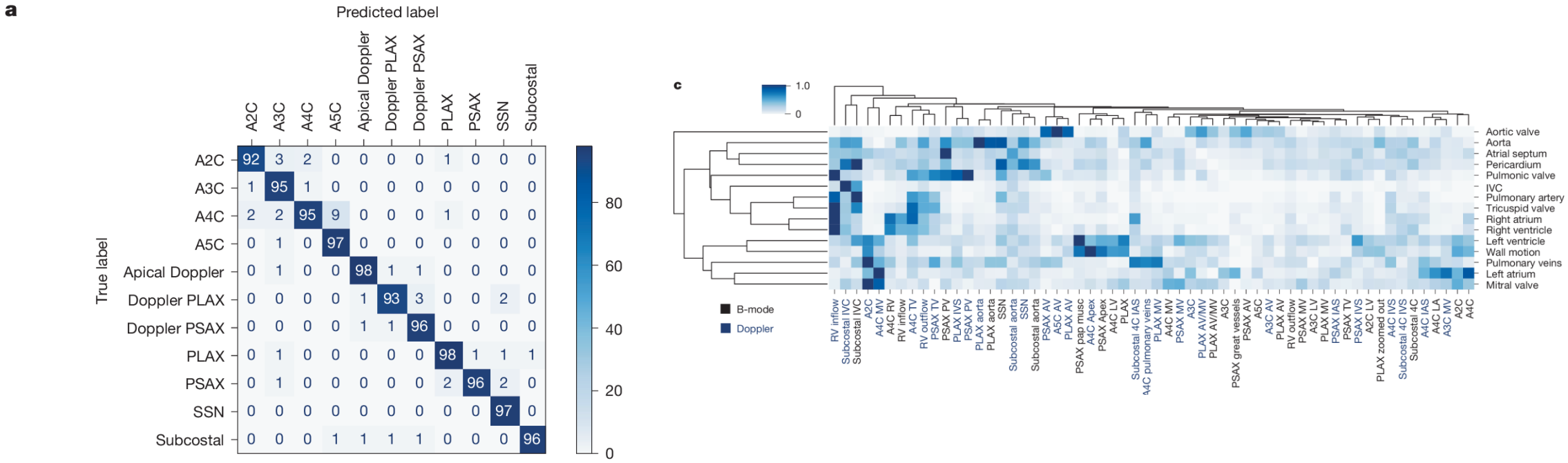
mViT: 16 frames at 224 × 224
BioMedBERT: up to 512 text tokens

A paired report can describe findings that one video cannot show.

Vukadinovic et al., Nature 2026 (online 2025). Fig. 1b detail and contrastive training Methods.

2. Ultrasound: case studies

EchoPrime (3/8): views and anatomical attention



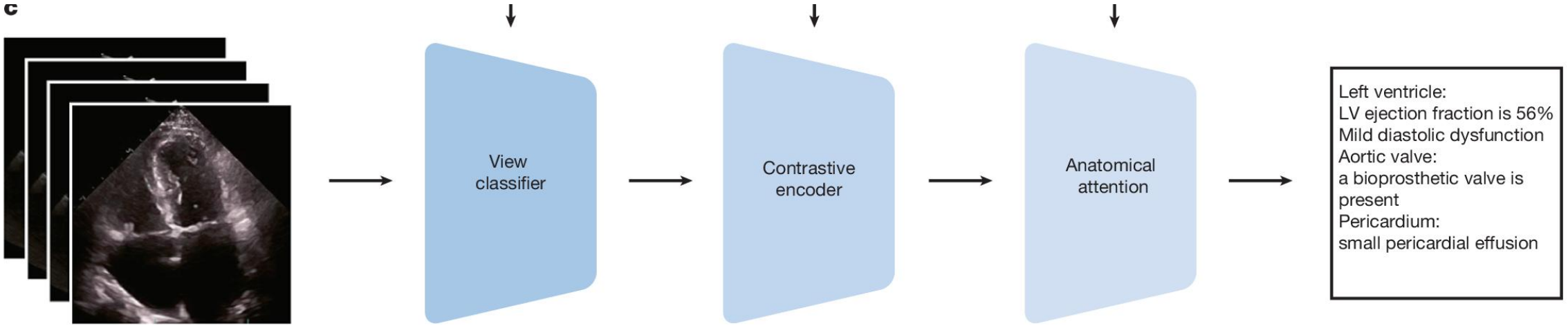
58 classes; AUC 0.997
4,000 internal test videos

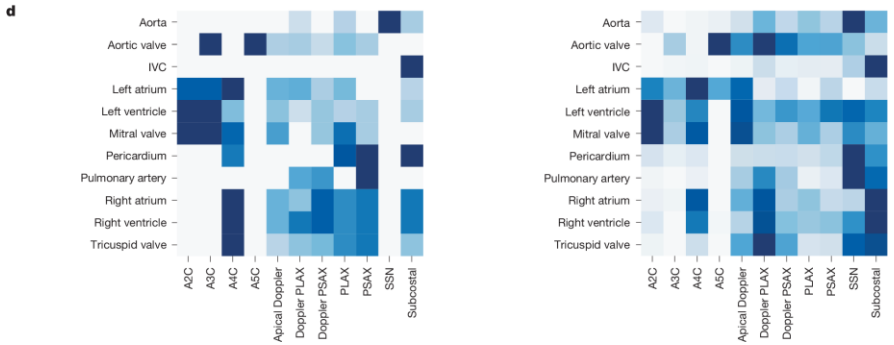
Each anatomical section learns different weights over videos.

Vukadinovic et al., Nature 2026 (online 2025). Fig. 3a,c. View classification and MIL are trained separately.

2. Ultrasound: case studies

EchoPrime (4/8): retrieval and study interpretation



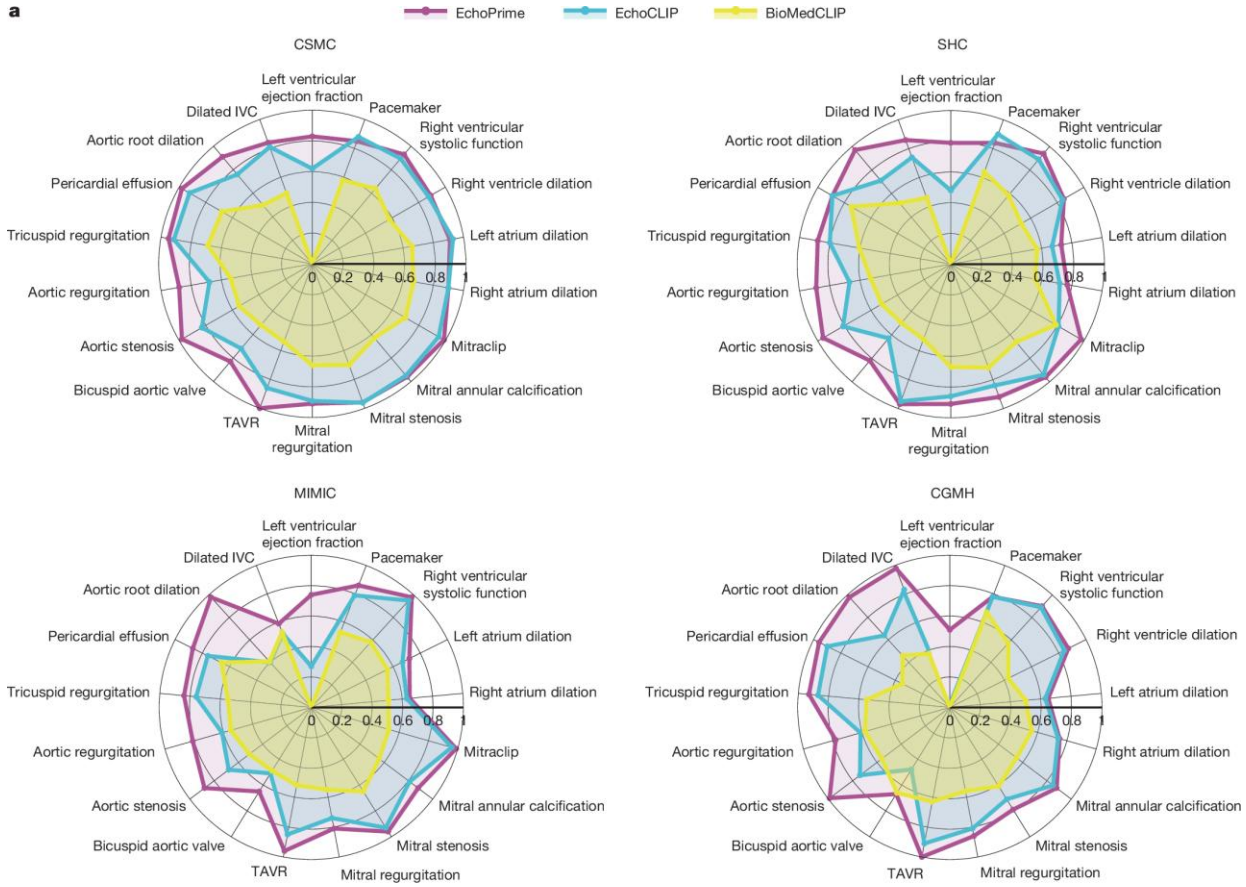


- Aggregate videos for each anatomy section
- Retrieve 50 candidate reports
- Combine extracted features for the final assessment

Vukadinovic et al., Nature 2026 (online 2025). Fig. 1c and Fig. 3d. Retrieved interpretation and view weighting.

2. Ultrasound: case studies

EchoPrime (5/8): external validation



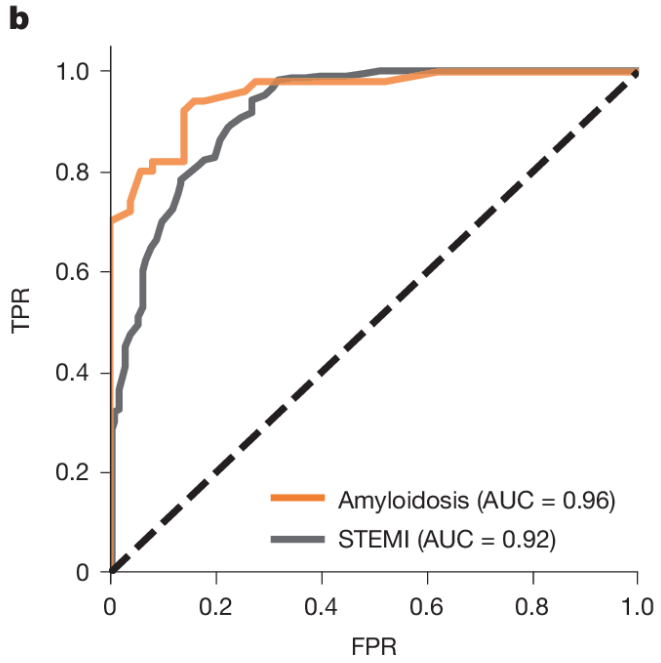
External cohort	Studies
SHC	1,792
MIMIC	2,431
CGMH	1,188
KP	4,891

Classification: AUROC
Regression: R^2
All four cohorts were retrospective.

Vukadinovic et al., Nature 2026 (online 2025). Fig. 2a excerpt and external-cohort Methods.

2. Ultrasound: case studies

EchoPrime (6/8): effect sizes and baselines



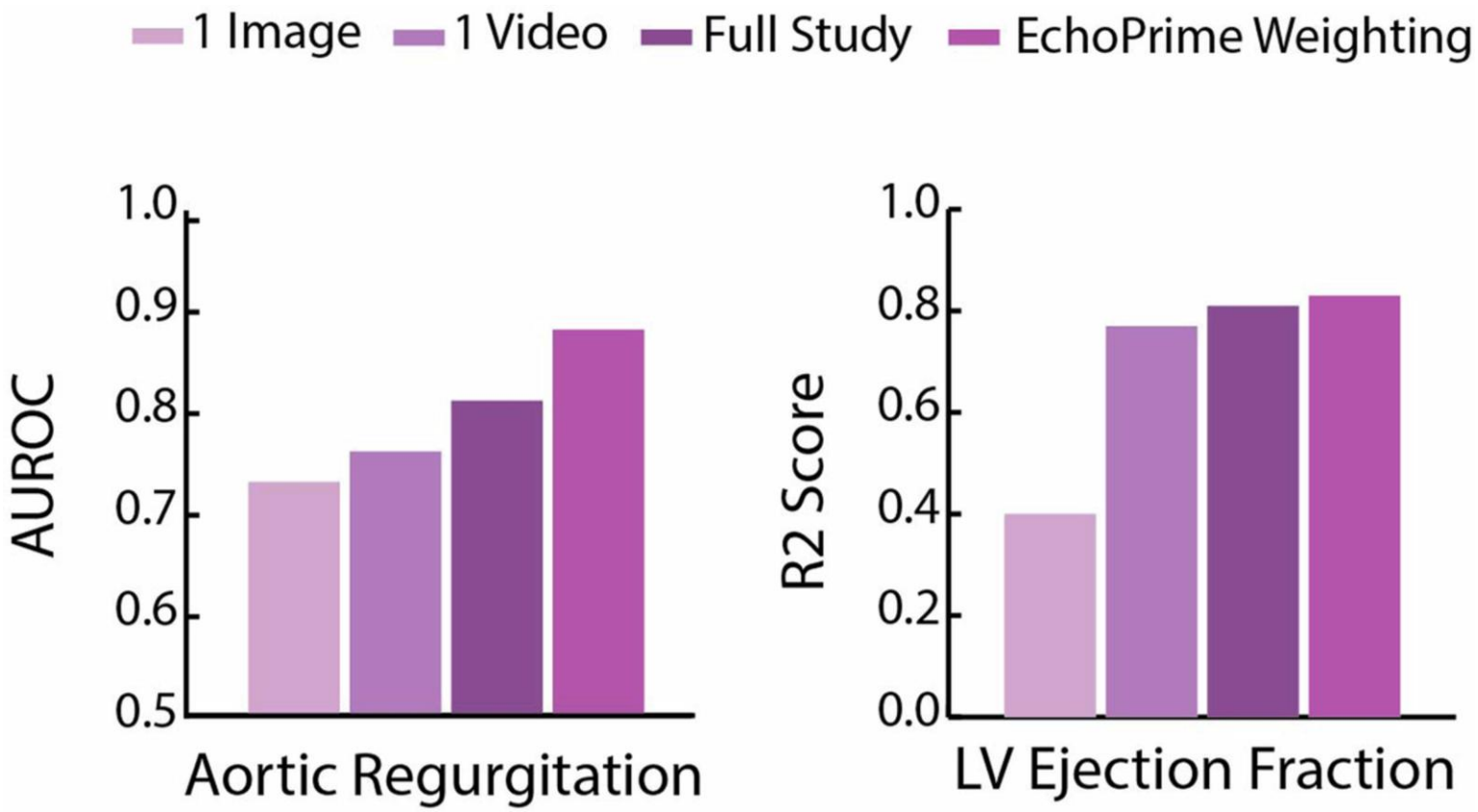
CSMC task	EchoPrime	EchoCLIP	BioMedCLIP
LVEF MAE ↓	4.79	7.00	26.93
Aortic regurg. AUC ↑	0.88	0.68	0.54

External LVEF MAE: 4.14–6.46
Left: kNN probing, amyloidosis 0.96 and STEMI 0.92 AUROC.

Vukadinovic et al., Nature 2026 (online 2025). Table 2 / Results and Fig. 2b. MAE is in LVEF percentage points.

2. Ultrasound: case studies

EchoPrime (7/8): what information helps?

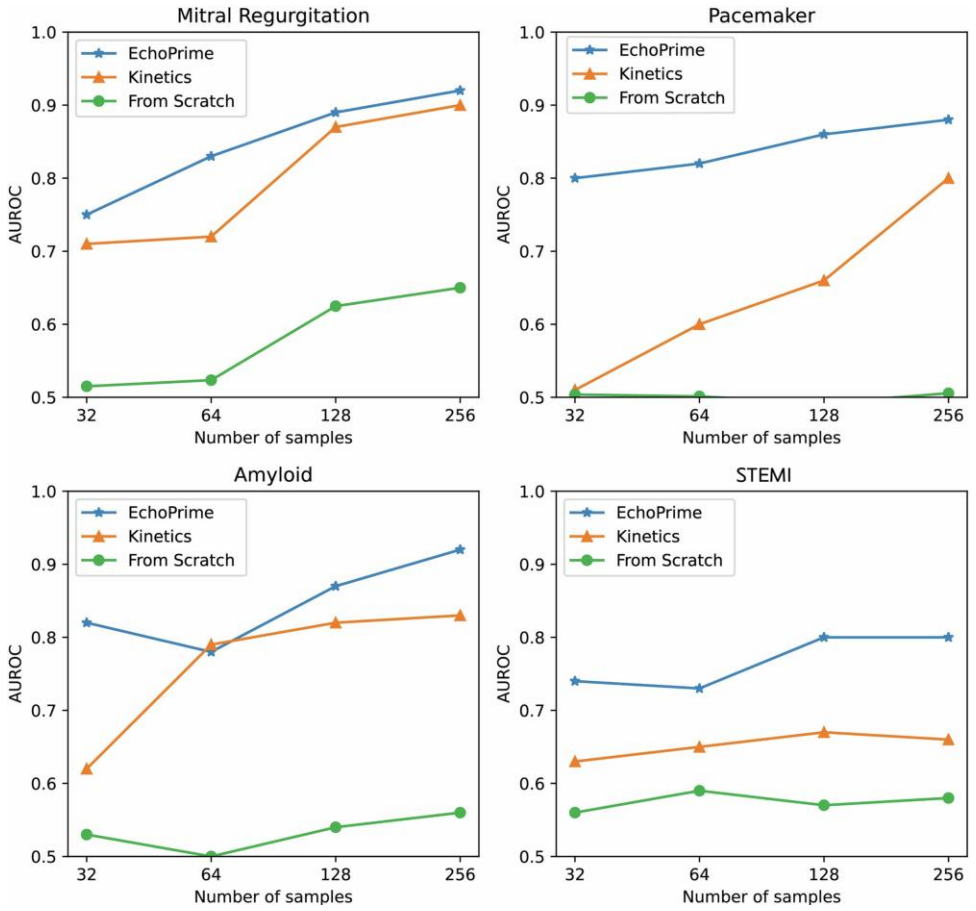


- A frame lacks temporal evidence
- Multiple videos add complementary views
- Anatomical weighting improves aggregation

Vukadinovic et al., Nature 2026 (online 2025). Extended Data Fig. 1. Input and aggregation ablation.

2. Ultrasound: case studies

EchoPrime (8/8): transfer and remaining questions

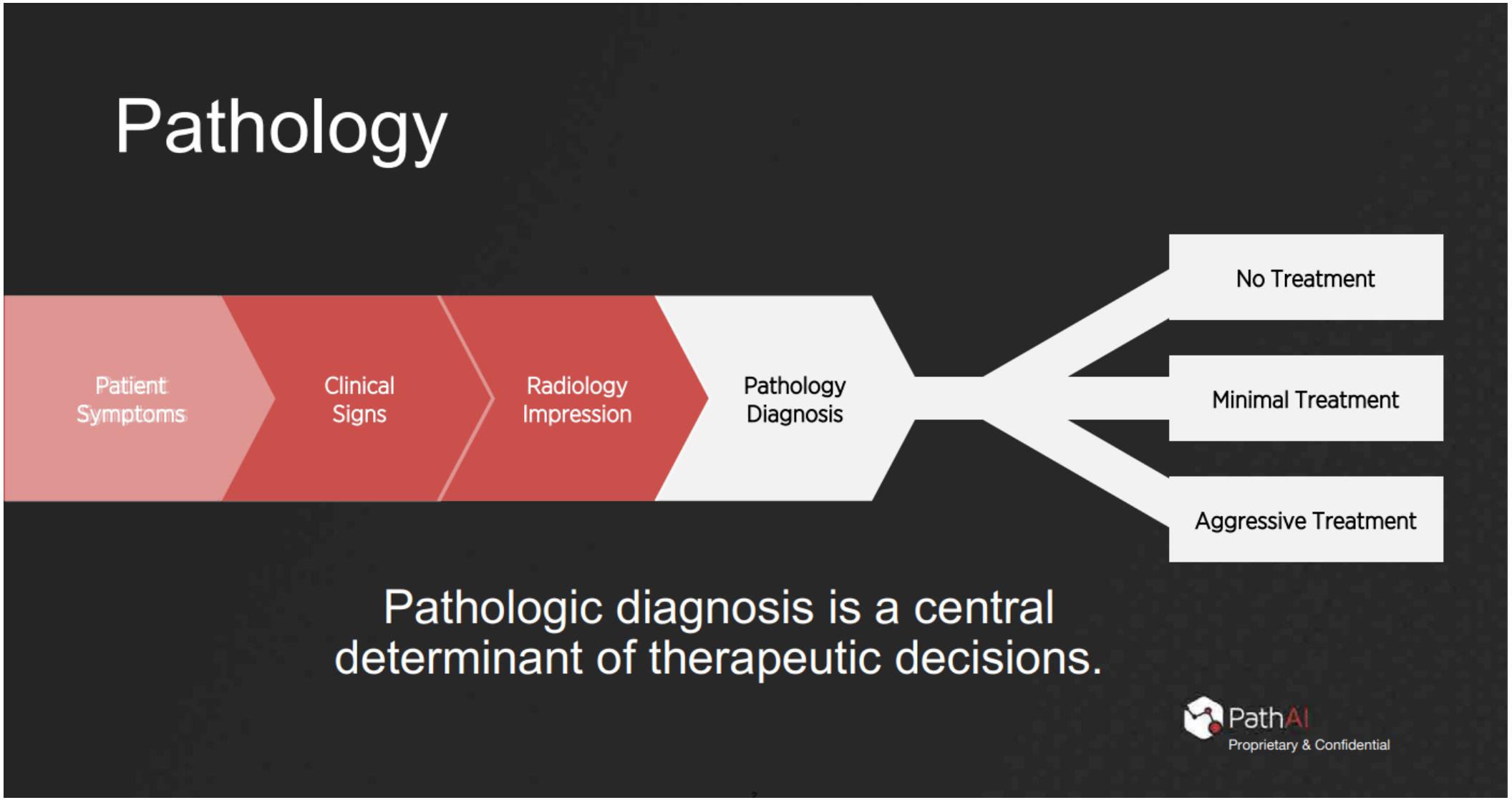


- Compare EchoPrime, Kinetics and random initialization
- Inspect 32, 64, 128 and 256 labelled examples
- Prospective workflow effects remain untested

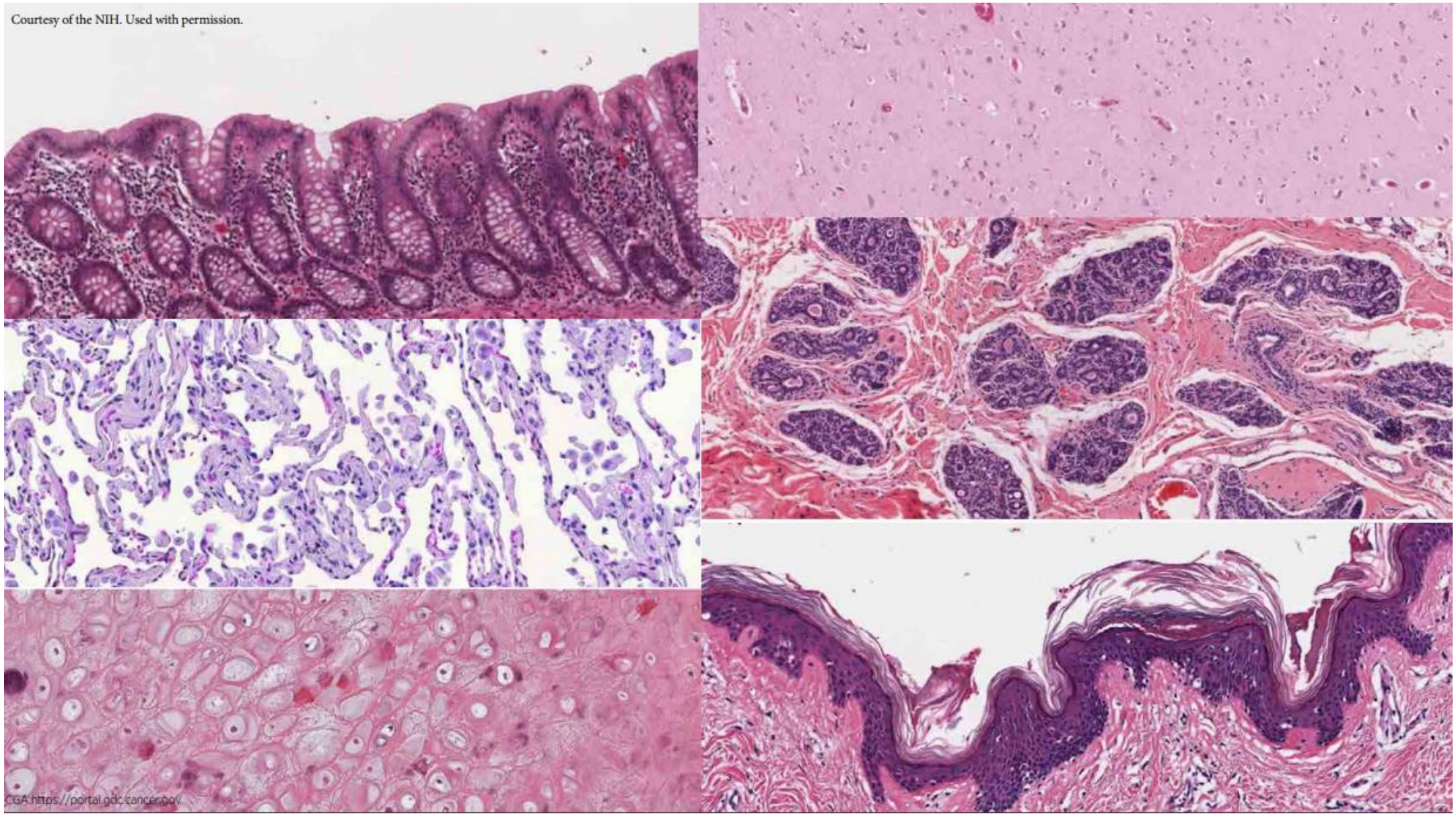
Missing inputs: spectral Doppler stills and M-mode.

Vukadinovic et al., Nature 2026 (online 2025). Extended Data Fig. 3 and Discussion.

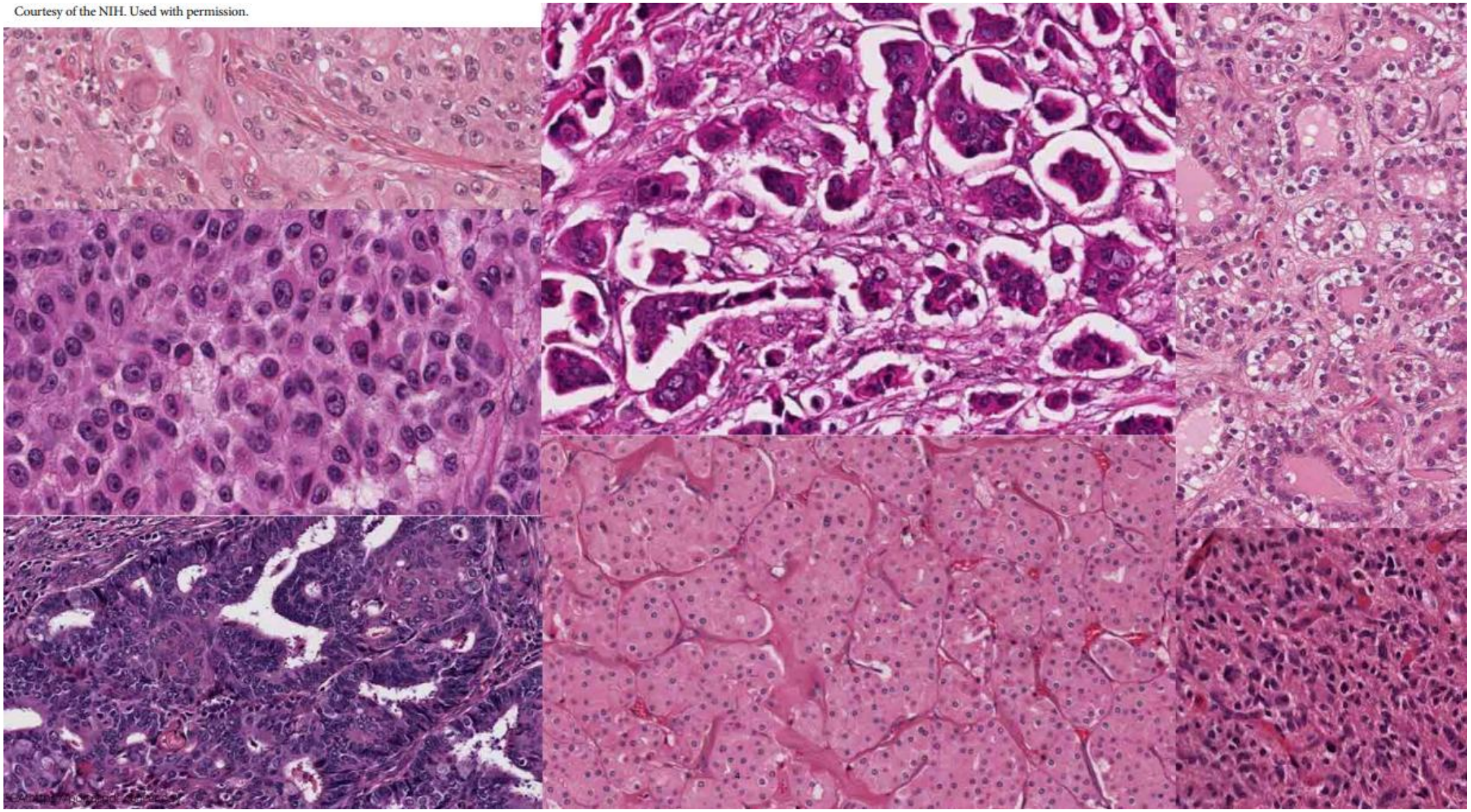
3. Computational pathology



3. Computational pathology

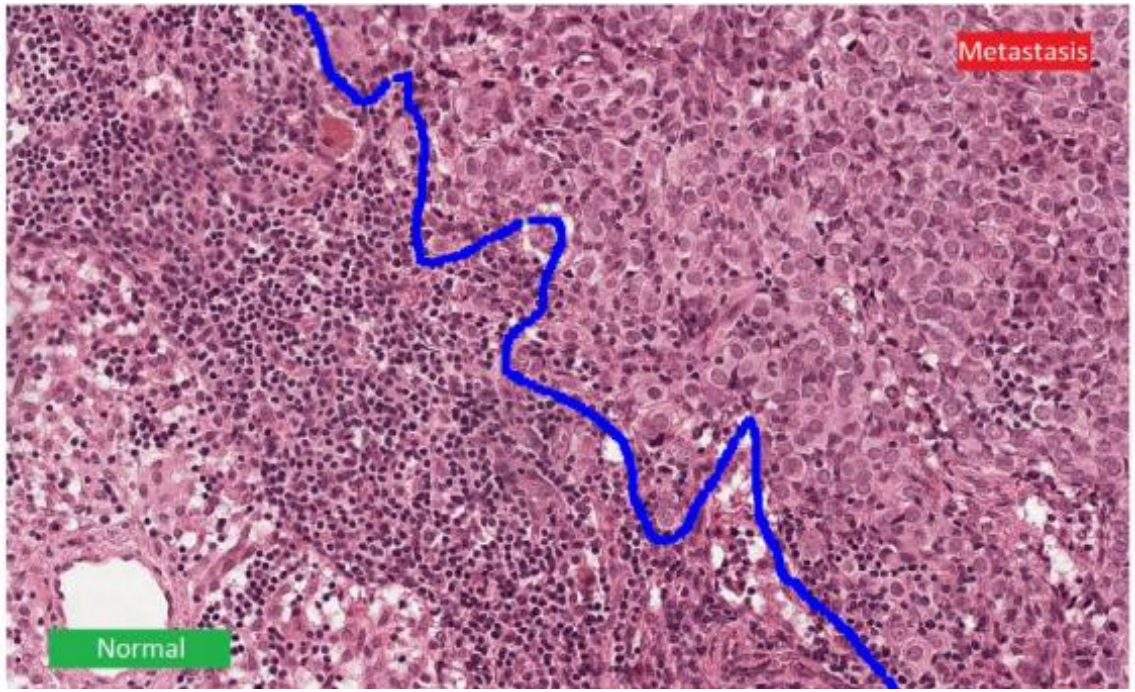


3. Computational pathology



3. Computational pathology

CAMELYON16: annotated whole-slide images



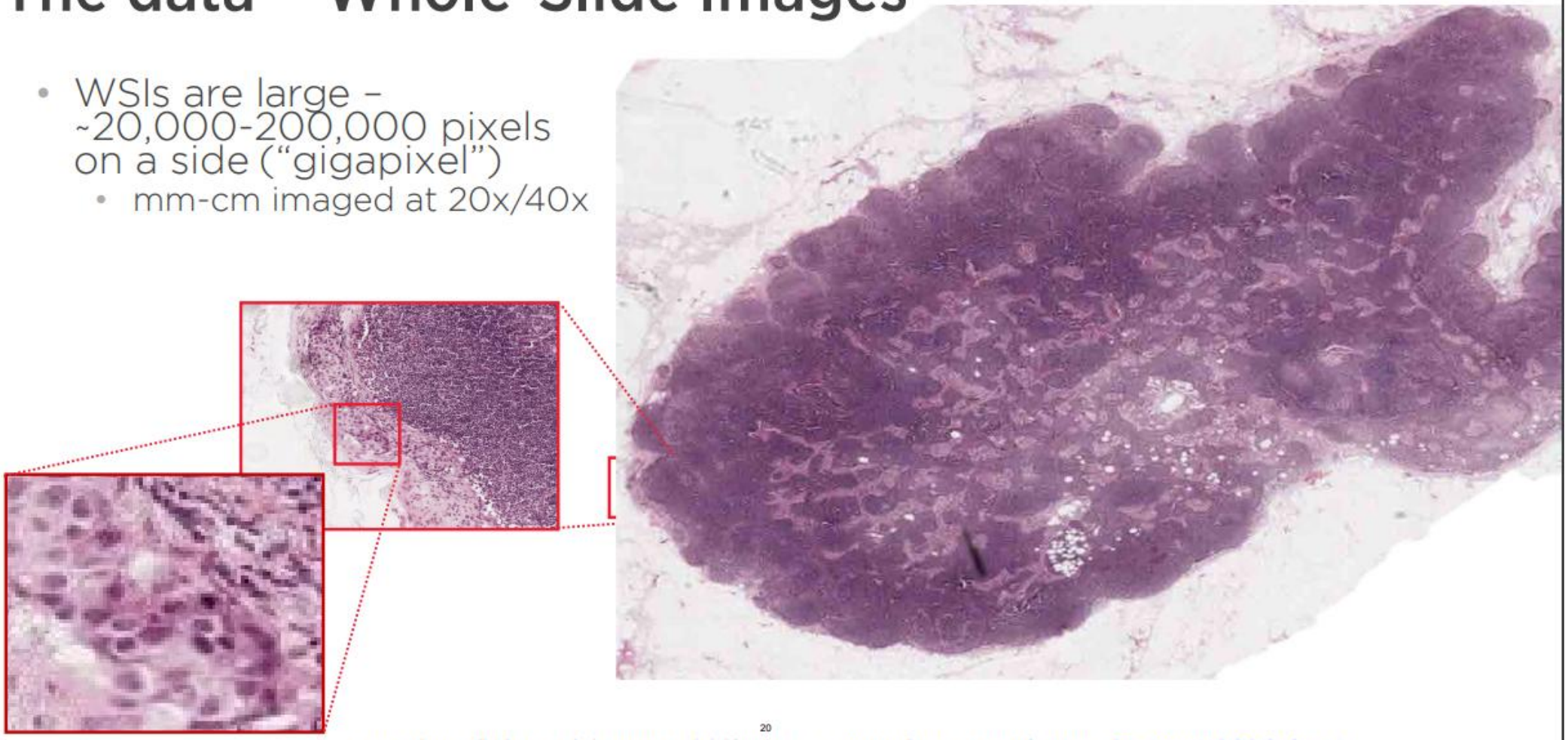
- 400 H&E whole-slide images
- 270 training slides; 130 test slides
- Tumour regions provide supervised patch labels
- Evaluate at both slide and lesion levels

Wang et al., arXiv:1606.05718v1 (2016). Annotation figure retained from the supplied lecture.

3. Computational pathology

The data – Whole-Slide Images

- WSIs are large – ~20,000-200,000 pixels on a side ("gigapixel")
  - mm-cm imaged at 20x/40x



20

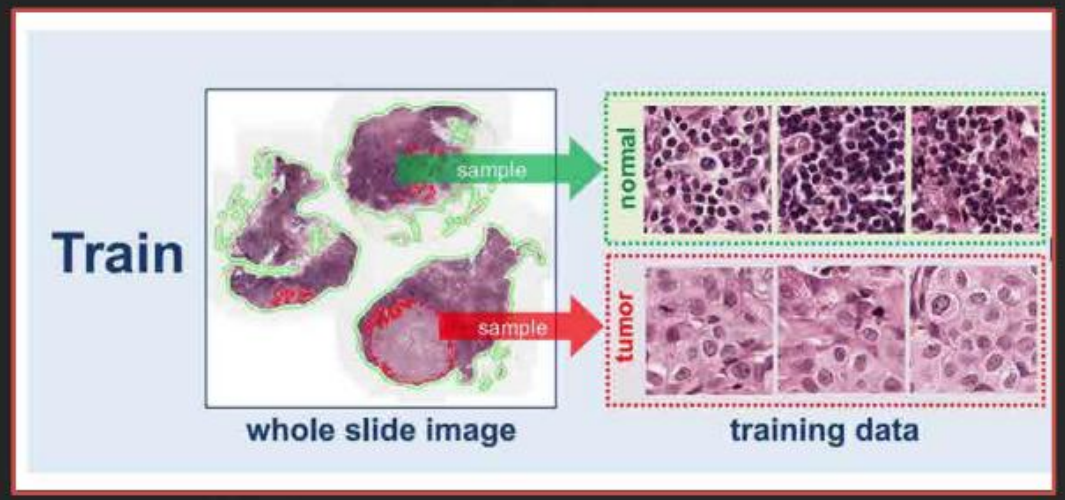
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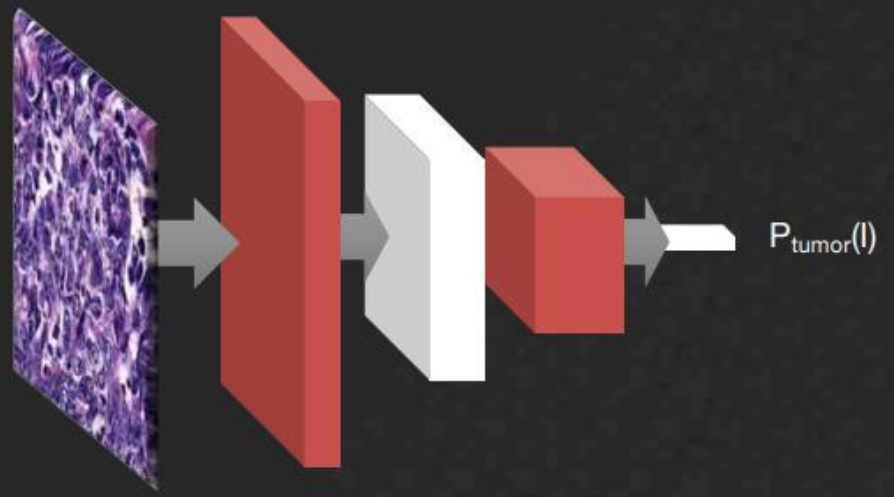
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Approach

Intelligent sampling to enrich the samples for hard cases

- Standard image classification approach needs a twist for WSIs: sampling



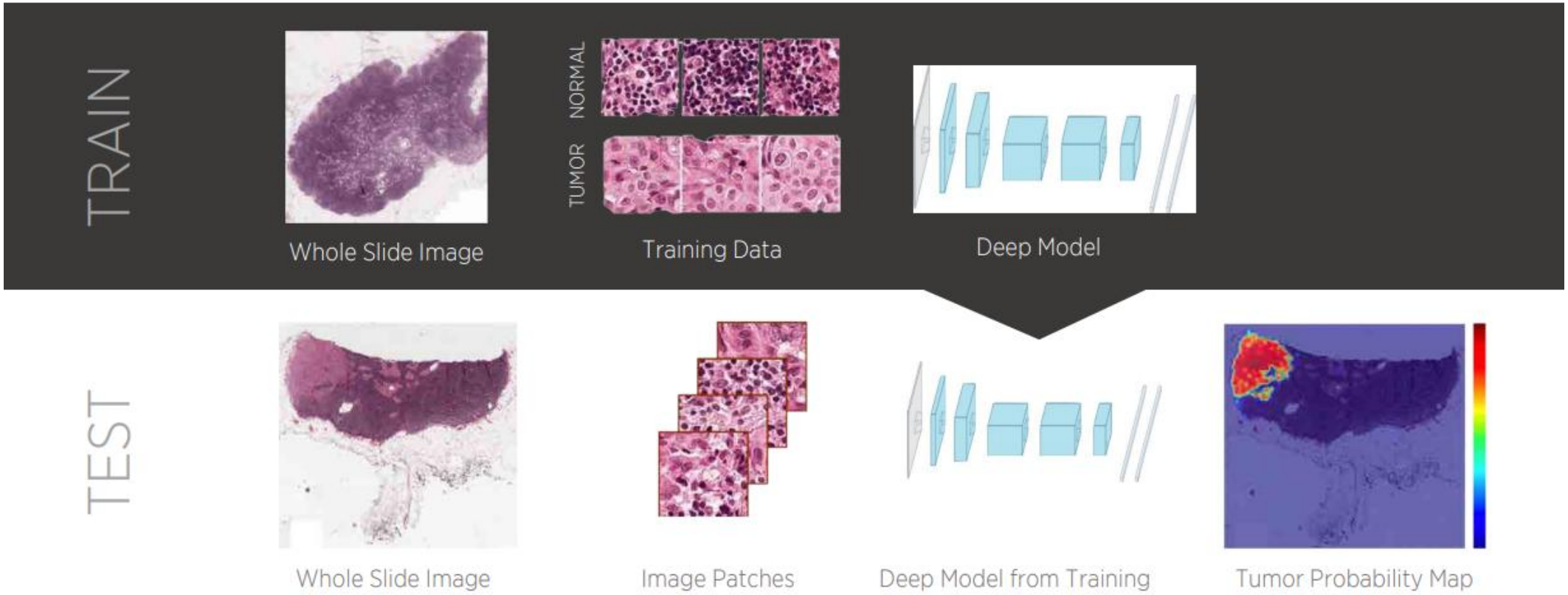


PathAI
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3. Computational pathology

Successfully applied deep learning approach to pathology

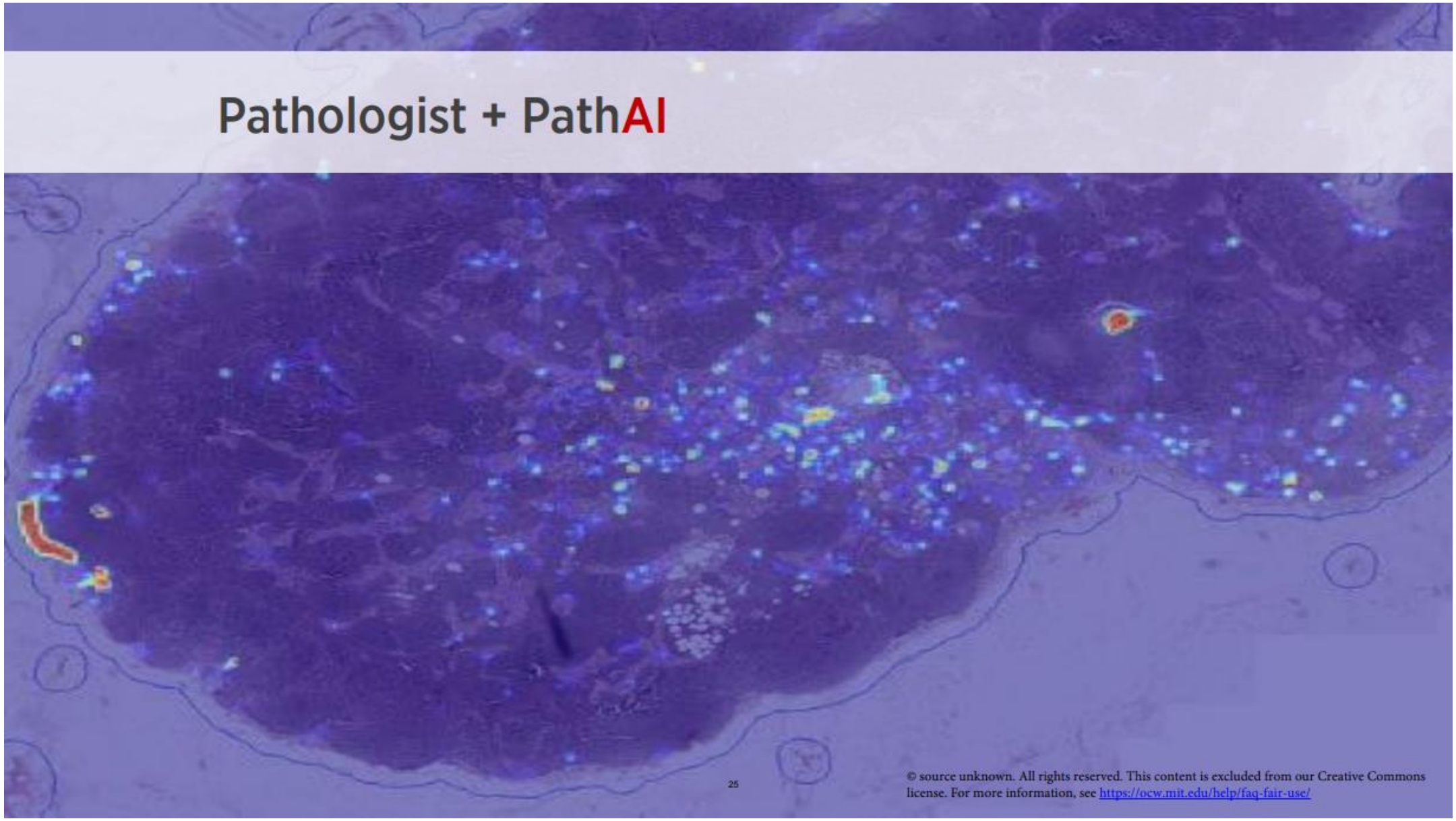
Our team won the Camelyon challenge in 2016, demonstrating outstanding initial performance in pathology



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Wang, D., Khosla, A., ... Beck, A.H., 2016. Deep learning for identifying metastatic breast cancer. arXiv preprint arXiv:1606.05718. JAMA. 2017 Dec 12;318(22):2199-2210.

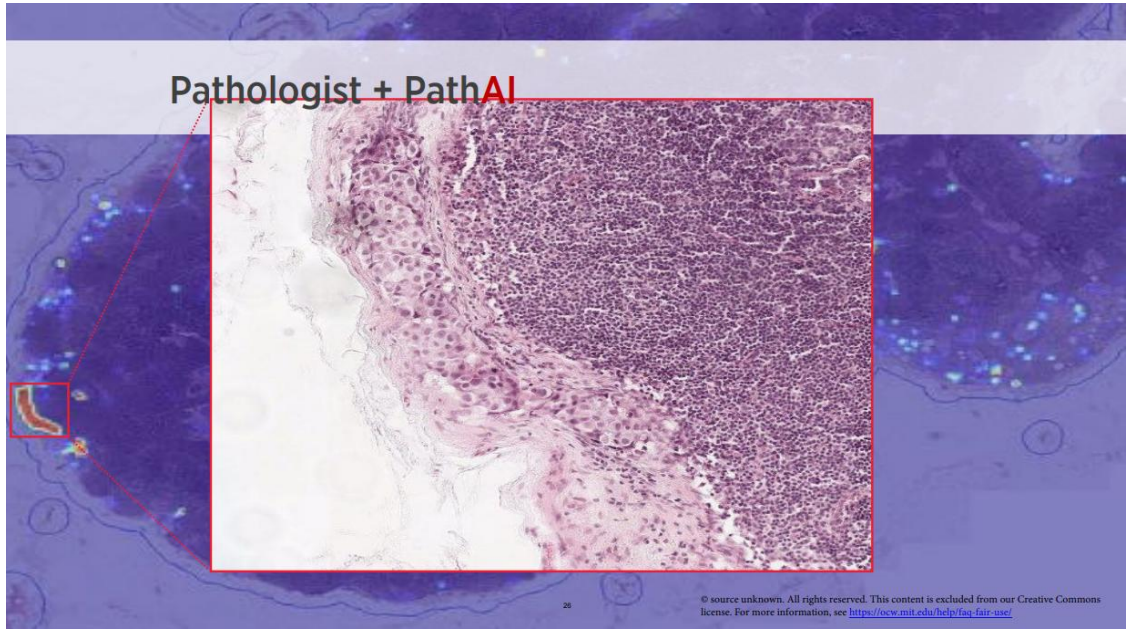
3. Computational pathology



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3. Computational pathology

CAMELYON16: tasks, metrics and reader combination

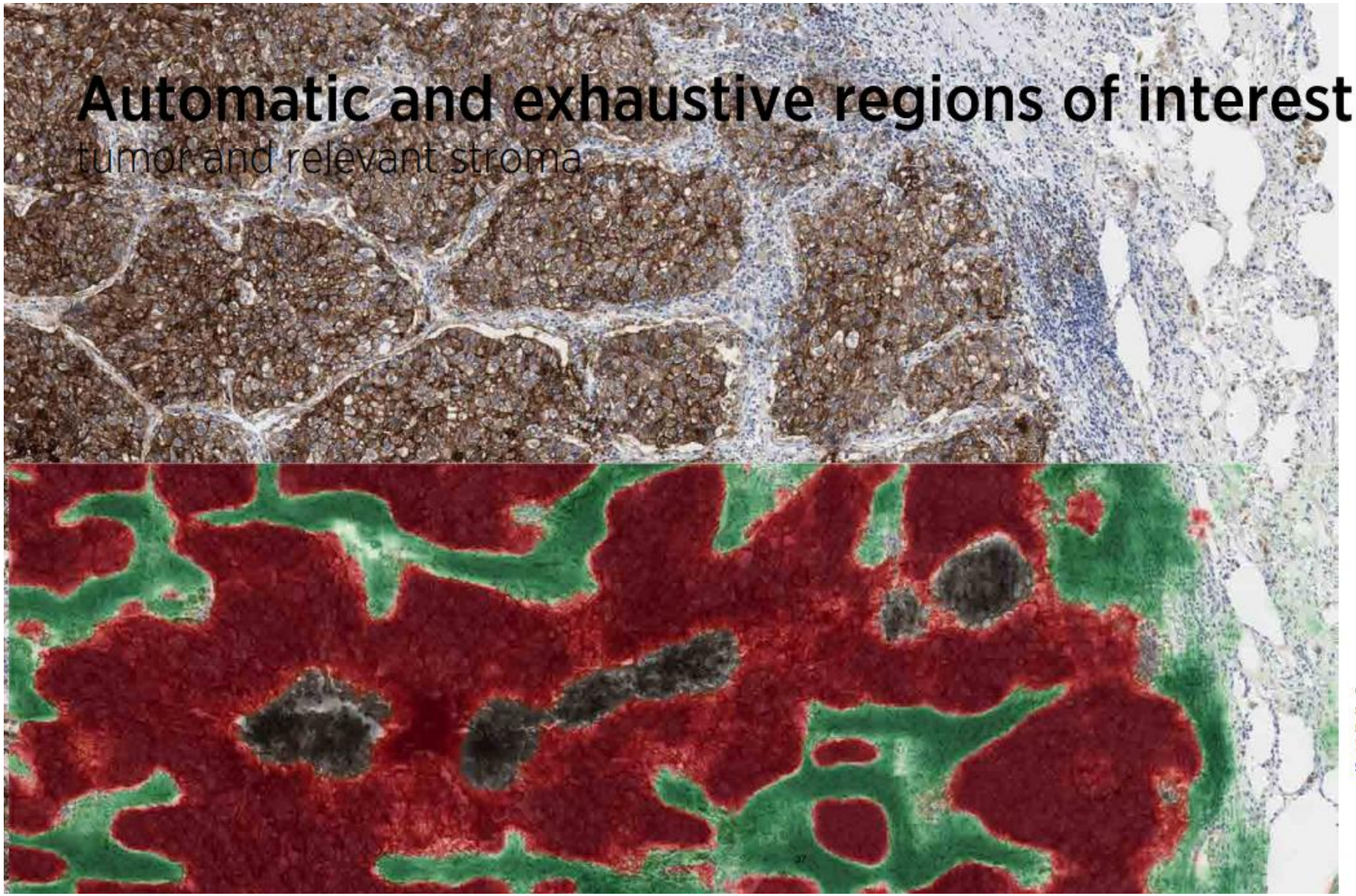


2016 paper, v1	Slide AUROC	Localization
AI system	0.925	0.7051
Study pathologist	0.966	0.733
Combined predictions	0.995	Not reported here

1 – AUROC is not the clinical misdiagnosis rate.

Wang et al., arXiv:1606.05718v1, 2016. Source WSI detail and abstract / evaluation protocol.

3. Computational pathology

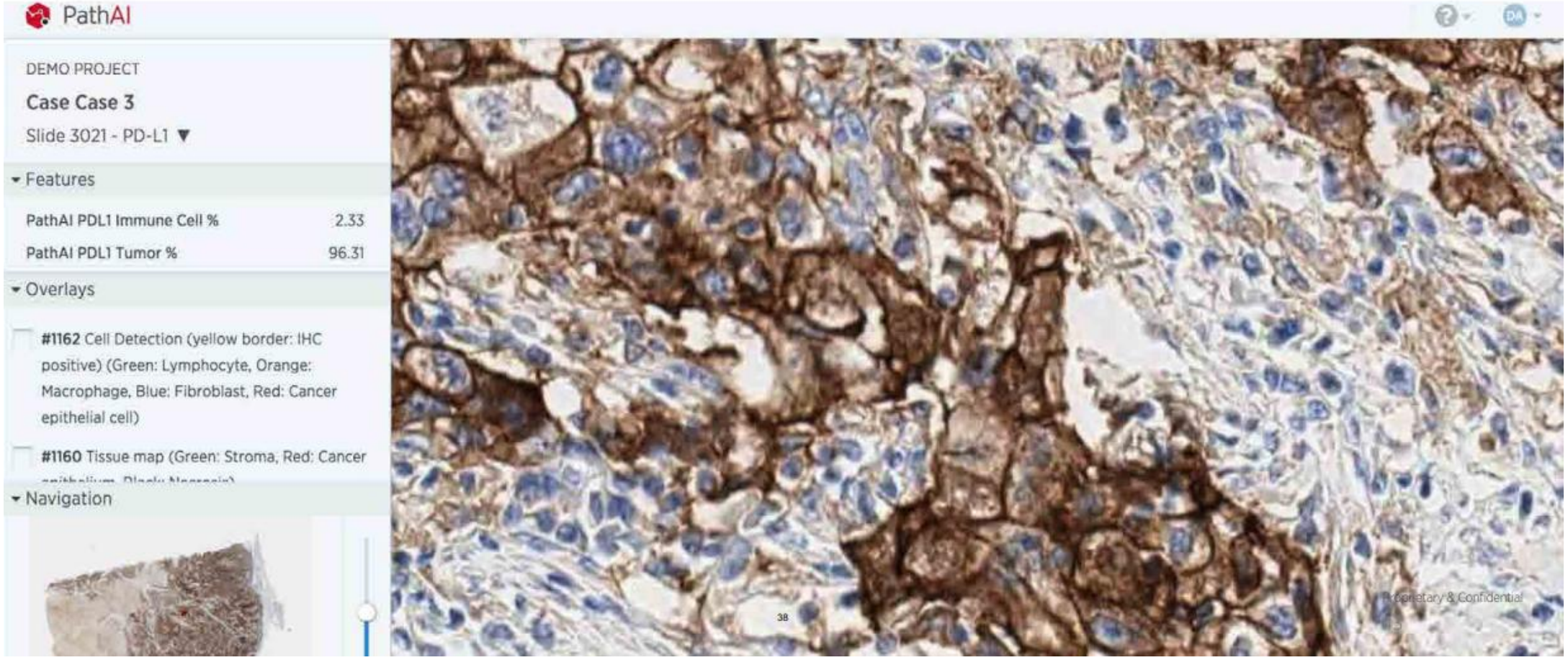


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3. Computational pathology

IHC expression difficult to detect

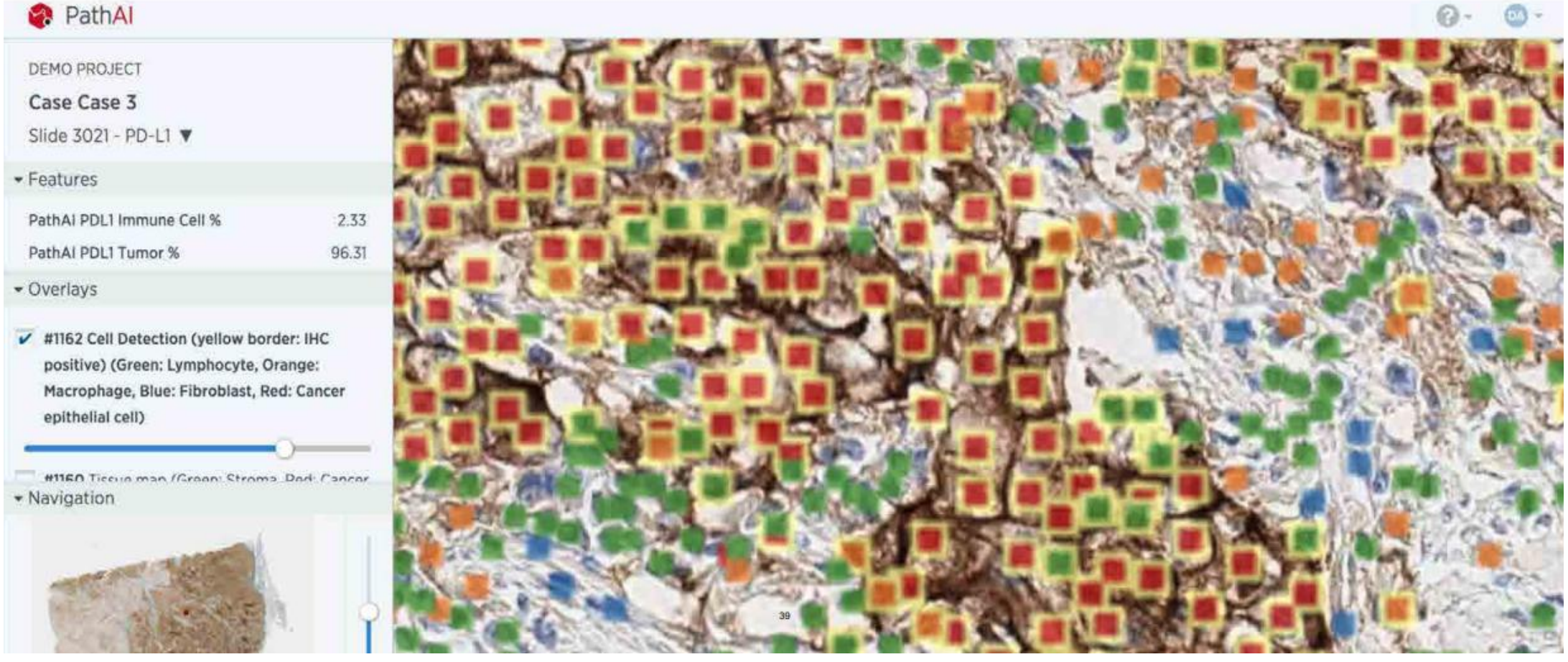
on immune and tumor cells



3. Computational pathology

Exhaustive automated classification

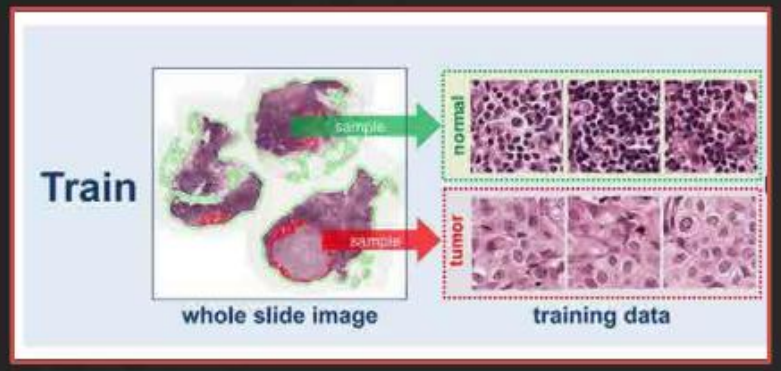
Cell type and cellular IHC positivity classification



3. Computational pathology

Taking it further
From quantitative assay to patient prediction

- PD-L1 scoring alone reduces billions of pixels to 1-2 numbers.
- Can we identify additional relevant information?
  - Using data from randomized controlled clinical trials
- However: Millions of patches, *hundreds* of patients



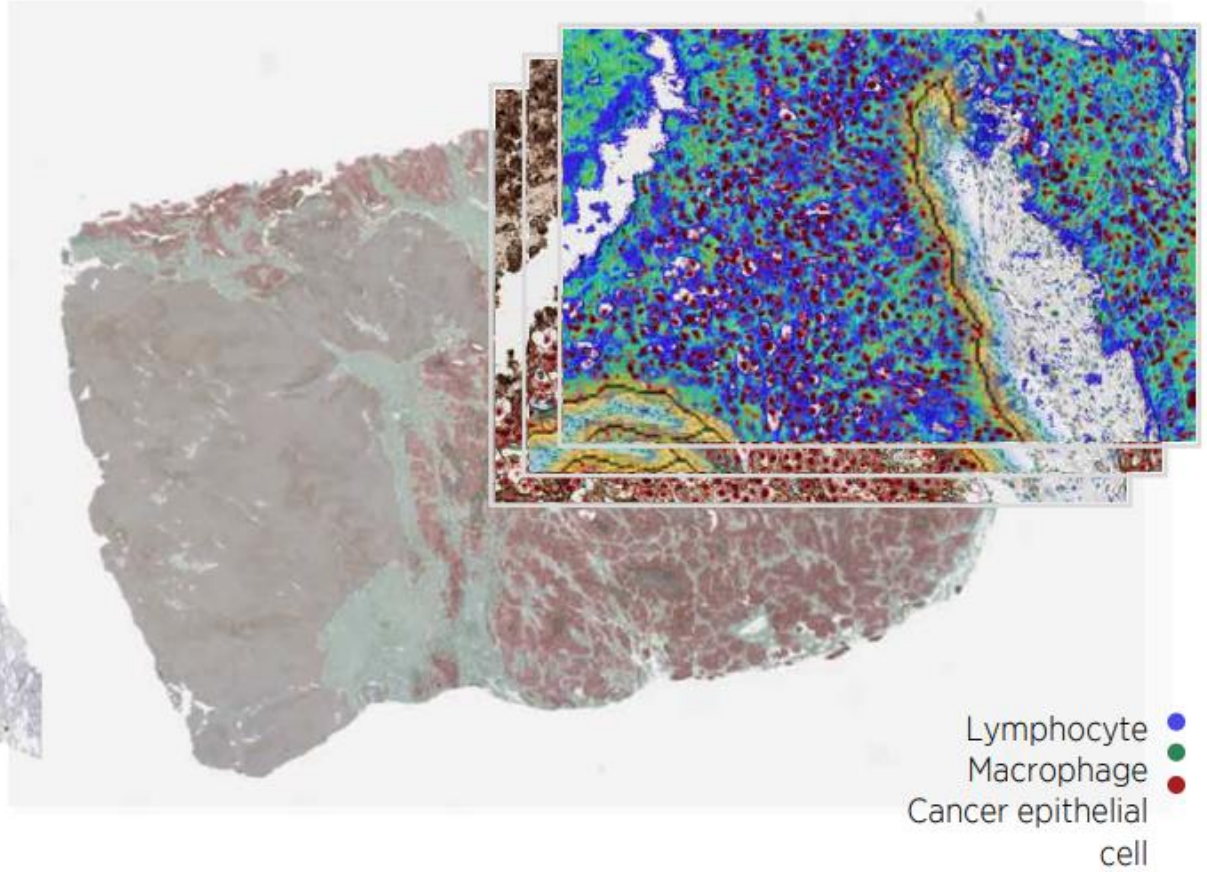


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Cell-type specific, tissue context-aware IHC-quantification





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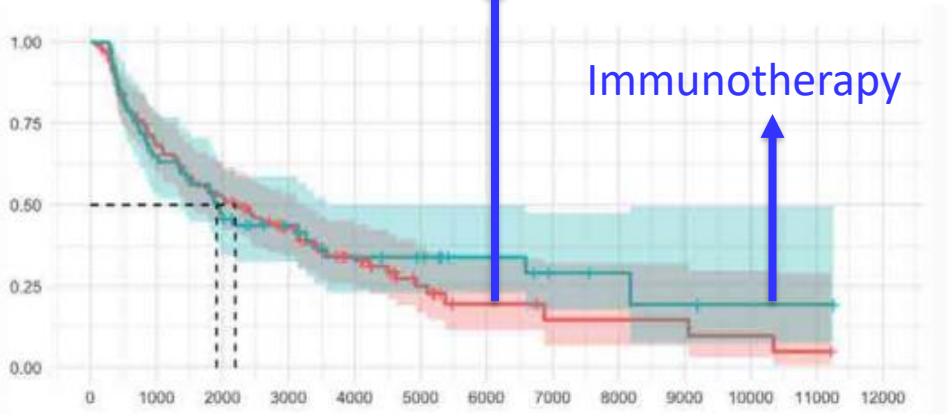
3. Computational pathology

Multivariate models predictive of IO response

- Low *n*, interpretability and measures of uncertainty valuable:
  - No deep learning (gasp!)
- Feature importance/selection in these models can provide disease insight
  - *Now we're doing things pathologists can't rather than automating / improving what they already can*

Traditional therapy

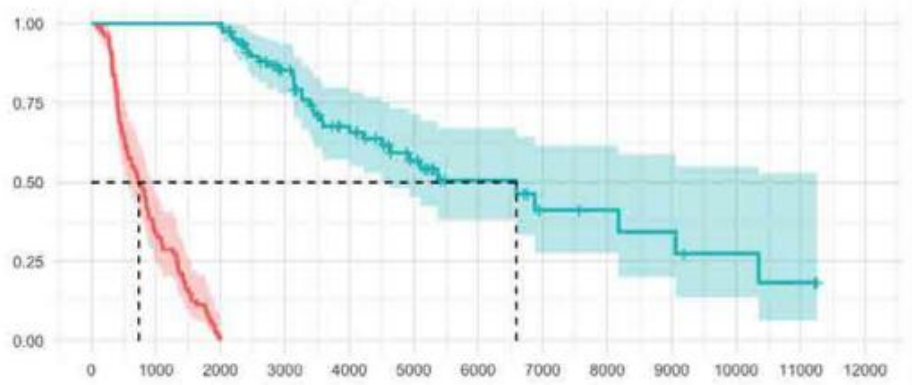
Immunotherapy



Variable	N	Hazard ratio	p
target	161	0.89 (0.60, 1.32)	0.6

0.7 0.8 0.9 1 1.1 1.2 1.3

Build a model to predict which patients actually benefit from the therapy.



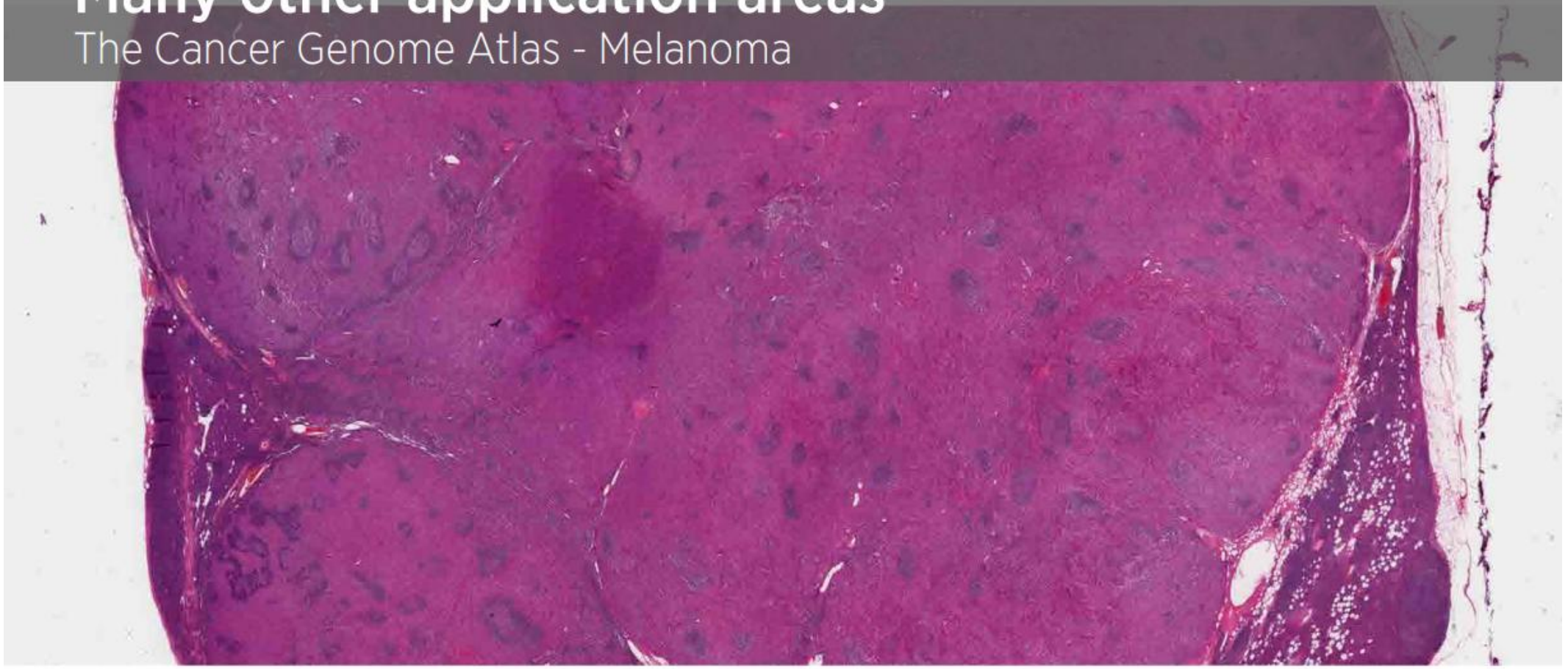
Note: KM curves for illustration only

48

3. Computational pathology

Many other application areas

The Cancer Genome Atlas - Melanoma



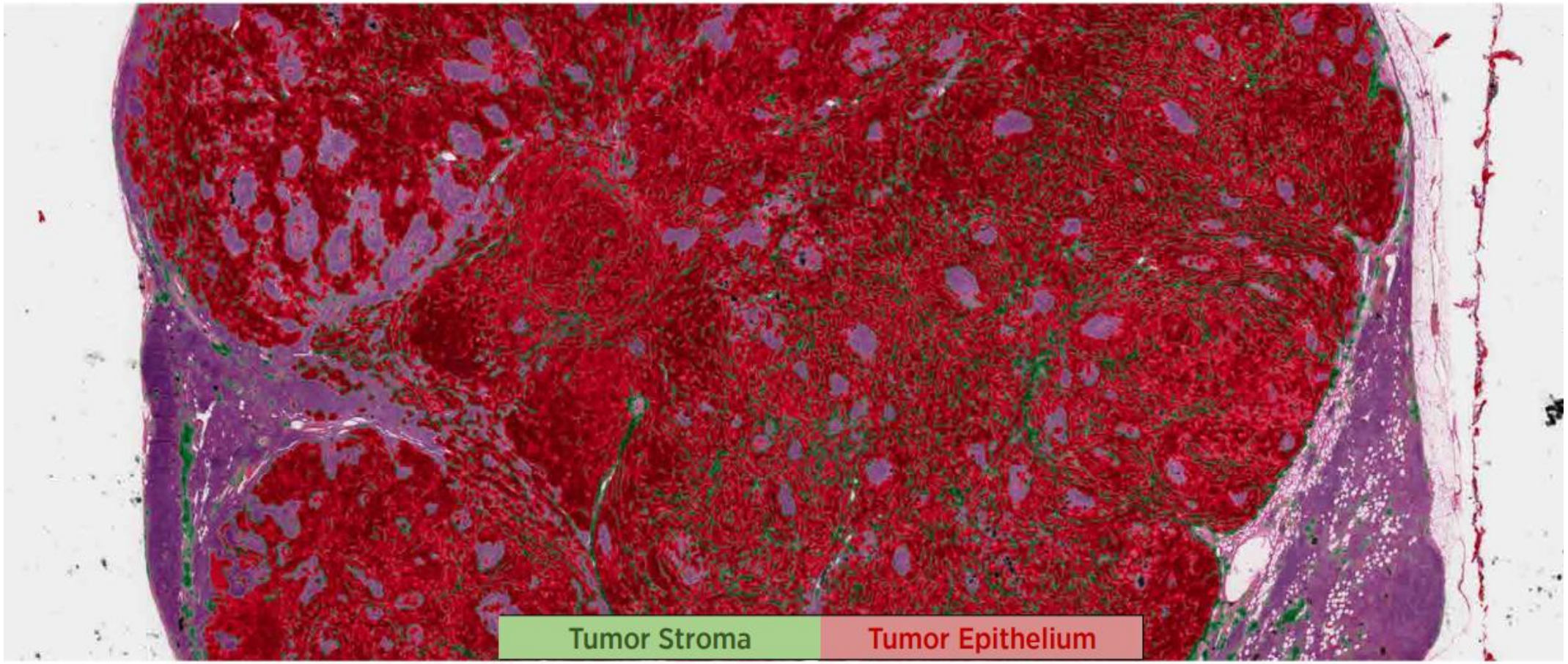
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TCGA-EE-A2GL, Malignant Melanoma

50

3. Computational pathology

Melanoma Tissue Map



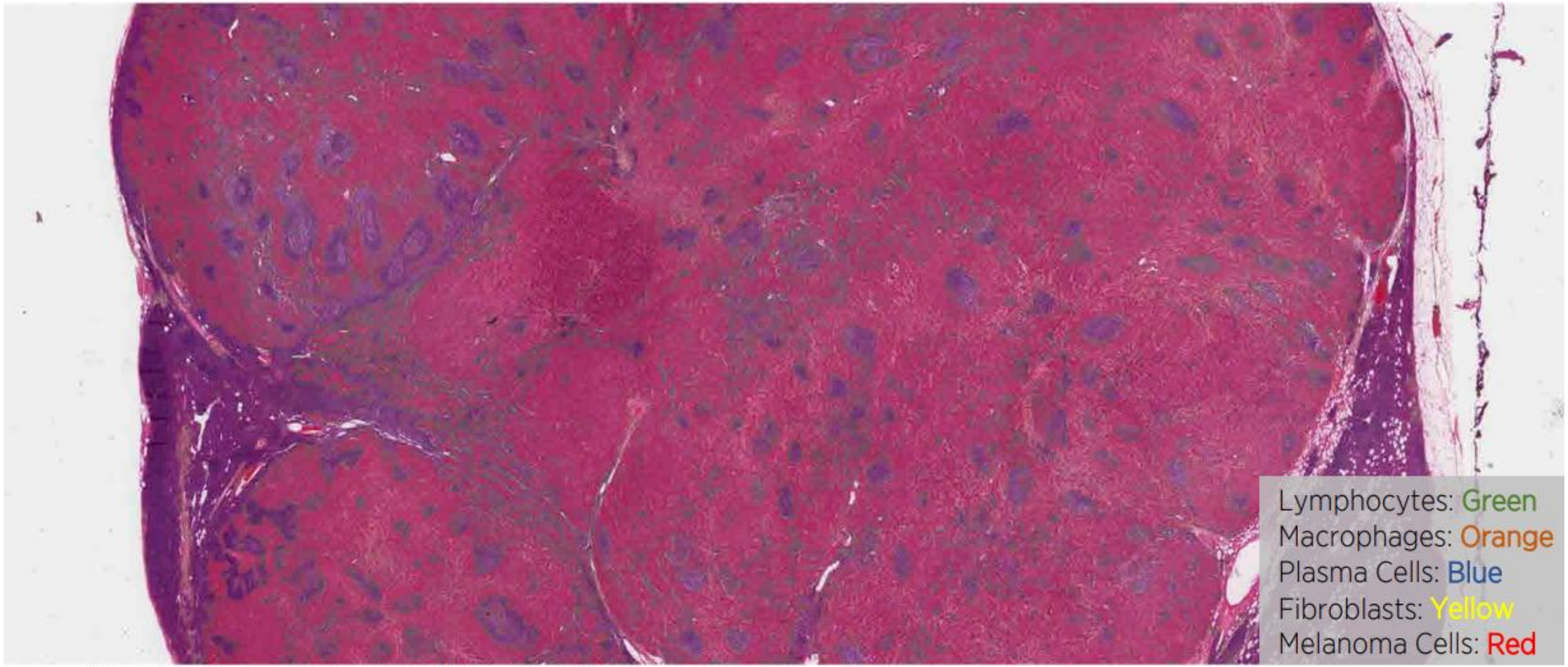
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TCGA-EE-A2GL, Malignant Melanoma

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Melanoma Cell Map



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TCGA-EE-A2GL, Malignant Melanoma

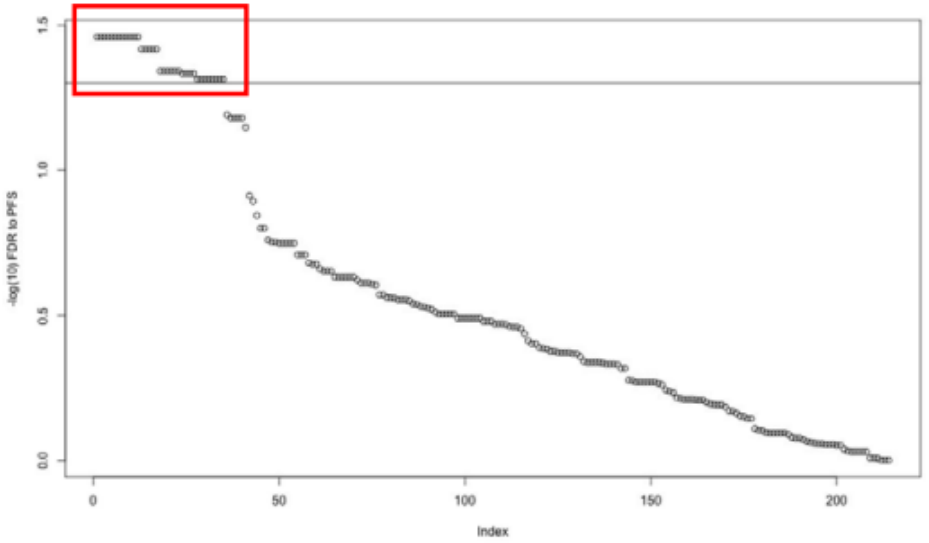
52

3. Computational pathology

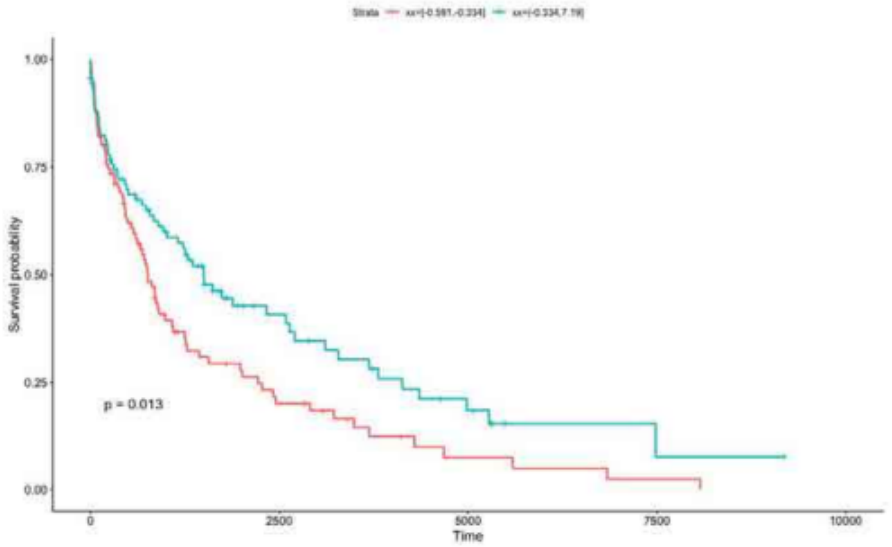
Exhaustive analysis of cellular features in TCGA to enable data-driven identification of pathological predictors of survival in malignant melanoma

Pathological phenotypes with FDR < 5% for association with Progression Free Survival

Rank features by correlation with survival



Increased area of stromal plasma cells associated with improved survival in melanoma

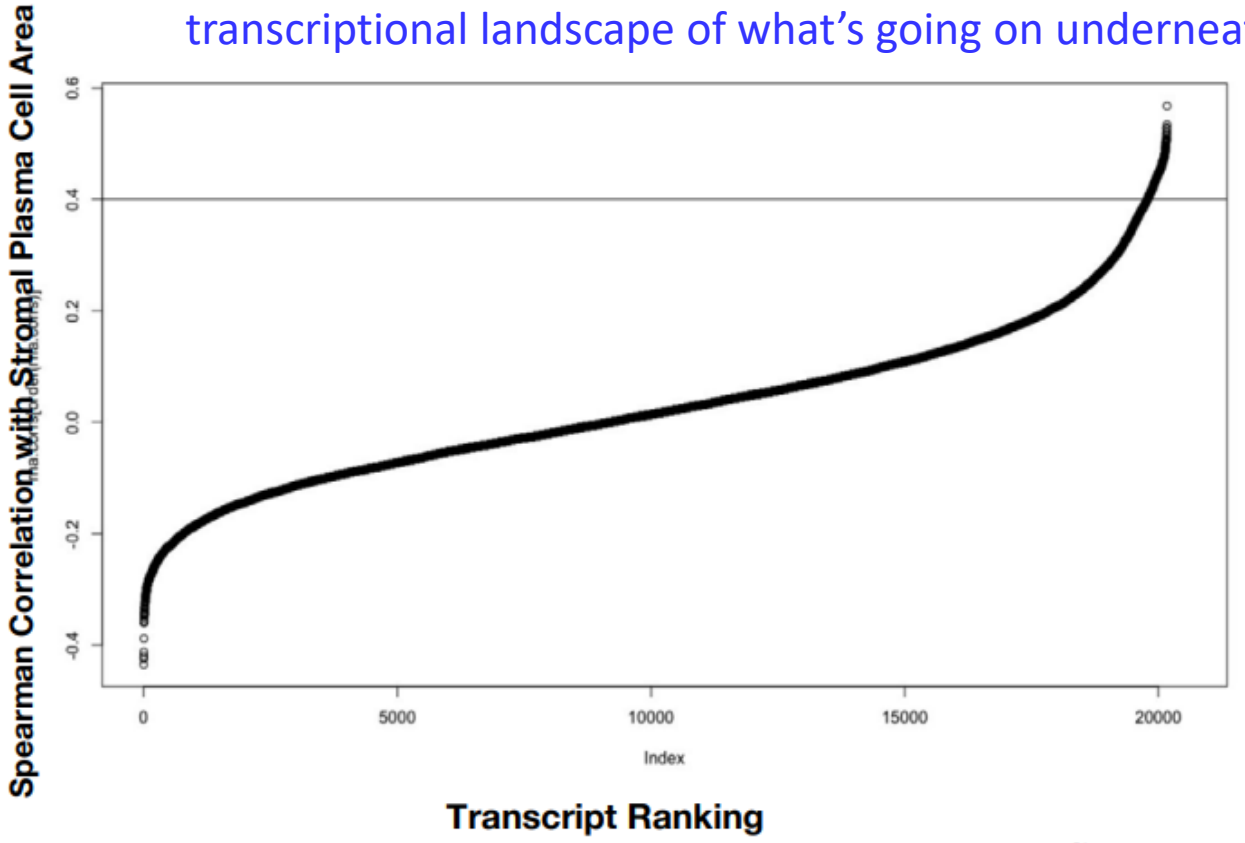


There are some small set of 15 features or so highly associated with survival.

3. Computational pathology

Data-driven identification of transcriptional signature underlying stromal area of plasma cells in melanoma

Pathology is telling you about cells and tissues. RNAs are telling you about the actual transcriptional landscape of what's going on underneath.

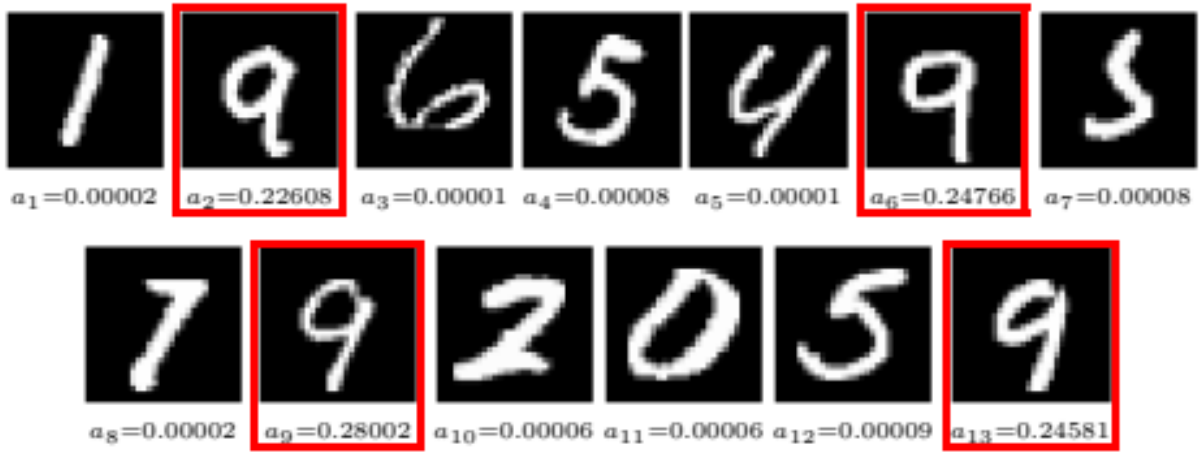


Top-ranking transcripts associated with stromal area of plasma cells

Gene	Correlation
REC8	0.57
GPR174	0.53
CD38	0.53
LAX1	0.53
TOX	0.53
AKAP5	0.53
C8orf80	0.52
JSRP1	0.52
IGJ	0.52
TNFRSF17	0.51
EAF2	0.51

4. Pathology: case studies

Attention MIL (1/5): labels on bags



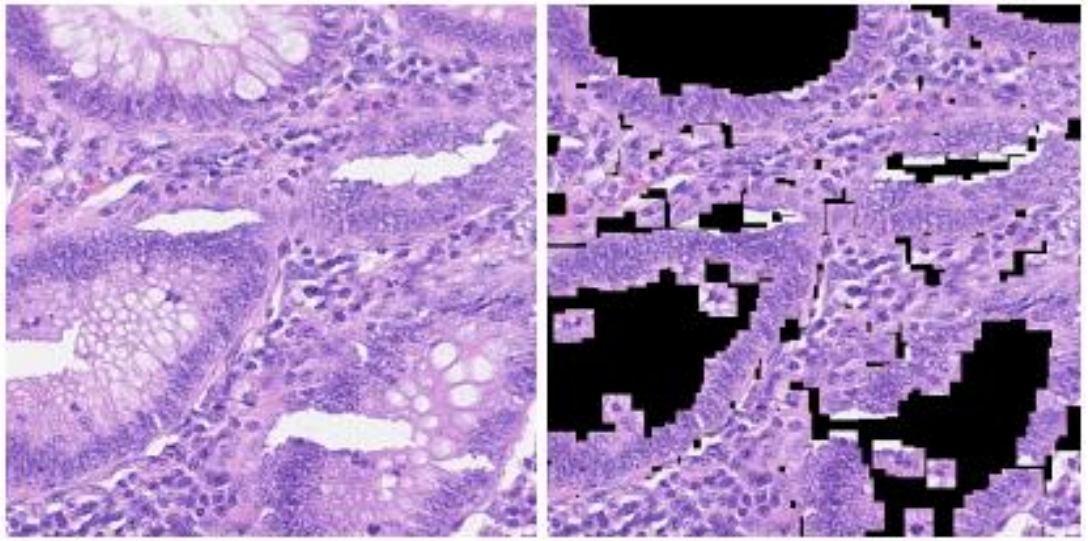
- A bag contains a variable number of instances
- In this experiment, a bag is positive if it contains a 9
- Only the bag label supervises training

The four highlighted digits contribute most of the attention mass.

Ilse et al., ICML 2018. Fig. 4. Positive MNIST bag and learned attention weights.

4. Pathology: case studies

Attention MIL (2/5): learned aggregation



$$h_i = f(x_i)$$

$$s_i = w^T \tanh(Vh_i)$$

$$a_i = \exp(s_i) / \Sigma_j \exp(s_j)$$

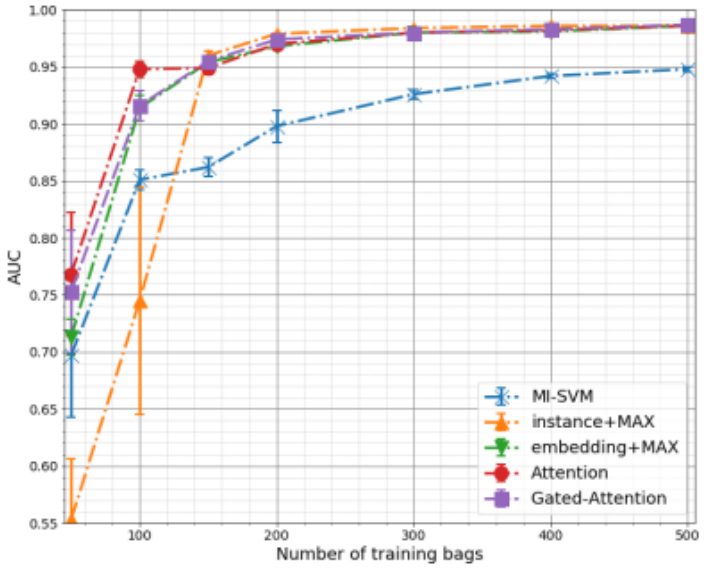
$$z = \Sigma_i a_i h_i \qquad \hat{y} = g(z)$$

A simultaneous permutation of patches and weights leaves z unchanged.

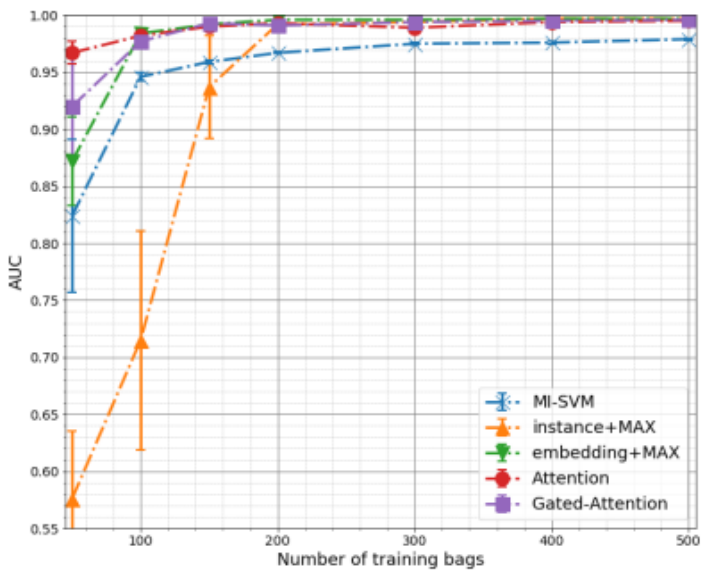
Ilse et al., ICML 2018. Fig. 5a,b and attention pooling equations.

4. Pathology: case studies

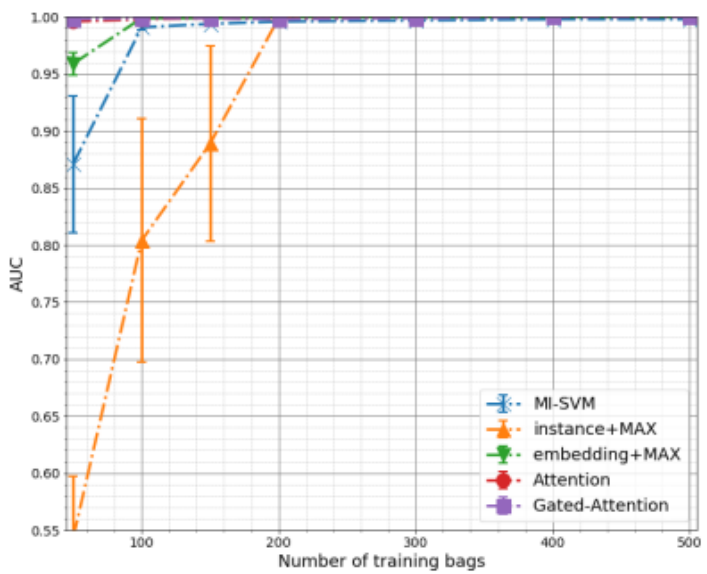
Attention MIL (3/5): sample efficiency



Mean bag size: 10



Mean bag size: 50



Mean bag size: 100

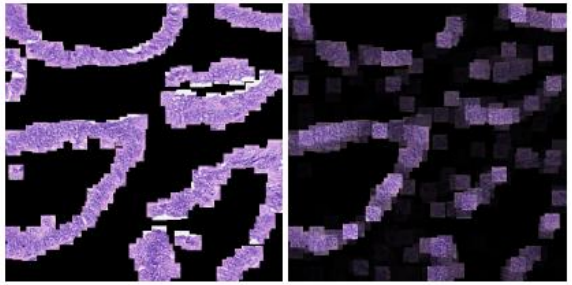
Same task, different training-bag counts. The largest gains occur in limited-data settings.

Ilse et al., ICML 2018. Figs. 1–3. MNIST-BAGS, 1,000 test bags.

4. Pathology: case studies

Attention MIL (4/5): pathology experiments

Pooling / level	Breast AUC	Colon AUC
Instance + max	0.612 ± 0.026	0.914 ± 0.010
Instance + mean	0.719 ± 0.019	0.866 ± 0.008
Embedding + max	0.650 ± 0.013	0.918 ± 0.010
Embedding + mean	0.796 ± 0.012	0.940 ± 0.010
Attention	0.775 ± 0.016	0.968 ± 0.009
Gated attention	0.799 ± 0.020	0.968 ± 0.010



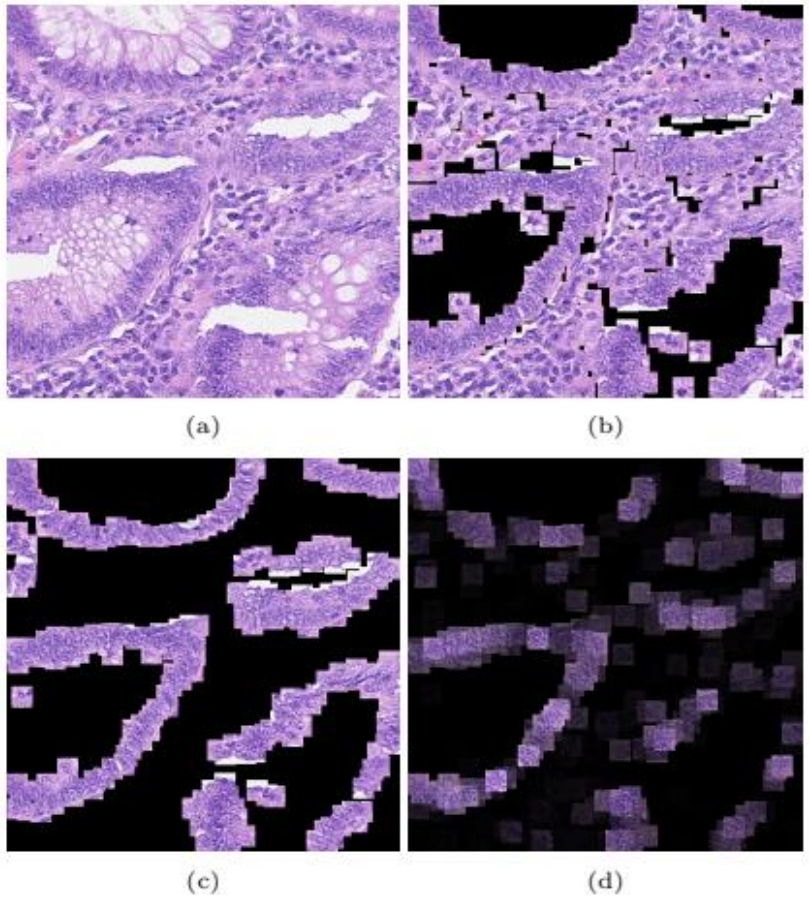
58 breast images
100 colon images
5 runs, mean ± SEM

Attention does not beat embedding + mean on every metric or dataset.

Ilse et al., ICML 2018. Tables 2–3, AUROC (mean ± SEM). Fig. 5c,d: colon example.

4. Pathology: case studies

Attention MIL (5/5): interpreting a heatmap



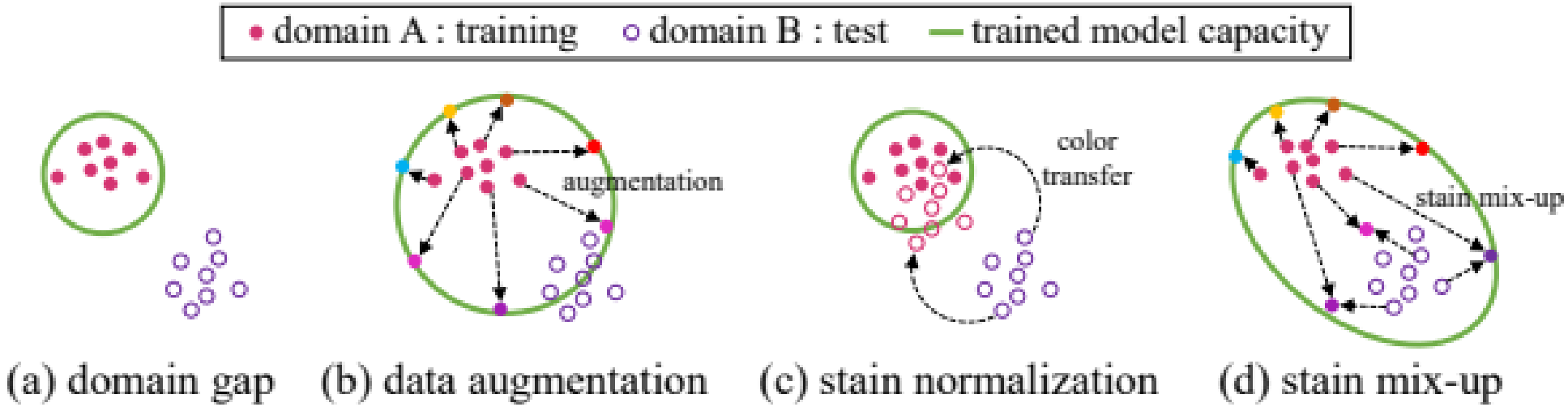
- Top left: original tissue
- Top right: extracted patches
- Bottom left: epithelial ground truth
- Bottom right: rescaled attention

What experiment would test whether the model relies on the intended tissue?

Ilse et al., ICML 2018. Fig. 5. Attention visualization and its interpretation.

4. Pathology: case studies

Stain Mix-up (1/5): the domain gap



The method uses labelled source images and unlabelled target-domain stain information.

Chang et al., MICCAI 2021. Fig. 1. Source and target information available during training.

4. Pathology: case studies

Stain Mix-Up: Unsupervised Domain Generalization for Histopathology Images

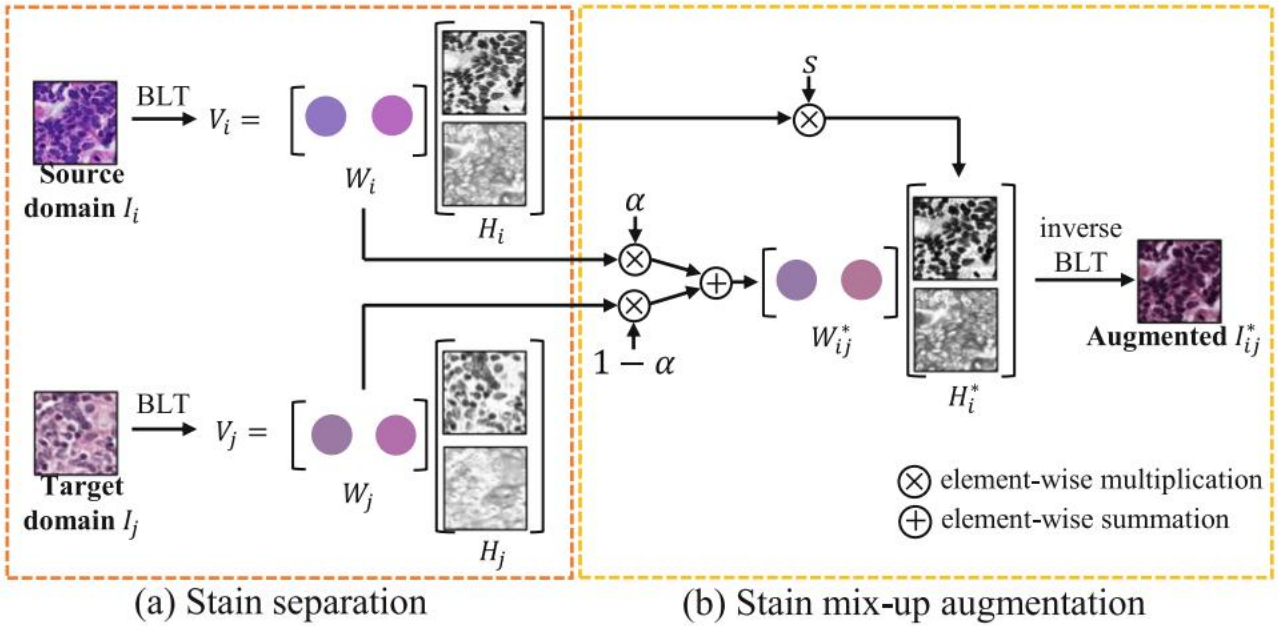
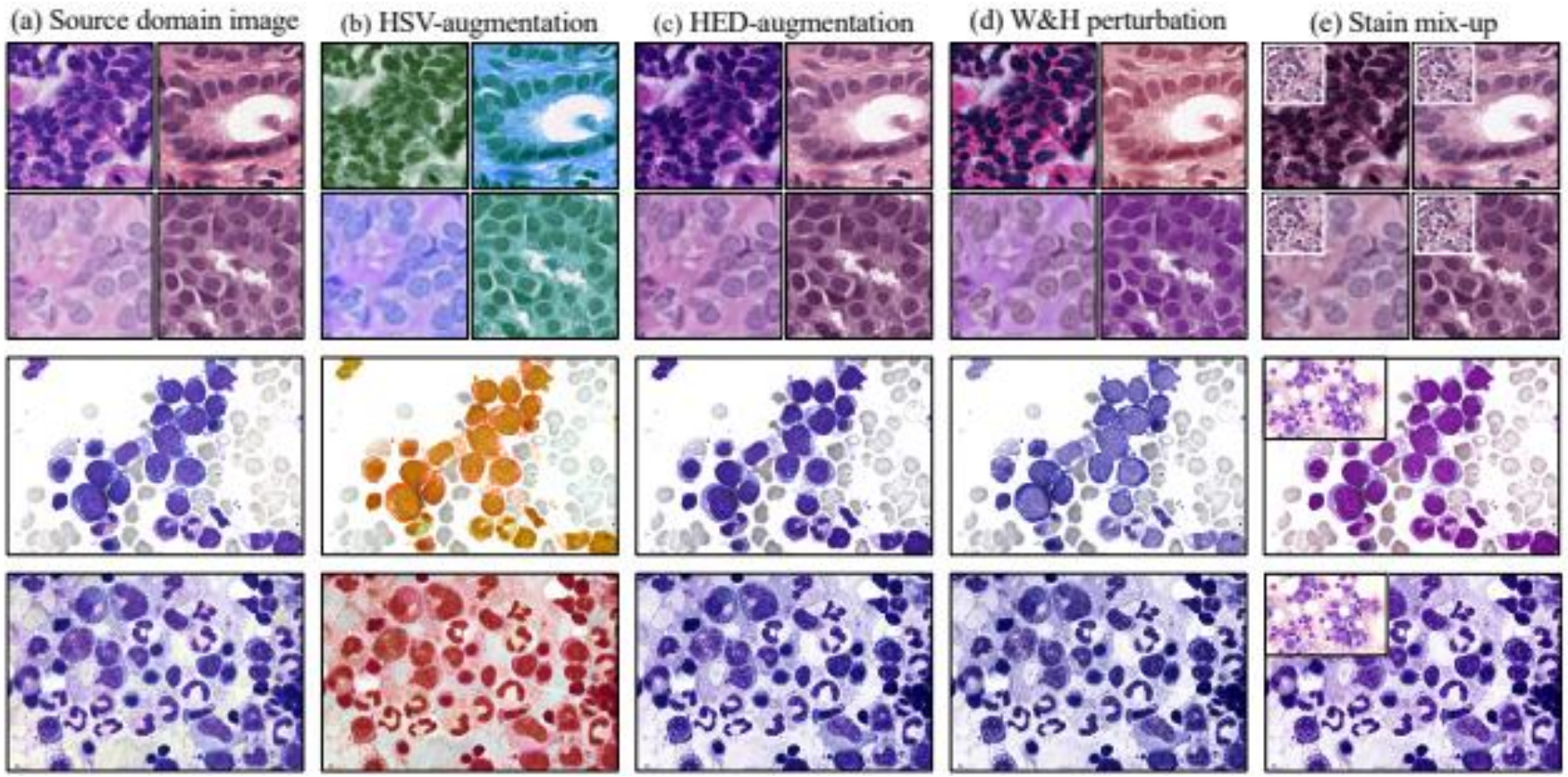


Fig. 2. Pipeline of the proposed stain mix-up augmentation. (a) Given a labeled image I_i from the source domain and an unlabeled image I_j from the target domain, stain separation decomposes the optical density of I_i derived via Beer-Lambert transformation (BLT), *i.e.* V_i, into its stain color matrix W_i and density map H_i. Similarly, we have W_j and H_j from V_j. (b) Stain mix-up augmentation is carried out by applying inverse BLT to a mixed stain color matrix W_{ij}^* and a perturbed density map H_i^*.

Chang et al. Stain Mix-Up: Unsupervised Domain Generalization for Histopathology Images, MICCAI, 2021

4. Pathology: case studies

Stain Mix-up (3/5): augmentation examples



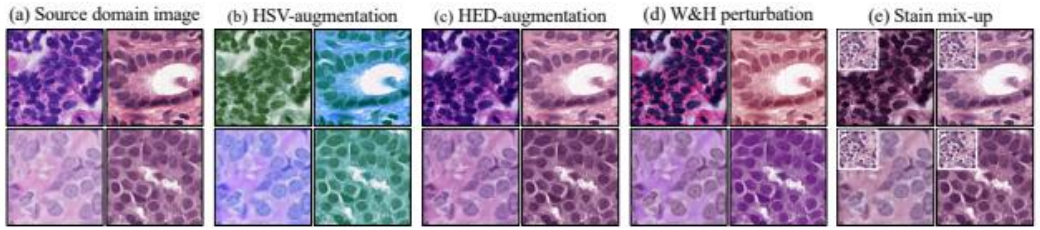
Compare colour changes while checking that cell and tissue morphology remains plausible.

Chang et al., MICCAI 2021. Fig. 3. All augmentation baselines and both datasets.

4. Pathology: case studies

Stain Mix-up (4/5): cross-centre classification

Method	C1	C2	C3	C4	C5	Mean
None	.947	.753	.854	.777	.861	.838
Vahadane	.943	.826	.898	.853	.707	.846
HSV	.984	.844	.954	.937	.845	.913
HED	.984	.925	.903	.861	.908	.916
Stain Mix-up	.979	.948	.946	.965	.942	.956

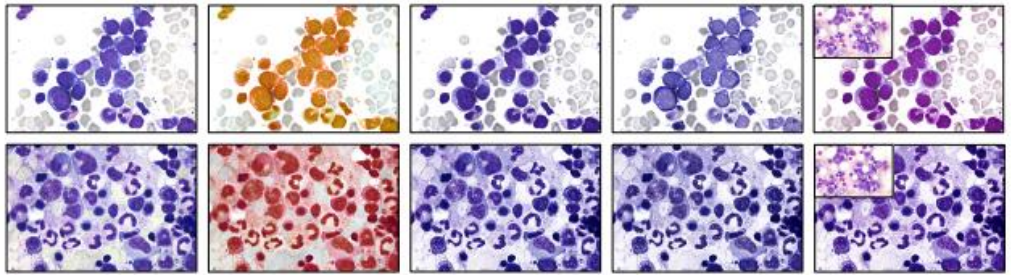


CAMELYON17, ResNet-50
Mean AUC: 0.956 (95% CI 0.953–0.959)
C1 and C3 illustrate why averages need detail.

Chang et al., MICCAI 2021. Table 1, AUROC. Fig. 3 H&E examples.

4. Pathology: case studies

Stain Mix-up (5/5): segmentation and ablation



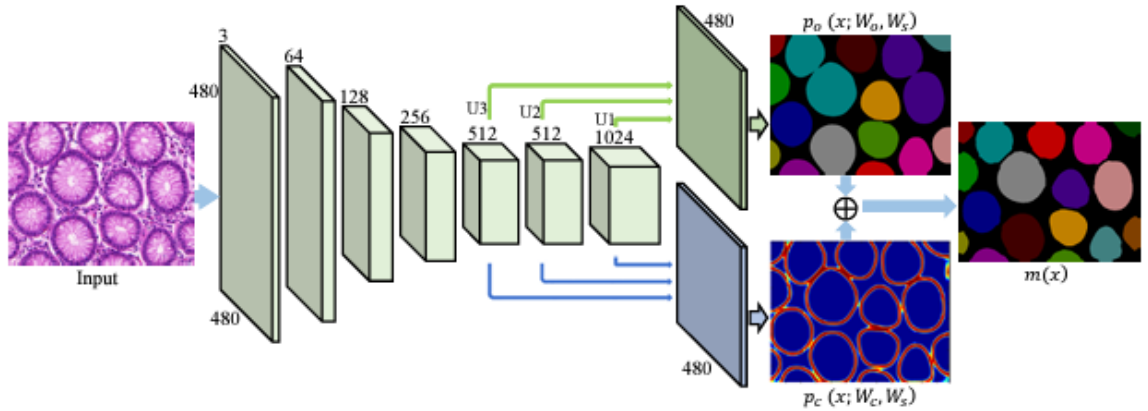
Hema, Mask R-CNN	Box AP	Mask AP
None	48.3	49.6
HSV	48.7	50.2
HED	48.1	49.5
Stain Mix-up	50.3	52.1

CAMELYON17 ablation	Mean AUC
H perturbation	0.916
W and H perturbation	0.948
W mix-up and H perturbation	0.956

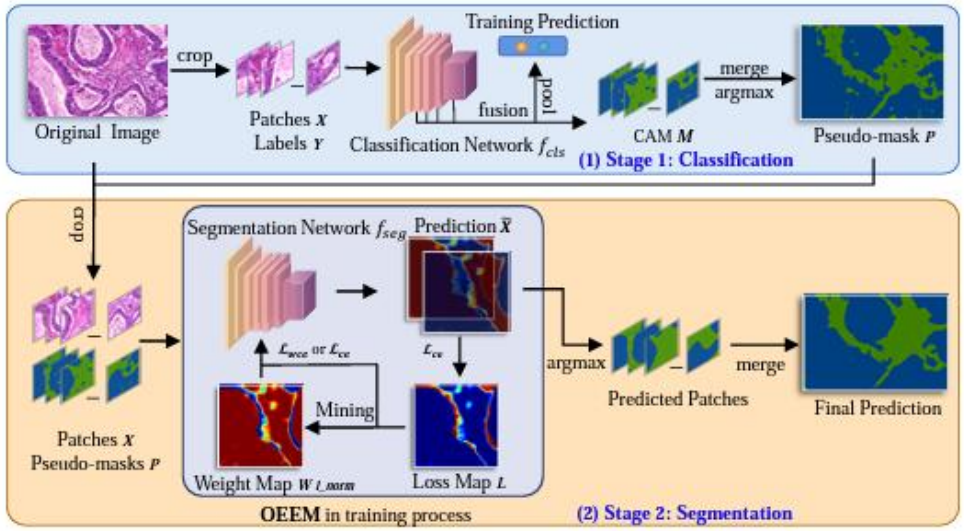
Hema: train at M1, test at M2.
Is the benefit colour mixing or simply more augmentation?

Chang et al., MICCAI 2021. Table 2, Table 1 ablation and Fig. 3 Hema examples.

I. Background



Fully-supervised Gland Segmentation



Weakly-supervised Gland Segmentation

- Fully-supervised Gland Segmentation utilizes pixel-level annotation for each gland image during the training of the segmentation network.

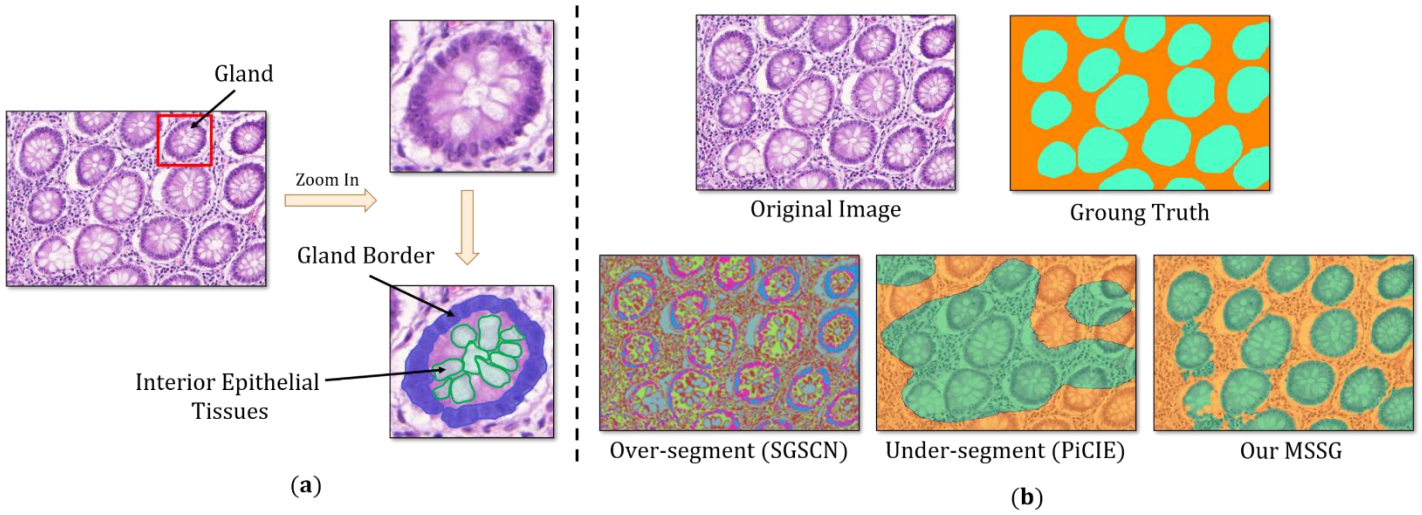
- Weakly-supervised Gland Segmentation utilizes weak annotation (e.g., patch tag and bound-box) for each gland image during the training of the segmentation network.

- **The annotated dataset is extremely expensive and requires specialized knowledge.**

- **A totally label-free method is appreciated**

II. Motivation

Current Unsupervised Semantic Segmentation methods for general Computer Vision[1] and medical image analysis[2] end up in Under-segment and Over-segment in gland images



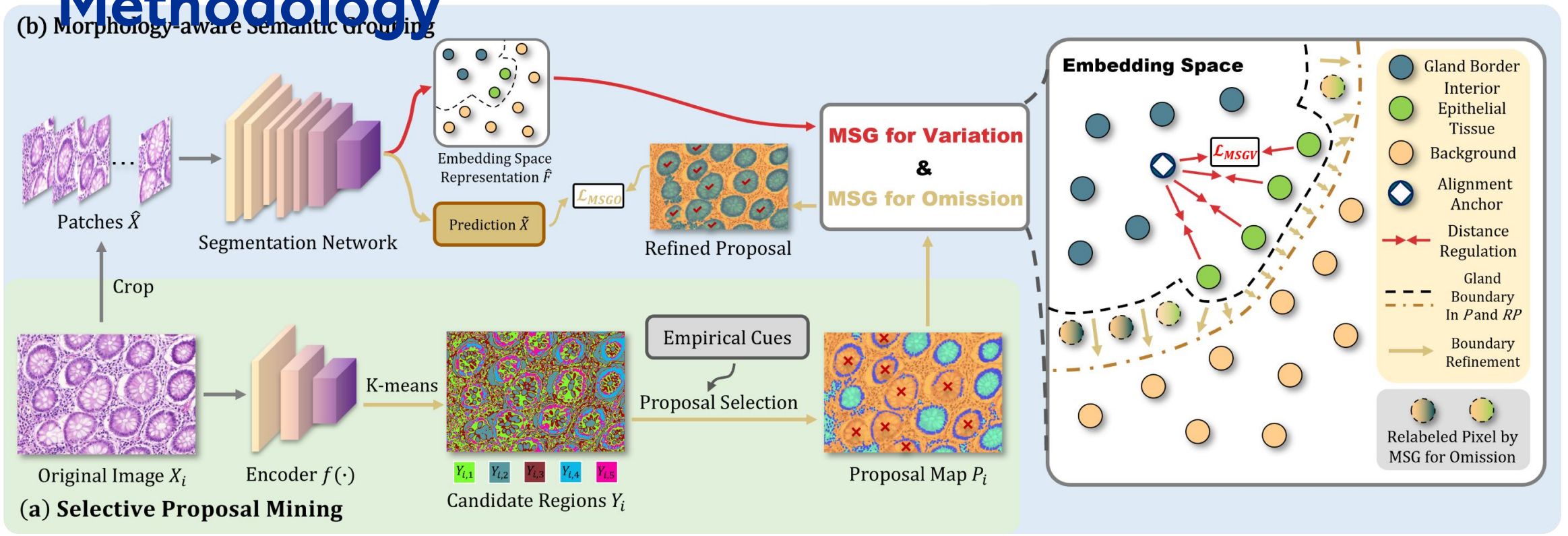
- Gland images present a unique challenge due to their highly dense and complex tissues with intricate color distribution, which leads to over-segmentation

- Glands are composed of different parts (i.e., gland border and interior epithelial tissues with significant variations in appearance, which leads to under-segmentation

[1] Jang Hyun Cho, et al. "PiCIE: Unsupervised Semantic Segmentation using Invariance and Equivariance in Clustering." CVPR, 2021

[2] Euijoon Ahn, et al. "A Spatial Guided Self-supervised Clustering Network for Medical Image Segmentation." MICCAI, 2021

III. Methodology

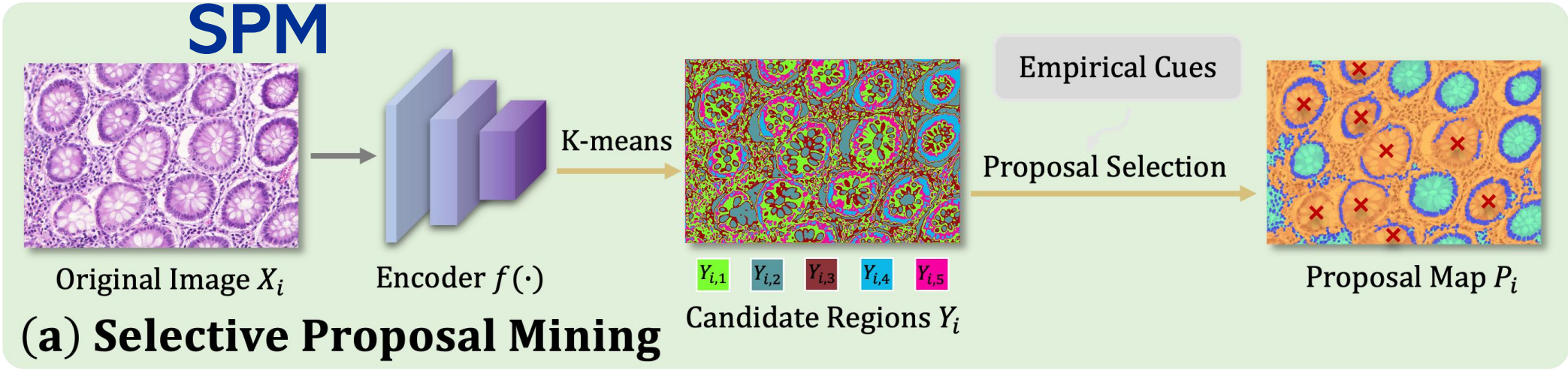


MSSG consists of two modules:

Selective Proposal Mining module: generate a proposal map highlighting the gland sub-regions

Morphology-aware Semantic Grouping module: summarize the overall information from sub-regions

III. Methodology:



$$\mathcal{L}_{SS}(F_i[:,h,w], C_i[:,h,w]) = -\sum_{d}^{D} C_i[d,h,w] \cdot \ln F_i[d,h,w]$$

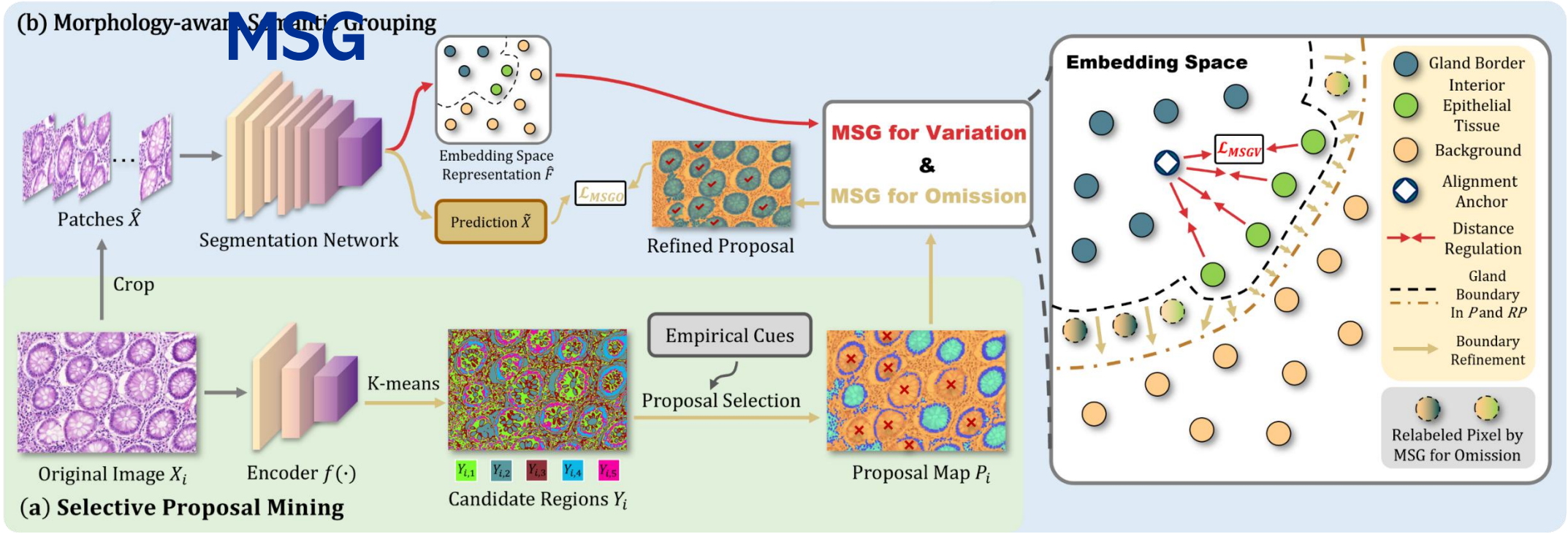
$$\mathcal{L}_{SC}(F_i) = \sum_{s,h,w}^{S,H-s,W-s} \left(F_i[:,h+s,w] - F_i[:,h,w]\right)^2 + \left(F_i[:,h,w+s] - F_i[:,h,w]\right)^2 .$$

Selective Proposal Mining for sub-region proposal mining:

- Train a shallow encoder to cluster pixels into 5 candidate regions
- Utilize an empirical cue to select gland sub-region proposals from the candidates
- High-level Idea: embed the morphology knowledge inside the empirical cue into the proposals

Empirical Cue: Each gland is comprised of a border region with high gray levels that surrounds the interior epithelial tissues

III. Methodology:



Morphology-aware Semantic Grouping (MSG):

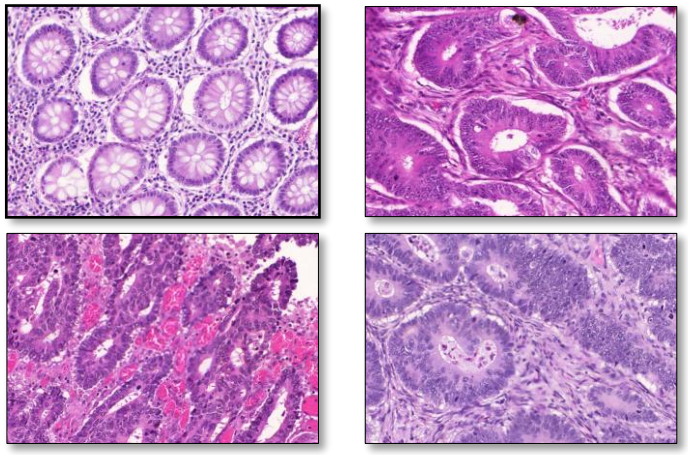
- **MSG for Variation** mitigates the adverse impact of appearance variation between the gland sub-regions

$$\mathcal{L}_{MSGV} = \frac{1}{I}\sum_{i\in I}\left(i - \frac{1}{G}\sum_{g\in G} g\right)^2.$$

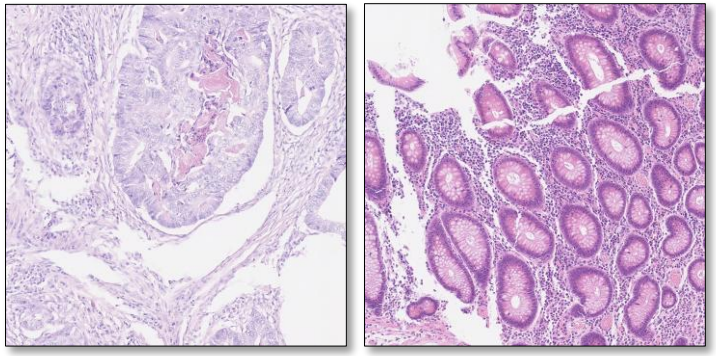
- **MSG for Omission** helps to overcome the problem of partial omission in the proposals

$$\mathcal{L}_{MSGO} = -\sum_{\hat{h},\hat{w}}^{\hat{H},\hat{W}} RP[:,\hat{h},\hat{w}] \cdot \ln \tilde{X}[:,\hat{h},\hat{w}].$$

IV. Experiment: Dataset



The Gland Segmentation challenge (GlaS) dataset contains **165** H&E-stained histopathology patches extracted from 16 WSIs. The patches are mostly 775 height and 522 width. The training set contains **85** patches, while testing set owns the rest **80**.



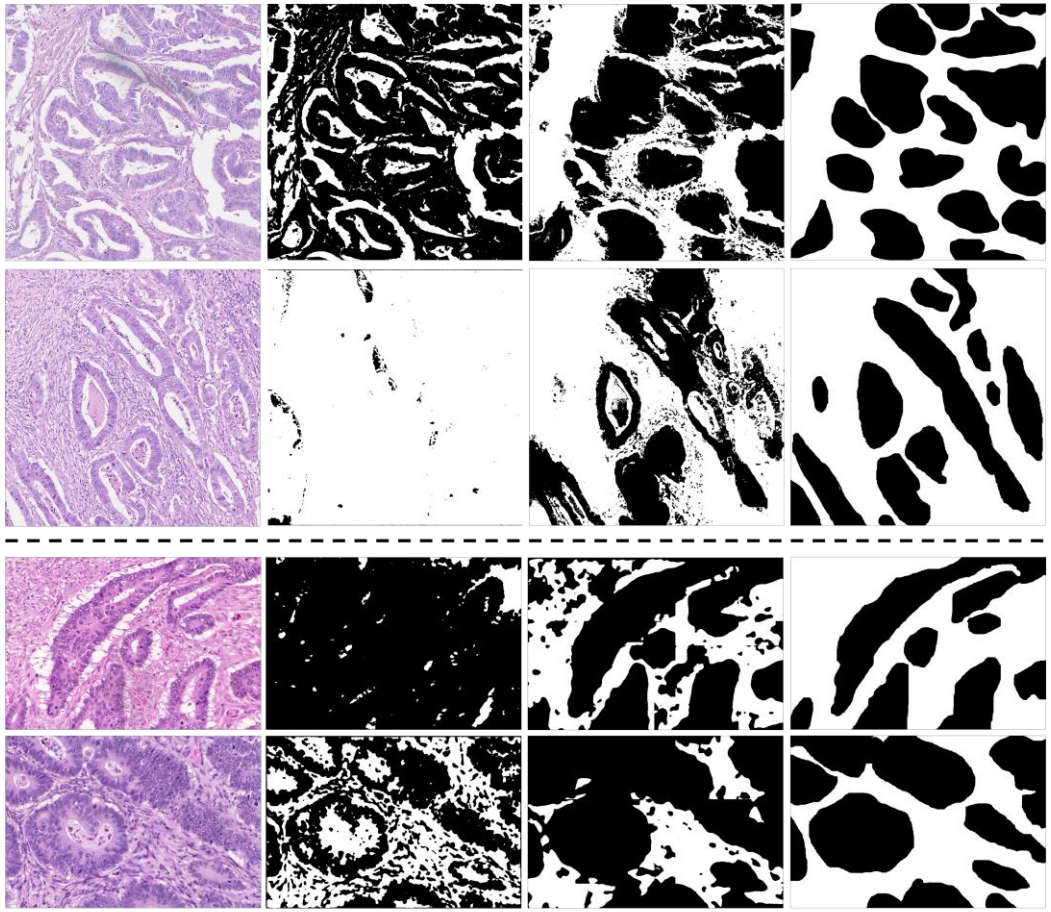
The Colorectal Adenocarcinoma Gland (CRAG) dataset owns **213** H&E-stained histopathology patches extracted from 38 WSIs. The patch size is 1512 × 1516 pixels. The training set contains **173** patches, and the testing set owns the rest **40.**

IV. Experiment: Results

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Table 1. Comparison results on GlaS and CRAG dataset. **Bold** and underline denote best and second-best results of the *unsupervised methods*.

Dataset	Method	Backbone	Supervision	F1	DICE	mIOU
GlaS Dataset	Unet [MICCAI 15']	U-Net	Fully	77.78%	79.04%	65.34%
	ResUNet [ITME 18']	U-Net	Fully	78.83%	79.48%	65.95%
	MedT [MICCAI 21']	Transformer	Fully	81.02%	82.08%	69.61%
	Randomly Initial	PSPNet	None	49.72%	48.63%	32.13%
	DeepCluster [ECCV 18']	PSPNet	None	57.03%	57.32%	40.17%
	PiCIE [CVPR 21']	PSPNet	None	64.98%	65.61%	48.77%
	DINO [ICCV 21']	PSPNet	None	56.93%	57.38%	40.23%
	DSM [CVPR 22']	PSPNet	None	68.18%	66.92%	49.92%
	SGSCN [MICCAI 21']	PSPNet	None	67.62%	68.72%	52.16%
	MSSG	PSPNet	None	**78.26%**	**77.09%**	**62.72%**
CRAG Dataset	Unet [MICCAI 15']	U-Net	Fully	82.70%	84.40%	70.21%
	VF-CNN [ICCV 17']	RotEqNet	Fully	71.10%	72.10%	57.24%
	MILDNet [MIA 19']	MILD-Net	Fully	86.90%	88.30%	76.95%
	PiCIE [CVPR 21']	PSPNet	None	67.04%	64.33%	52.06%
	DSM [CVPR 22']	PSPNet	None	67.22%	66.07%	52.28%
	SGSCN [MICCAI 21']	PSPNet	None	69.29%	67.88%	55.31%
	MSSG	PSPNet	None	**77.43%**	**77.26%**	**65.89%**

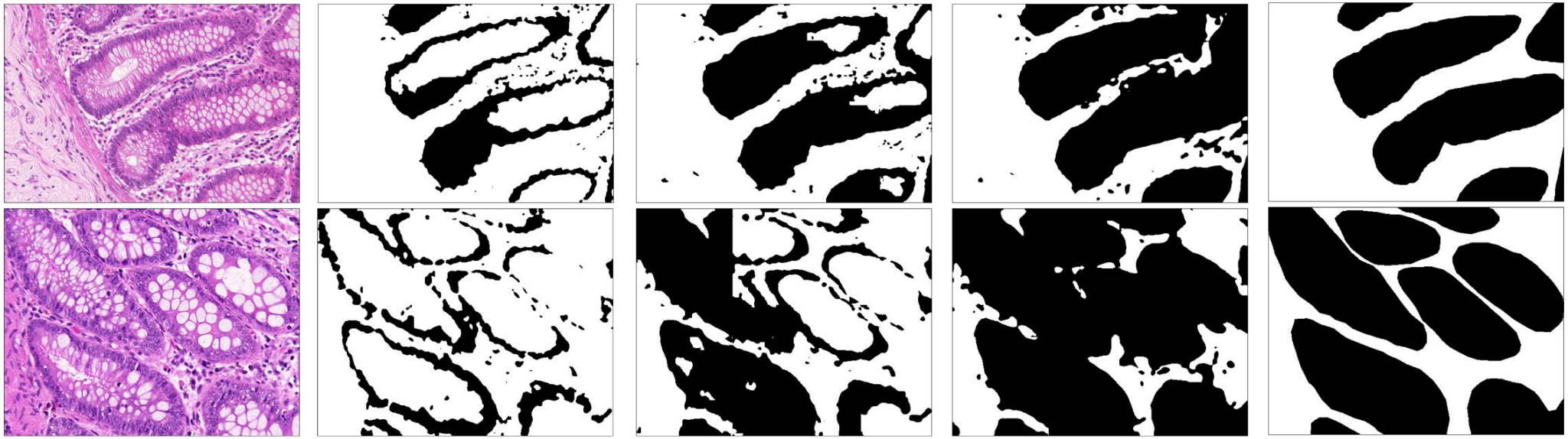


IV. Experiment: Ablations

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Performance gains with MSG modules

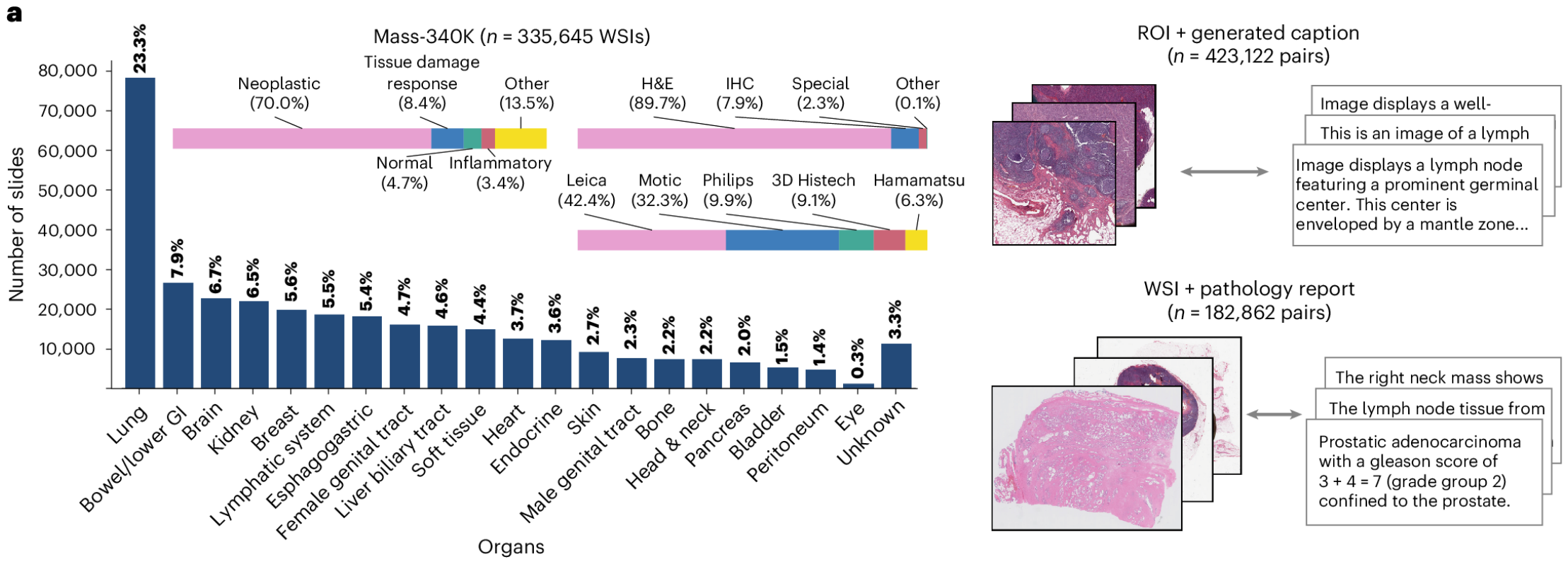
MSG variation	MSG omission	mIoU (%)	Gain (pp)
No	No	48.42	—
Yes	No	56.12	+7.70
No	Yes	50.18	+1.76
Yes	Yes	62.72	+14.30



Original Images | Ours w/o MSG for Variation and MSG for Omission | Ours w/o MSG for Omission | Ours w/ MSG for Variation and MSG for Omission | Ground Truth

4. Pathology: case studies

TITAN (1/8): whole-slide pretraining data

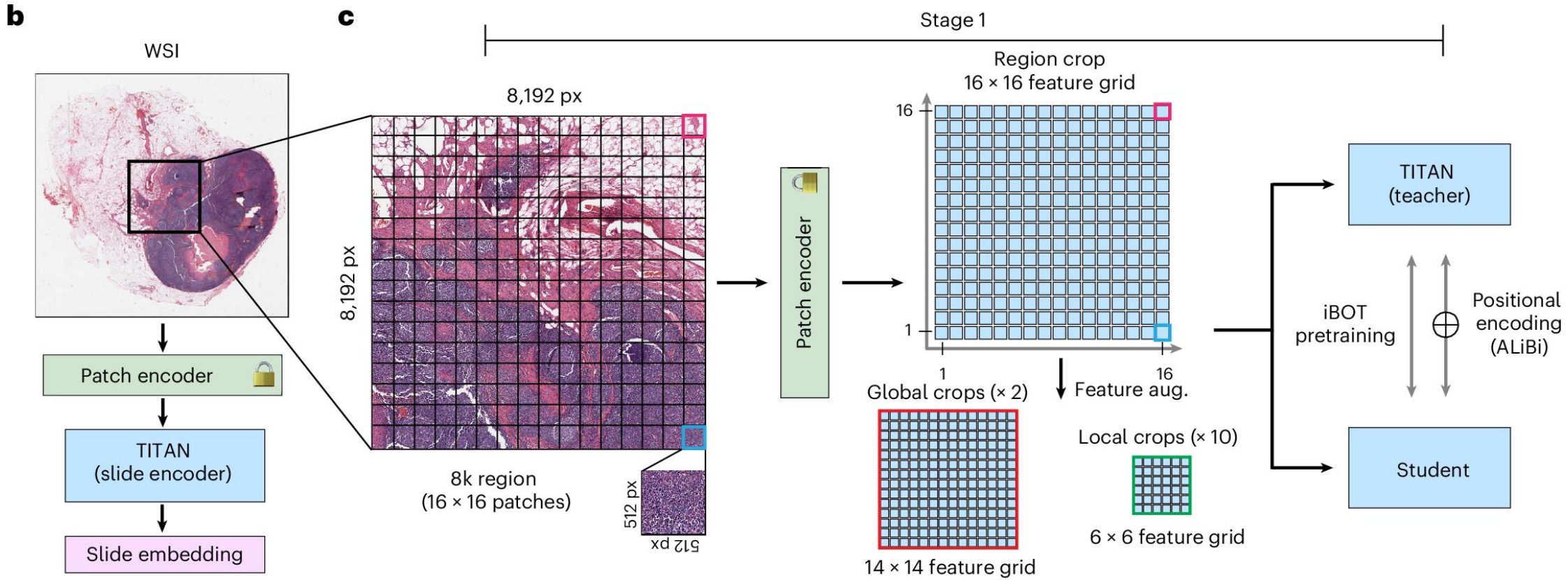


335,645 WSIs • 423,122 ROI–caption pairs • 182,862 slide–report pairs

Ding et al., Nature Medicine 2025. Fig. 1a. Mass-340K dataset composition.

4. Pathology: case studies

TITAN (2/8): spatial self-supervision

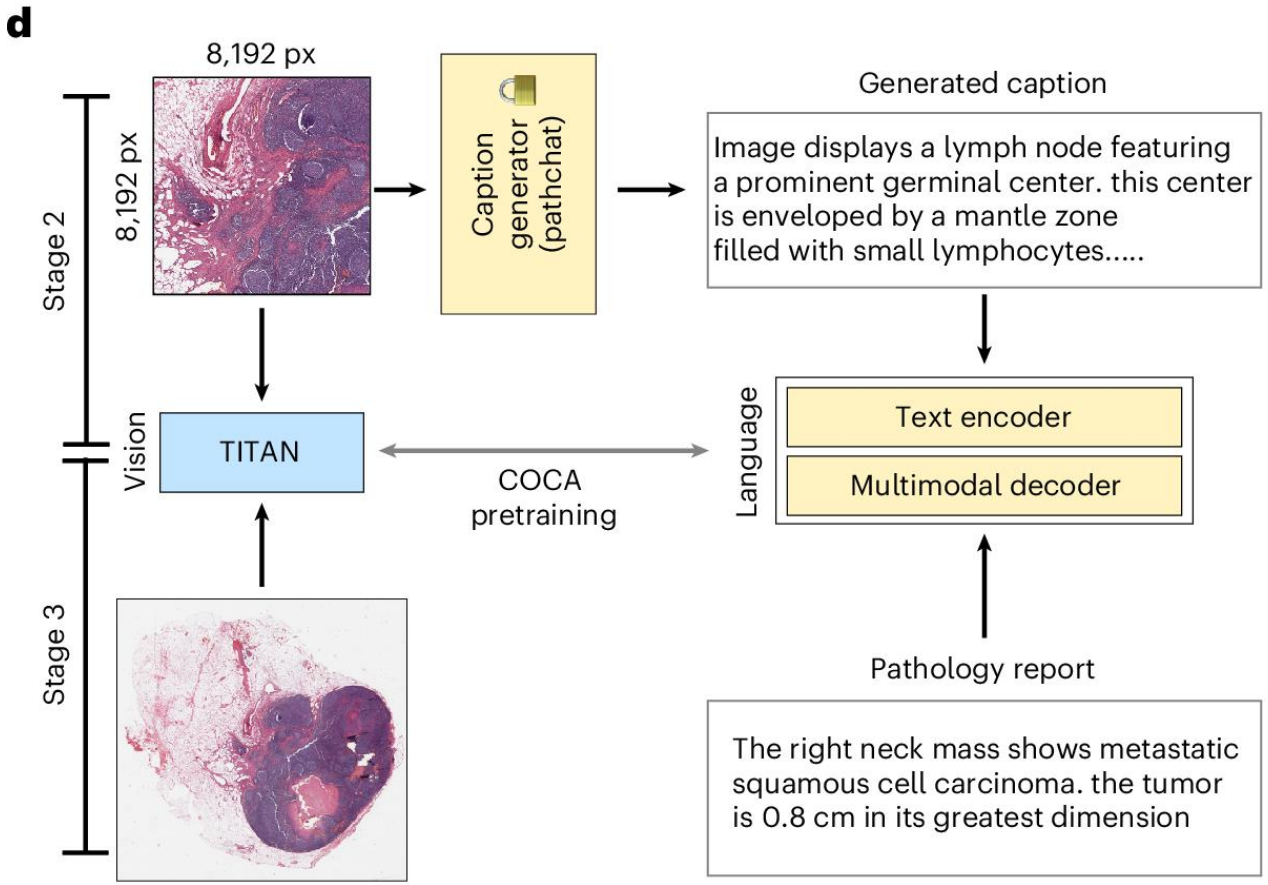


Frozen patch features preserve location. A slide encoder learns from global and local crops.

Ding et al., Nature Medicine 2025. Fig. 1b,c. Stage 1, patch features and slide encoder.

4. Pathology: case studies

TITAN (3/8): language supervision at two scales



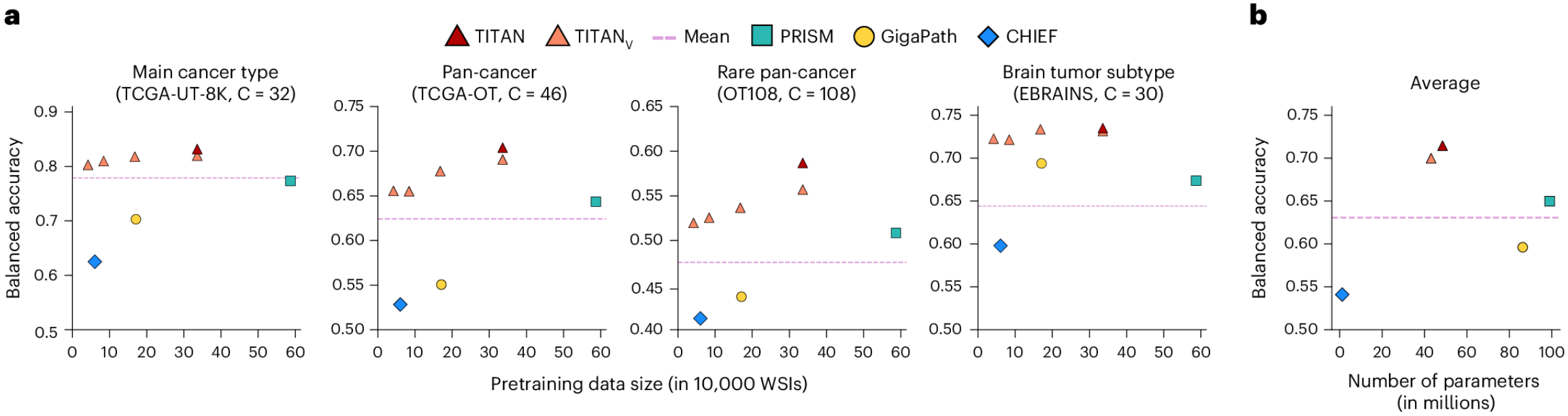
- Stage 2 aligns regions with generated captions
- Stage 3 aligns slides with pathology reports
- Contrastive and captioning objectives provide different signals

Synthetic captions can introduce systematic errors.

Ding et al., Nature Medicine 2025. Fig. 1d. Stages 2–3, CoCa pretraining.

4. Pathology: case studies

TITAN (4/8): data and parameter scaling



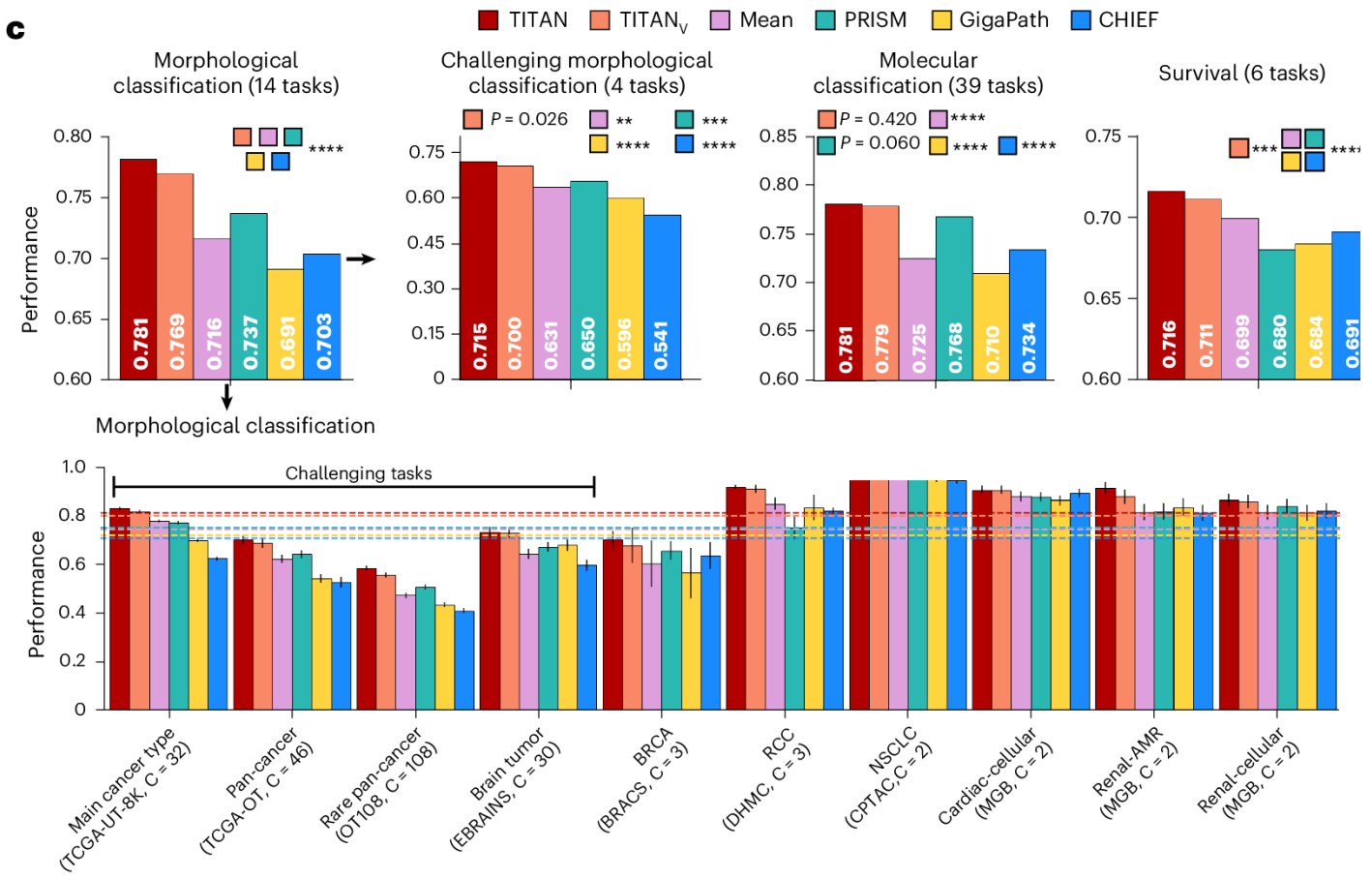
Compare data size within TITAN, then compare encoders across model families.

Across-family comparisons also change training data, objectives and architecture.

Ding et al., Nature Medicine 2025. Fig. 2a,b. Balanced accuracy and pretraining scale.

4. Pathology: case studies

TITAN (5/8): downstream tasks and baselines



Read each task's metric:

Multiclass: balanced accuracy

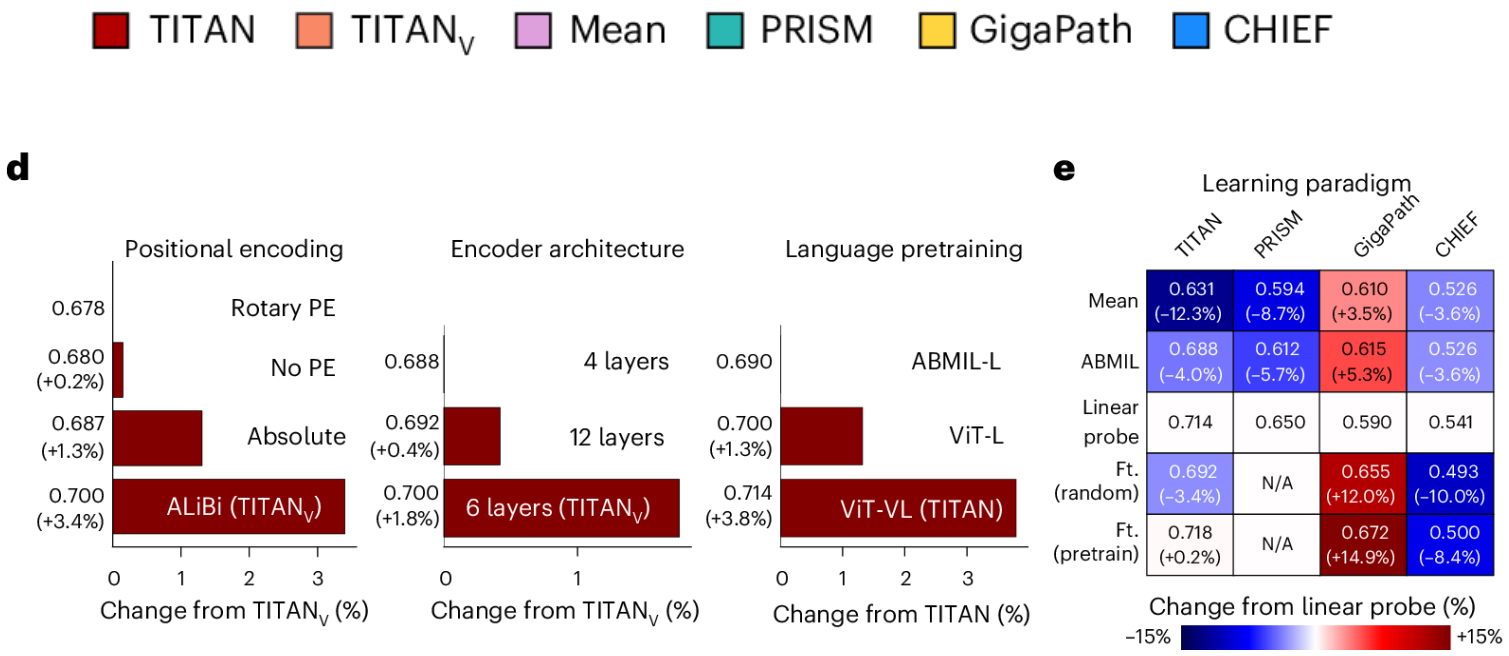
Binary: AUROC

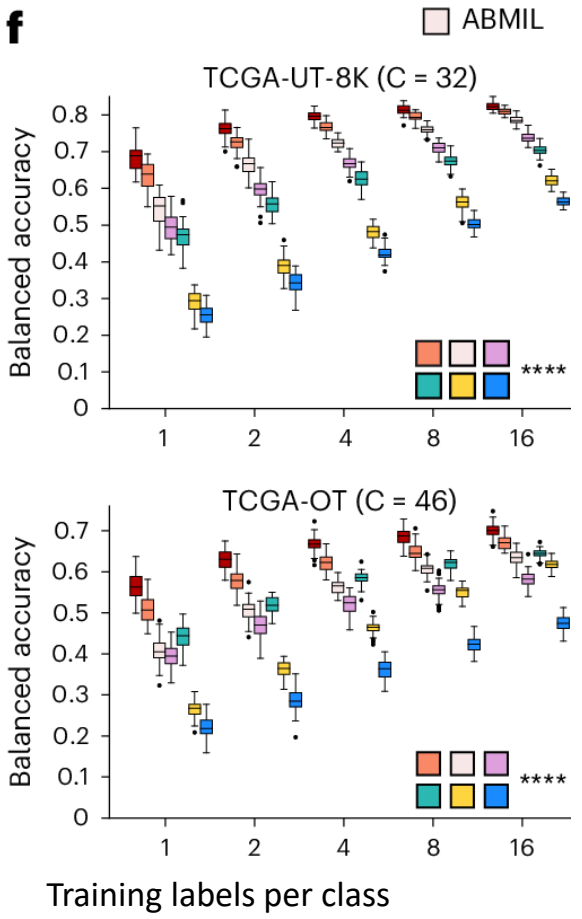
Survival: C-index

Ding et al., Nature Medicine 2025. Fig. 2c. Preserve task-specific metrics and reported uncertainty.

4. Pathology: case studies

TITAN (6/8): ablations and limited labels



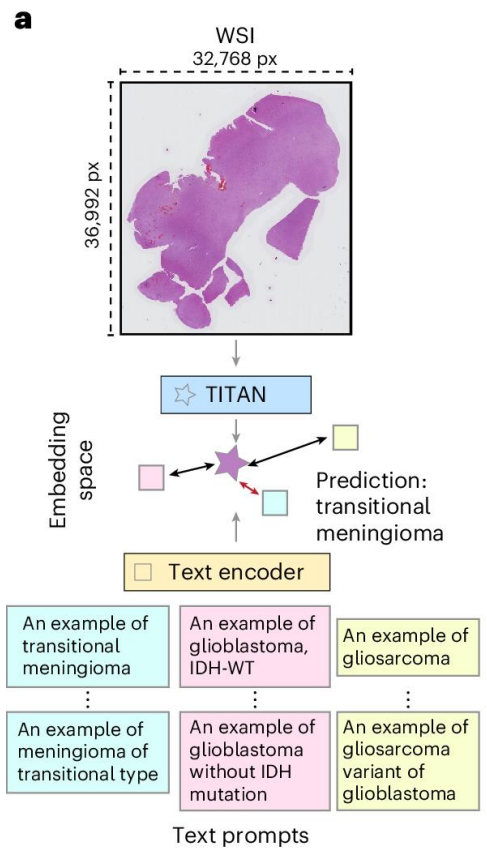


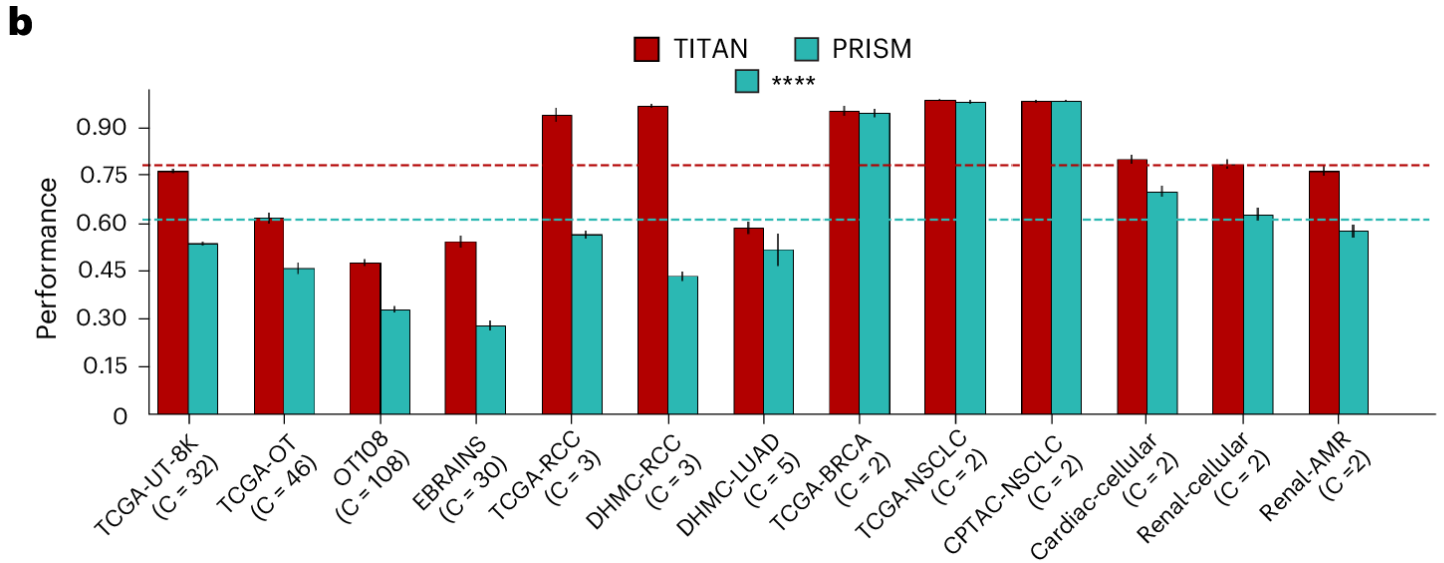
ALiBi, encoder depth and language supervision contribute differently.

Ding et al., Nature Medicine 2025. Fig. 2d–f. Architecture ablations and few-shot learning.

4. Pathology: case studies

TITAN (7/8): zero-shot classification



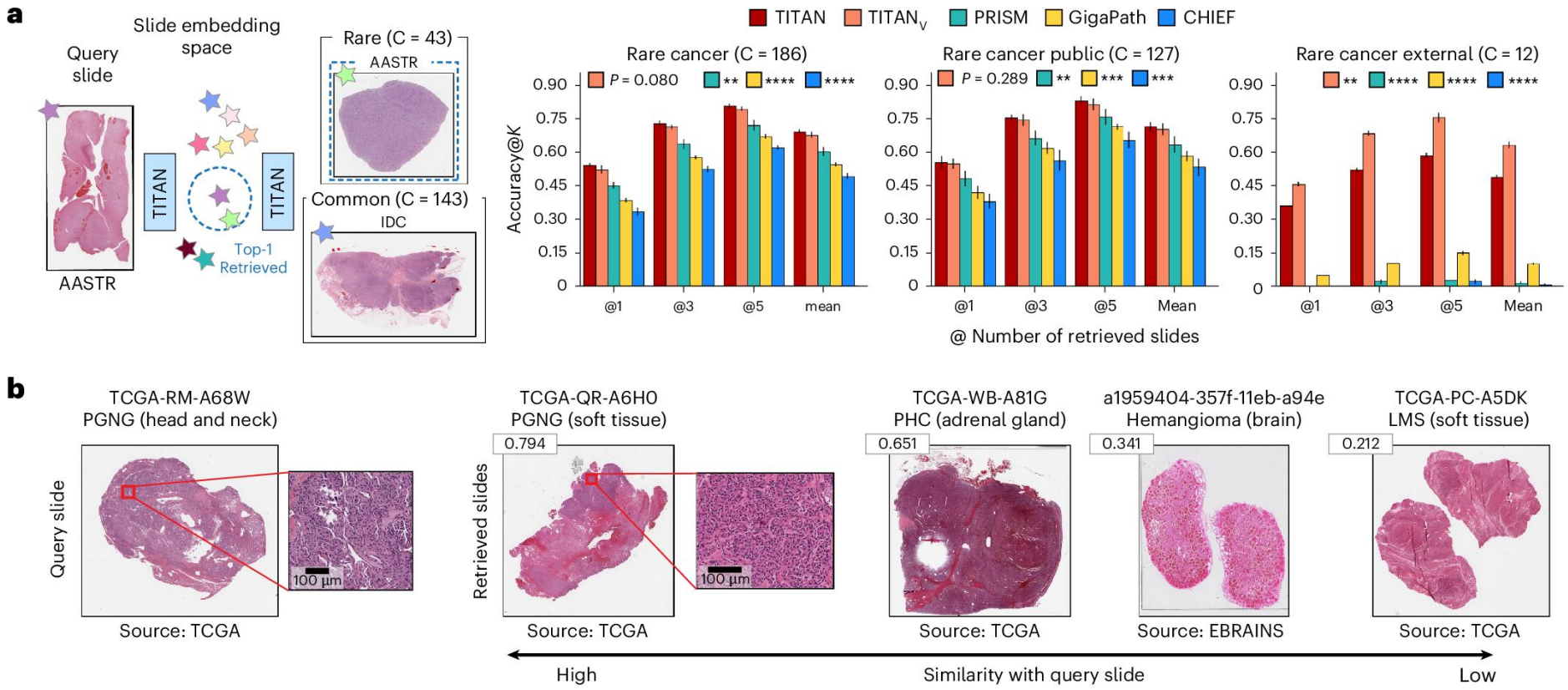


Text prompts define candidate classes. The nearest text embedding gives the prediction.

Ding et al., Nature Medicine 2025. Fig. 3a,b. Prompt-based prediction and zero-shot results.

4. Pathology: case studies

TITAN (8/8): retrieval and its limits

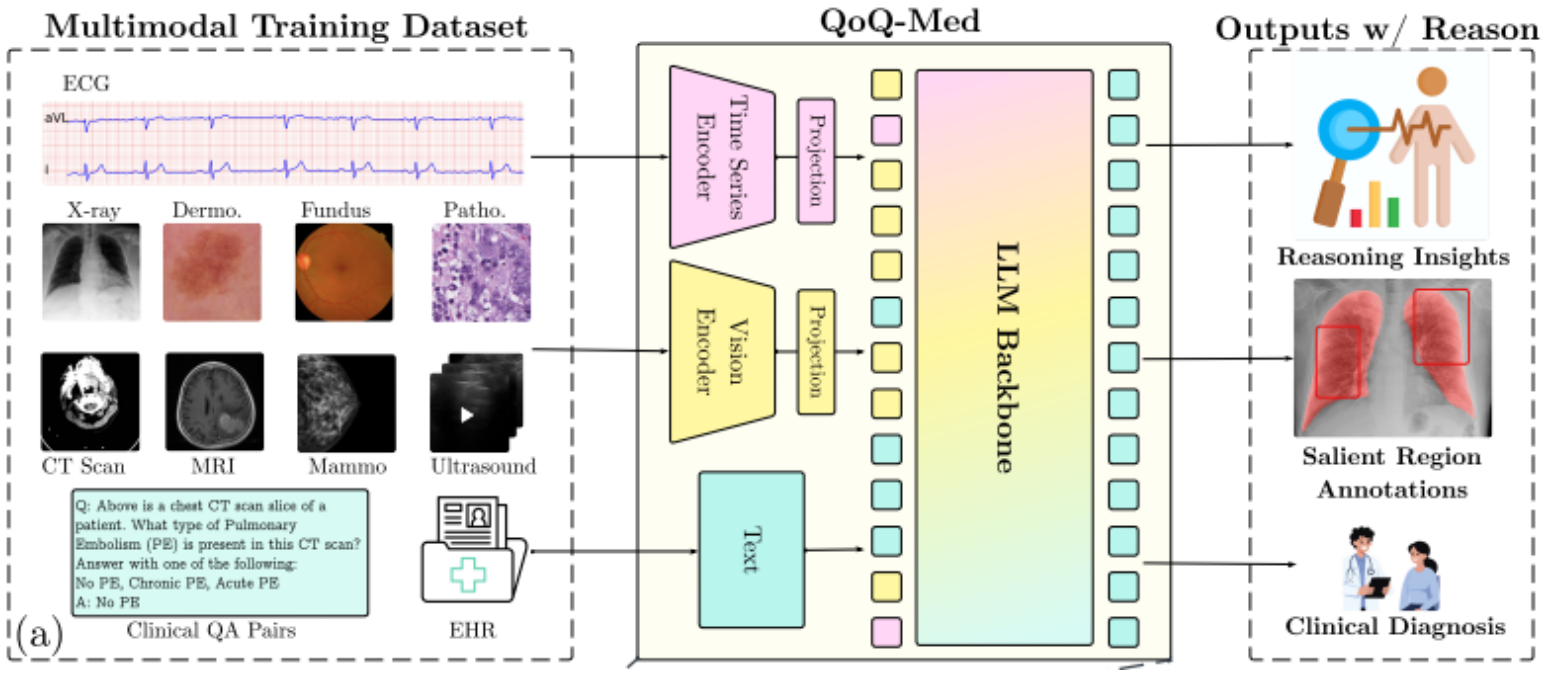


Nearest slides can share morphology. Inspect external rare-cancer results as well as the mean.

Ding et al., Nature Medicine 2025. Fig. 4a,b. Retrieval accuracy and a ranked histology example.

5. Medical model post-training

QoQ-Med (1/5): multimodal clinical questions



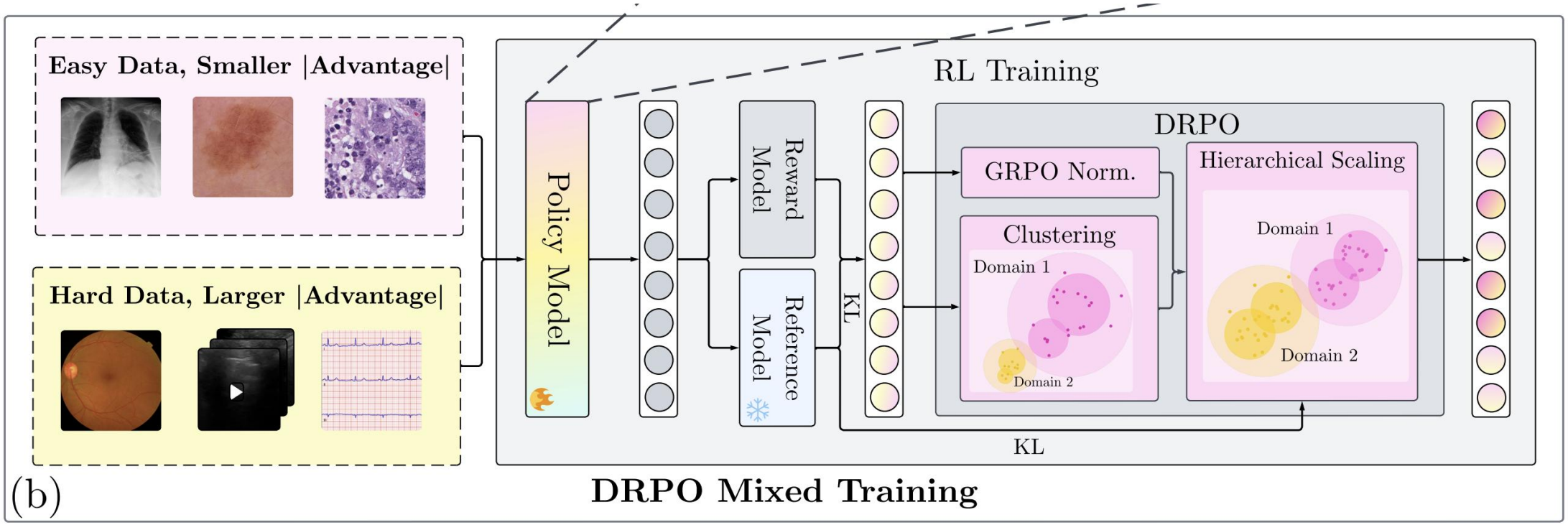
Vision and time-series encoders connect to an LLM through projections.

The study trains on 2.61 million instruction pairs across clinical domains.

Dai et al., NeurIPS 2025. Fig. 1a. Model inputs, projections and outputs.

5. Medical model post-training

QoQ-Med (2/5): GRPO and domain imbalance



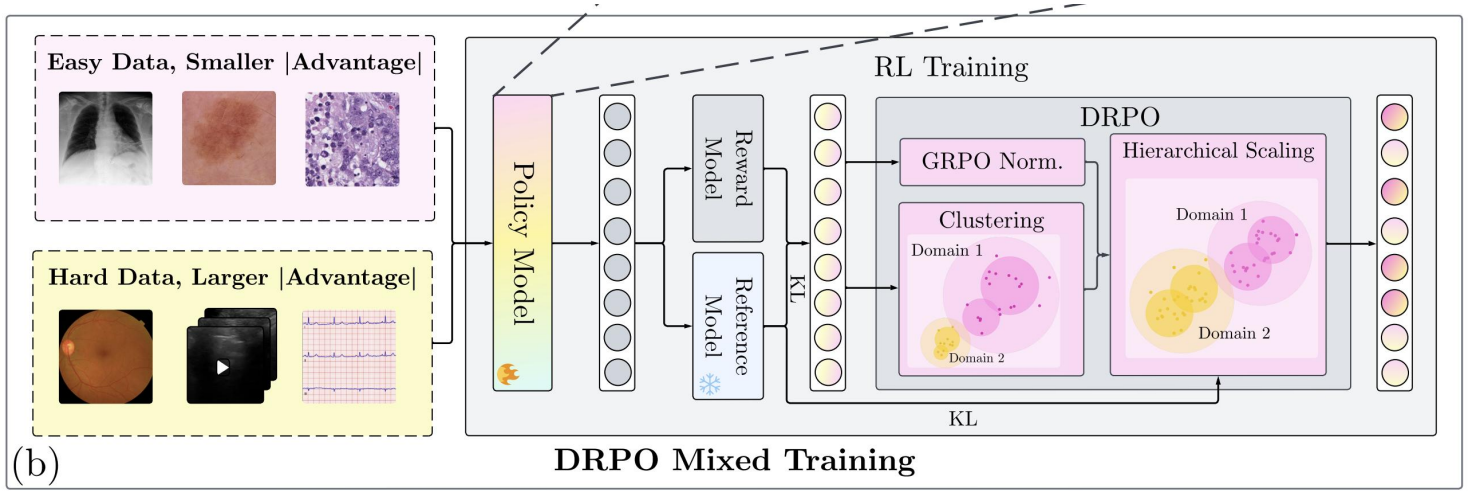
$$A_i = (r_i - \text{mean}(r)) \,/\, (\text{std}(r) + \varepsilon)$$

Several responses to the same question form one group.

Dai et al., NeurIPS 2025. Fig. 1b, equation (1), reward design.

5. Medical model post-training

QoQ-Med (3/5): domain-aware scaling



$$T = \max(\sqrt{(N \cdot \mu)}, \varepsilon)$$

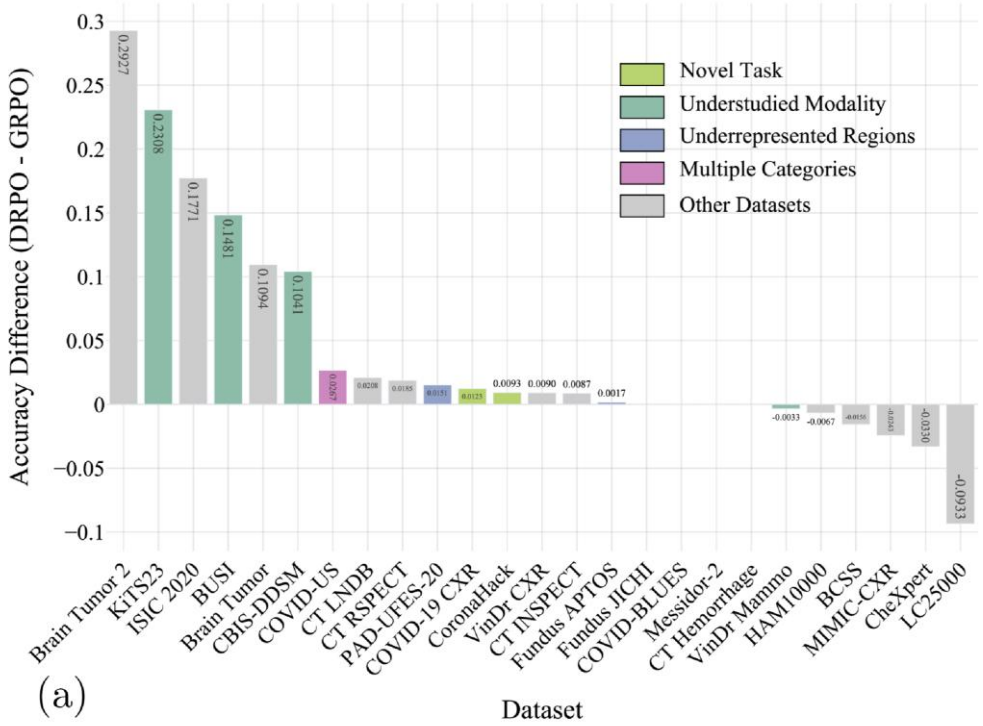
- Cluster reward vectors within each domain
- Smaller or harder groups receive greater weight
- Rescale advantages before the policy update

The same diagnosis reward can produce different training emphasis across domains.

Dai et al., NeurIPS 2025. Fig. 1b detail and equations (2)–(3).

5. Medical model post-training

QoQ-Med (4/5): overall scores and exceptions



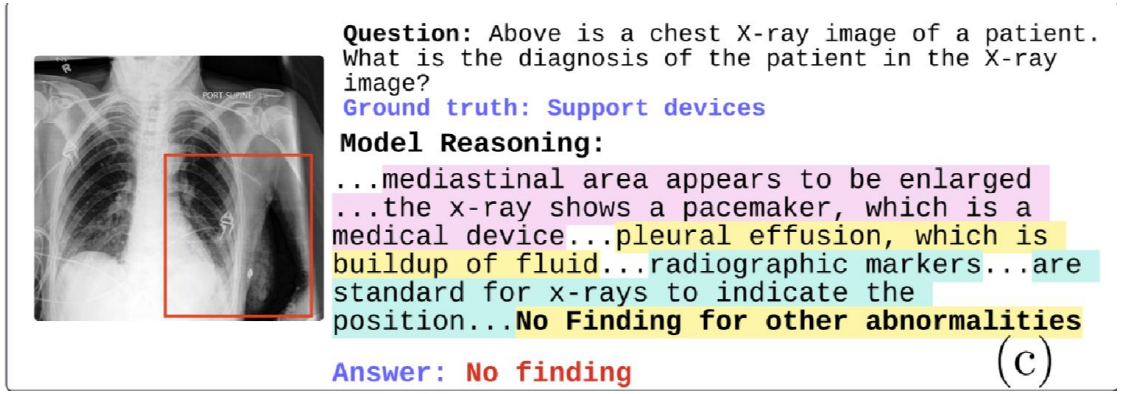
Training	Overall F1	US F1	Path. F1
SFT	.139	.235	.083
GRPO	.193	.146	.286
DRPO domain only	.237	.098	.388
DRPO no KL	.283	.228	.300
DRPO	.295	.223	.265

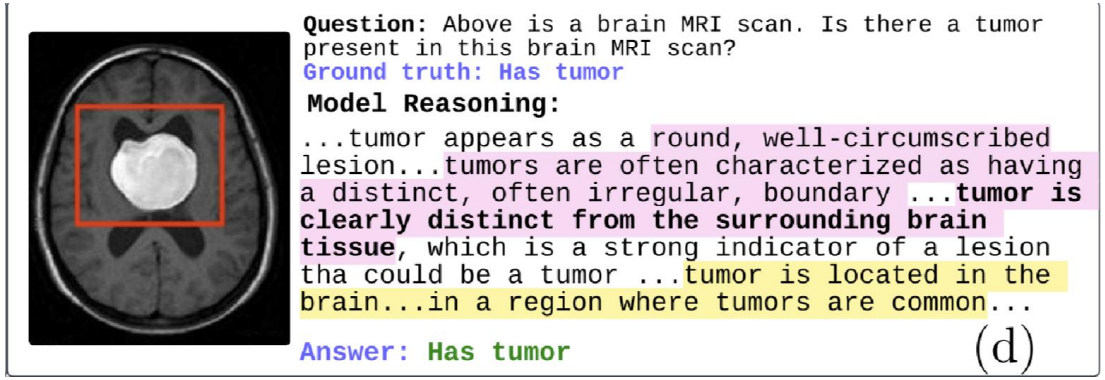
DRPO improves mean F1 over GRPO. Ultrasound is below SFT, and pathology is below GRPO.

Dai et al., NeurIPS 2025. Fig. 2 and Table 2. Selected columns, mean of four runs.

5. Medical model post-training

QoQ-Med (5/5): evidence and a failure case





Chest X-ray: support device in the label, but “No finding” in the model answer.

Brain MRI: a matching answer and region still require systematic validation.

Dai et al., NeurIPS 2025. Fig. 4c,d. The paper’s examples include an incorrect final answer.

Summary

Questions that connect the methods



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- What information and supervision does the model use?
- Which baseline isolates the claimed contribution?
- Which experiment reveals a failure or limitation?

Imaging physics, data quality and evaluation remain central as model families change.